Implicit-Euler Value Iteration: Long-Horizon Planning via Stable Integration of the Bellman Residual Flow

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Abstract

Value Iteration (VI) is the unit-step explicit-Euler discretization of the continuous-time Bellman residual flow $\dot{V} = \mathrm{BR}(V)$, and its linear dependence on the effective horizon $1/(1-\gamma)$ reflects the step-size restriction that explicit integration imposes on stiff dynamics. We propose Implicit-Euler Value Iteration (IE-VI), a planning routine based on the implicit-Euler discretization of the same flow. An exact implicit step is unconditionally contractive for any step size, but requires solving an equation in the true Bellman operator. IE-VI approximates this step by iterated defect correction: a short sequence of cheap planning rounds in an approximate model $\hat{\mathcal{P}} \approx \mathcal{P}$ (a simulator or a learned dynamics model), with the true model queried once per round to refresh a residual correction term. In Policy Evaluation and Control, suitable choices of the step size and number of planning rounds ensure that IE-VI converges to the true value function of $\mathcal{P}$ for every Markov surrogate $\hat{\mathcal{P}}$, however inaccurate. This avoids both the asymptotic bias of planning with $\hat{\mathcal{P}}$ alone and the small-error restrictions of prior fixed-point-preserving methods. For sufficiently accurate surrogates, true-model query complexity grows polylogarithmically with the effective horizon rather than linearly; for arbitrary surrogates, it stays within a logarithmic factor of VI's bound. Finally, we introduce Adaptive IE-VI, which removes the need to know the model-error scale, and IE-DYNA, a variant for tabular RL.

1 Introduction

The value function is a central object in Reinforcement Learning (RL) and Planning problems. In the Planning problem, when the model $\mathcal{P}$ is known, the Value Iteration (VI) is a commonly used approach to compute the value function. It also underlies many value-based RL algorithms, such as Temporal Difference Learning [Sutton, 1988], Fitted Value Iteration [Ernst et al., 2005, Munos and Szepesvári, 2008, Farahmand et al., 2009, Tosatto et al., 2017, Chen and Jiang, 2019], Deep Q-Network and variants [Mnih et al., 2015, Van Hasselt et al., 2016]. For long planning horizons, however, VI and its RL counterparts require many queries to (or samples from) $\mathcal{P}$. With discount factor γ, VI converges at rate $O(\gamma^k)$ after k iterations, so a fixed accuracy costs approximately $O(\frac{1}{1-\gamma})$ queries to $\mathcal{P}$, a large count when $\gamma \approx 1$. Long planning horizons are especially impractical when access to $\mathcal{P}$ is expensive. Examples include querying $\mathcal{P}$ through a slow simulator or sampling from $\mathcal{P}$ through physical interaction, as in robotics or healthcare.

Model-based RL attempts to address this challenge using an approximate surrogate model $\hat{\mathcal{P}}$ [Sutton, 1990, Deisenroth and Rasmussen, 2011, Schrittwieser et al., 2020, Hafner et al., 2020]. A prototypical MBRL agent learns a model $\hat{\mathcal{P}}$ from environment interactions, then uses $\hat{\mathcal{P}}$ instead of the true model $\mathcal{P}$. Relying on $\hat{\mathcal{P}}$ reduces the number of interactions needed. The downside, however, is that

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relying on the surrogate model $\hat{\mathcal{P}}$ alone leaves a residual error in the order of $\frac{\gamma\|\mathcal{P}-\hat{\mathcal{P}}\|_\infty}{1-\gamma}$. With finite samples, statistical errors make $\|\mathcal{P}-\hat{\mathcal{P}}\|$ non-zero. Even asymptotically, model approximation error can persist whenever the surrogate's model class is smaller than the true environment, i.e., the big world hypothesis [Javed and Sutton, 2024]. This leaves an important question: *Can an approximate surrogate model $\hat{\mathcal{P}}$ help compute the value function of $\mathcal{P}$ with fewer queries to $\mathcal{P}$ than VI requires?* Here, we seek a procedure that converges to the true value function of $\mathcal{P}$, rather than to the value function of either $\hat{\mathcal{P}}$ or a mixture of the two models, as many existing MBRL algorithms do.

This work proposes and analyzes *Implicit-Euler Value Iteration* (IE-VI), a method that uses a surrogate model $\hat{\mathcal{P}}$ to converge to the true value function of $\mathcal{P}$, no matter how inaccurate $\hat{\mathcal{P}}$ is, with suitable parameter choices (Theorem 2). Our analysis shows how a more accurate surrogate permits a stronger contraction and fewer queries to $\mathcal{P}$. Concretely, for $0 < \gamma < 1$, when the model error $\|\mathcal{P}-\hat{\mathcal{P}}\|_\infty$ is smaller than $C(1-\gamma)\log(1/(1-\gamma))/\gamma$, for a constant $C > 0$ independent of γ, IE-VI's true-model query complexity is polylogarithmic in $1/(1-\gamma)$, improving on VI's standard linear dependence (Theorem 2). For arbitrary surrogates, its worst-case query bound remains within a logarithmic factor of VI's.

The idea behind IE-VI comes from interpreting VI as the *explicit (or forward)* Euler discretization of the continuous-time system $\dot{V}(t) = \mathrm{BR}(V) = TV - V$, where BR is the Bellman Residual (BR) and T is the Bellman operator, both defined later. We call this Ordinary Differential Equation (ODE) the *Bellman residual flow*. Conventional VI corresponds to $V_{k+1} = V_k + h\,\mathrm{BR}(V_k) = V_k + h(TV_k - V_k)$ with step size $h = 1$. Explicit Euler, however, limits the step size h: large h can destabilize the discrete-time dynamics. This restriction limits progress along the Bellman residual flow towards its equilibrium: the true value function, the Bellman equation's fixed point. This is also how the discount factor slows convergence (Section 3 and Section 3.1).

IE-VI is instead based on the *implicit (or backward)* Euler discretization of the BR flow: $V_{k+1} = V_k + h\,\mathrm{BR}(V_{k+1})$. The exact implicit step has no stability restriction on h. In the limit $h \to \infty$, one implicit-Euler step of the BR flow solves for the true value function. This is the first key idea behind IE-VI.

This step, however, is expensive in queries to $\mathcal{P}$: it is equivalent to solving a Planning problem with reward $\frac{1}{1+h}V_k + \frac{h}{1+h}r^\pi$ and effective discount $\gamma_h := \frac{h}{1+h}\gamma$ using the same expensive dynamics $\mathcal{P}$.

The second key idea behind IE-VI is to solve each step's Planning problem with the surrogate model $\hat{\mathcal{P}}$ instead of $\mathcal{P}$. To mitigate the effect of error in $\hat{\mathcal{P}}$ relative to $\mathcal{P}$, IE-VI uses *iterated defect correction* [Stetter, 1978, Böhmer and Stetter, 1984] from numerical analysis. This technique replaces an expensive operator solve with successive cheaper surrogate equations, each grounded in the true operator by a residual correction (Section 3.2 and Section 4).

Once we approximate the implicit step using $\hat{\mathcal{P}}$, however, the unconditional stability of the exact step is no longer guaranteed, so we cannot take $h \to \infty$ without accounting for surrogate error. In fact, taking $h \to \infty$ with a single defect-correction round recovers Operator-Splitting Value Iteration (OS-VI; Rakhsha et al., 2022), the closest prior work to ours. This infinite-step limit contracts toward the true value function when the surrogate is sufficiently accurate, but can diverge for inaccurate surrogates. Choosing a suitable finite step size and sufficiently many correction rounds allows IE-VI to guarantee convergence for any Markov surrogate. Our analysis also establishes acceleration over VI in regimes beyond OS-VI's standard contraction guarantee (Section 4.2).

The step-size choices in our analysis depend on the model error $\|\mathcal{P}-\hat{\mathcal{P}}\|_\infty$. When this error is unknown, Adaptive IE-VI (Section 4.3) adjusts h using the true Bellman residual. We also adapt the update to tabular RL through IE-DYNA (Section 4.4), which estimates the residual correction from sampled transitions instead of evaluating exact expectations under $\mathcal{P}$. Section 5 empirically evaluates the planning and RL variants.

2 Discounted MDP setup and value iteration

We consider a discounted Markov Decision Process (MDP) $(\mathcal{X}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$ with discount factor $\gamma \in [0, 1)$, following standard conventions [Bertsekas and Tsitsiklis, 1996, Sutton and Barto, 2018, Szepesvári, 2010, Meyn, 2022]. Value functions belong to the Banach space $\mathcal{B}(\mathcal{X})$ of bounded

measurable functions on $\mathcal{X}$ under the supremum norm

$$\|V\|_\infty := \sup_{x\in\mathcal{X}} |V(x)|.$$

Appendix A contains the measure-theoretic details for general state spaces; our empirical evaluation uses finite MDPs.

For each action $a \in \mathcal{A}$ and stationary policy π, the true transition kernel $\mathcal{P}$ and approximate Markov kernel $\hat{\mathcal{P}}$ induce Markov operators $\mathcal{P}^a, \hat{\mathcal{P}}^a$ and $\mathcal{P}^\pi, \hat{\mathcal{P}}^\pi$ on $\mathcal{B}(\mathcal{X})$, each averaging next-state values under its model. Let $R^a(x)$ denote the expected immediate reward for taking action a in state x, and $r^\pi(x) := \int_{\mathcal{A}} R^a(x)\,\pi(da \mid x)$ the expected reward under π.

Policy Evaluation seeks the value function V^π of policy π; Control seeks the optimal value function $V^\star$. For $x \in \mathcal{X}$, define the Bellman operator T^π and Bellman optimality operator $T^\star$:

$$(T^\pi V)(x) := r^\pi(x) + \gamma(\mathcal{P}^\pi V)(x), \qquad (T^\star V)(x) := \sup_{a\in\mathcal{A}} \{R^a(x) + \gamma(\mathcal{P}^a V)(x)\}. \tag{1}$$

These operators are γ-contractions in the supremum norm. By the Banach fixed-point theorem, their unique fixed points are V^π and $V^\star$, respectively. Let $\pi^\star$ denote an arbitrary policy greedy for $V^\star$.

Value Iteration (VI) computes these value functions by repeated Bellman updates using $\mathcal{P}$:

$$V_{k+1} = T^\pi V_k \quad \text{(Policy Evaluation)}, \qquad V_{k+1} = T^\star V_k \quad \text{(Control)}. \tag{2}$$

Since both operators are γ-contractions, value error decays as $O(\gamma^k)$. Reaching ϵ-accuracy takes $O(\ln(1/\epsilon)/(1-\gamma))$ iterations, each applying $\mathcal{P}$ once. As γ approaches 1, the growing effective horizon $1/(1-\gamma)$ makes VI expensive.

3 Bellman residual flow and its numerical integration

We take the Bellman residual—the increment in a VI update—as the rate of change of a continuous-time value estimate, defining the Bellman residual flow. We recover VI as the flow's unit-step explicit-Euler discretization for Policy Evaluation and Control. Section 3.1 motivates implicit integration through the notion of stiffness. Section 3.2 shows how to approximate implicit steps using surrogate-based defect correction.

For a fixed policy π, define the linear operator $L^\pi := I - \gamma\mathcal{P}^\pi$. The policy-evaluation Bellman residual of a value estimate V is

$$\mathrm{BR}^\pi(V) := T^\pi V - V = r^\pi - L^\pi V.$$

For Control, the nonlinear Bellman optimality residual is

$$\mathrm{BR}^\star(V) := T^\star V - V.$$

The Bellman residual flows satisfy

$$\dot{V}(t) = \frac{\mathrm{d}V(t)}{\mathrm{d}t} = \begin{cases} \mathrm{BR}^\pi(V(t)) & \text{(Policy Evaluation)} \\ \mathrm{BR}^\star(V(t)) & \text{(Control)}. \end{cases} \tag{3}$$

At equilibrium, $\dot{V}(t) = 0$, giving the Bellman fixed-point equations $T^\pi V = V$ and $T^\star V = V$. The γ-contractive Bellman operators ensure unique solutions V^π and $V^\star$.

The explicit (forward) and implicit (backward) Euler methods discretize $\dot{V}(t) = f(V(t))$ on $\mathcal{B}(\mathcal{X})$ with step size $h > 0$ as

$$V_{k+1} = V_k + hf(V_k) \quad \text{(explicit Euler)}, \qquad V_{k+1} = V_k + hf(V_{k+1}) \quad \text{(implicit Euler)}.$$

Applying explicit Euler to the Bellman residual flow for Policy Evaluation gives

$$V_{k+1} = V_k + h\,\mathrm{BR}^\pi(V_k) = V_k + h(T^\pi V_k - V_k).$$

With step size $h = 1$, explicit Euler gives $V_{k+1} = T^\pi V_k$ (Policy Evaluation) and $V_{k+1} = V_k + \mathrm{BR}^\star(V_k) = T^\star V_k$ (Control), recovering conventional VI.

3.1 Implicit integration for long-horizon stiffness

To understand why explicit Euler, and therefore VI, struggles with long planning horizons, we draw on the notion of numerical stiffness from numerical analysis. Stiffness arises when fast-decaying components restrict the stable step size of an explicit integrator, while slow components require a long integration time to approach equilibrium. Here, these components represent different patterns of value-estimation error across states.

For Policy Evaluation, the error $e(t) := V(t) - V^\pi$ evolves according to $\dot{e}(t) = -L^\pi e(t)$. Because $\mathcal{P}^\pi \mathbf{1} = \mathbf{1}$, a uniform error shared across all states satisfies $L^\pi \mathbf{1} = (1-\gamma)\mathbf{1}$ and decays at rate $1-\gamma$, requiring continuous time of order $1/(1-\gamma)$ to contract by a fixed factor. In contrast, non-uniform error differences across states can change on a timescale that remains bounded as $\gamma \to 1$. Under explicit Euler, $e_{k+1} = (I - hL^\pi)e_k$, these fast components can restrict the step size h needed to keep the discrete-time dynamics contractive in $\|\cdot\|_\infty$ (for instance, requiring $0 < h < 2/(1+\gamma)$ when states alternate deterministically; see Appendix B.2 for a closed-form two-state example). When contractive explicit step sizes remain bounded as $\gamma \to 1$, reducing a uniform error by a fixed factor requires $\Omega((1-\gamma)^{-1})$ steps. VI, corresponding to $h = 1$, attains this order.

We quantify stiffness using a norm-based ratio. For Policy Evaluation, define

$$\rho_{\text{stiff}}(L^\pi) := \|L^\pi\|_\infty \|(L^\pi)^{-1}\|_\infty.$$

The first factor measures the largest instantaneous change in error per unit error, while the inverse norm captures the horizon scale. Appendix B establishes the exact identities

$$\|(L^\pi)^{-1}\|_\infty = \frac{1}{1-\gamma}, \qquad \|L^\pi\|_\infty = 1 + \gamma - 2\gamma p_{\text{inf}},$$

where $p_{\text{inf}} := \inf_{x \in \mathcal{X}} \mathcal{P}^\pi(\{x\} \mid x)$ is the infimum of the self-transition probabilities. Hence

$$\rho_{\text{stiff}}(L^\pi) = \frac{1 + \gamma - 2\gamma p_{\text{inf}}}{1-\gamma}.$$

For a fixed transition kernel with $p_{\text{inf}} < 1$, this ratio diverges as $\gamma \to 1$. Appendix B also derives the analogous nonlinear stiffness ratio for Control, with the infimum over states and actions.

Implicit Euler allows large steps while damping fast components, a standard approach to stiff dynamics [Hairer and Wanner, 1996, Ch. IV]. For Policy Evaluation, its error update is

$$e_{k+1} = (I + hL^\pi)^{-1} e_k, \qquad e_k := V_k - V^\pi.$$

Proposition 6 in Appendix A establishes that the exact implicit step is unconditionally contractive:

$$\|e_{k+1}\|_\infty \le \frac{1}{1 + h(1-\gamma)} \|e_k\|_\infty$$

for every $h > 0$. For fixed h, this contraction factor approaches one as $\gamma \to 1$. Scaling h proportionally to $1/(1-\gamma)$ instead keeps the factor fixed. In particular, choosing $h = 1/(1-\gamma)$ gives $\|e_{k+1}\|_\infty \le \frac{1}{2}\|e_k\|_\infty$, so each exact implicit step at least halves the error norm, independently of the horizon.

Realizing this benefit requires computing the implicit step efficiently. Writing $K(h) := I + hL^\pi$, each step solves

$$K(h) V_{k+1} = V_k + h r^\pi,$$

a Policy Evaluation problem with

$$\text{reward: } \frac{1}{1+h} V_k + \frac{h}{1+h} r^\pi, \qquad \text{discount: } \gamma_h := \frac{h\gamma}{1+h},$$

still under the true model $\mathcal{P}^\pi$. Solving it exactly may be as expensive as the original planning problem. Section 3.2 approximates this step using inexpensive surrogate planning and true-model residual evaluations.

3.2 Iterated defect correction of the Bellman resolvent

To approximate the implicit step $K(h)V_{k+1} = V_k + hr^\pi$ without inverting $K(h)$, we use *iterated defect correction* (type A) [Stetter, 1978, Böhmer and Stetter, 1984]. Consider a linear operator equation $AU = b$ where A is expensive to invert but cheap to evaluate, and suppose an approximate operator $\hat{A} \approx A$ is cheap to both evaluate and invert. Given a current estimate U_j, defect correction evaluates the residual $b - AU_j$ under the true operator and uses $\hat{A}^{-1}$ to compute a corrective displacement:

$$U_{j+1} = U_j + \hat{A}^{-1}(b - AU_j), \qquad \text{or equivalently,} \qquad \hat{A}U_{j+1} = b + (\hat{A} - A)U_j. \tag{4}$$

Each refinement requires one forward evaluation of A and one solve with the cheap surrogate $\hat{A}$.

Holding the outer iterate V_k fixed and initializing at $U_0 := V_k$, we apply Equation (4) to the implicit-step equation with $A = K(h) = I + hL^\pi$, $b = V_k + hr^\pi$, and the surrogate implicit-step operator $\hat{A} = \hat{K}(h) := I + h\hat{L}^\pi$, where $\hat{L}^\pi := I - \gamma\hat{\mathcal{P}}^\pi$. Because the operator discrepancy is $\hat{K}(h) - K(h) = h(\hat{L}^\pi - L^\pi) = h\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)$, the right-hand form of Equation (4) becomes

$$\hat{K}(h)U_{j+1} = V_k + h\big(r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_j\big). \tag{5}$$

Factoring $\hat{K}(h) = (1+h)(I - \gamma_h\hat{\mathcal{P}}^\pi)$ with the effective discount $\gamma_h = \frac{h\gamma}{1+h}$ from Section 3.1 and dividing Equation (5) by $1 + h$ yields a standard Bellman evaluation equation:

$$U_{j+1} = \underbrace{\frac{1}{1+h}\Big(V_k + h\big(r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_j\big)\Big)}_{\text{modified reward } \tilde{r}_j^\pi} + \underbrace{\gamma_h\hat{\mathcal{P}}^\pi}_{\text{surrogate dynamics}} U_{j+1}. \tag{6}$$

Equation (6) turns each defect-correction round into an exact Policy Evaluation problem in a surrogate MDP with transition dynamics $\hat{\mathcal{P}}^\pi$, effective discount γ_h, and modified reward $\tilde{r}_j^\pi$. Only the local reward correction $\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_j$ queries the true model $\mathcal{P}$, so m rounds require m true-model queries. Solving for U_{j+1} uses only $\hat{\mathcal{P}}^\pi$, under an effective discount γ_h that increases from 0 toward γ as h grows.

4 Implicit-Euler Value Iteration

IE-VI interleaves outer implicit-Euler steps with m inner defect-correction rounds of Equation (6), setting $V_{k+1} := U_m$. Viewed from the outer loop, this m-round procedure acts as a hybrid operator—blending $\mathcal{P}$ and $\hat{\mathcal{P}}$—to approximate the exact implicit step, creating a trade-off between the step size h and the correction depth m. A larger h advances further along the Bellman residual flow per outer step, reducing the number of outer iterations, but also widens the discrepancy $\hat{K}(h) - K(h) = h\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)$, which can weaken the outer update's contraction toward V^π. Increasing m makes the hybrid update a more faithful approximation of the true implicit step—ensuring it remains contractive along the original flow—at the price of one query to $\mathcal{P}$ per inner round.

Section 4.1 formalizes IE-VI for Policy Evaluation and extends it to nonlinear Control. Section 4.2 bounds the outer-step contraction as a function of h and m, yielding parameter schedules that guarantee convergence for arbitrary surrogates $\hat{\mathcal{P}}$ and acceleration over VI when $\hat{\mathcal{P}}$ is sufficiently accurate. Section 4.3 introduces Adaptive IE-VI, which selects h online using the true Bellman residual when the model error is unknown, and Section 4.4 extends the update to reinforcement learning via IE-DYNA.

4.1 The algorithm

Each outer IE-VI update approximates an implicit-Euler step of size $h > 0$ using m defect-correction rounds, starting at $U_0 := V_k$. Each round $j = 1, \ldots, m$ implements Equation (6): true-model defect evaluation, reward shifting, then exact surrogate planning (Lines 5–7 in Algorithm 1). After m rounds, the final inner iterate becomes the next outer estimate (Algorithm 1).

Algorithm 1 IE-VI for policy evaluation

1: **Require:** Initial value function V_0, fixed policy π, step size $h > 0$, inner depth $m \geq 1$
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: $\quad U_0 \leftarrow V_k$
4: $\quad$ **for** $j = 1, \ldots, m$ **do**
5: $\quad\quad \Delta^\pi(U_{j-1}) \leftarrow r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_{j-1}$ ▷ Line 5: one true-model query
6: $\quad\quad \tilde{r}_j^\pi \leftarrow \frac{1}{1+h} V_k + \frac{h}{1+h}\Delta^\pi(U_{j-1})$
7: $\quad\quad$ Solve $U_j = \tilde{r}_j^\pi + \gamma_h \hat{\mathcal{P}}^\pi U_j$ exactly ▷ Line 7: exact surrogate planning (free)
8: $\quad$ **end for**
9: $\quad V_{k+1} \leftarrow U_m$
10: **end for**

Line 5 alone queries $\mathcal{P}$, measuring the local discrepancy between $\mathcal{P}^\pi$ and $\hat{\mathcal{P}}^\pi$ at inner iterate U_{j-1}. This discrepancy shifts the original reward to $\tilde{r}_j^\pi$. With this reward fixed, Line 7 solves the surrogate MDP for U_j, using transitions $\hat{\mathcal{P}}^\pi$ and discount $\gamma_h := h\gamma/(1+h) \leq \gamma$, strictly smaller when $\gamma > 0$.

Extension to control. Control keeps the outer-inner structure, but its surrogate step solves a Bellman optimality equation. Defect evaluation and reward shifting occur per action, before the supremum:

$$\Delta^a(U_{j-1}) := R^a + \gamma(\mathcal{P}^a - \hat{\mathcal{P}}^a)U_{j-1}, \qquad \tilde{r}_j^a := \frac{1}{1+h} V_k + \frac{h}{1+h}\Delta^a(U_{j-1}), \qquad \forall a \in \mathcal{A}. \tag{7}$$

With these shifted rewards, the surrogate step solves

$$U_j(x) = \sup_{a \in \mathcal{A}} \left\{ \tilde{r}_j^a(x) + \gamma_h(\hat{\mathcal{P}}^a U_j)(x) \right\}. \tag{8}$$

Each inner round therefore issues one action-wise true-model query. Control preserves the true Bellman-optimality fixed point and shares Policy Evaluation's contraction properties; see Appendix D.

4.2 Error dynamics and iteration complexity

We define a **true-model query** as one full operator application $V \mapsto \mathcal{P}^\pi V$ for Policy Evaluation, or $V \mapsto \{\mathcal{P}^a V\}_{a \in \mathcal{A}}$ for Control. Because each defect-correction round queries $\mathcal{P}$ once, N outer steps of depth m require Nm true-model queries. Our theoretical bounds count only these true-model queries, treating surrogate planning in $\hat{\mathcal{P}}$ as free; Section 5 complements this oracle model by evaluating weighted operator costs and wall-clock runtimes when surrogate computation is charged.

Let $V^\dagger \in \{V^\pi, V^*\}$ denote the target value function, and define the worst-case surrogate error

$$E_\infty := \begin{cases} \gamma \|\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi\|_\infty, & \text{Policy Evaluation,} \\ \gamma \sup_{a \in \mathcal{A}} \|\mathcal{P}^a - \hat{\mathcal{P}}^a\|_\infty, & \text{Control.} \end{cases} \tag{9}$$

Theorem 1 characterizes how the m-round hybrid update contracts toward $V^\dagger$, separating the exact implicit-Euler contraction rate from the error left by m defect-correction rounds (proof in Appendix D).

Theorem 1 (Implicit-Step Error Decomposition)**.** *For every step size $h > 0$ and integer depth $m \geq 1$, the IE-VI updates for both Policy Evaluation and Control satisfy*

$$\|V_{k+1} - V^\dagger\|_\infty \leq \left(\frac{1}{1 + h(1-\gamma)} + \rho(h)^m \frac{h(1 + \gamma + h(1-\gamma))}{(1+h)(1+h(1-\gamma))} \right) \|V_k - V^\dagger\|_\infty, \tag{10}$$

where $\rho(h) := \frac{hE_\infty}{1+h(1-\gamma)}$ bounds the inner-loop error-propagation factor per defect-correction round.

Unlike planning in $\hat{\mathcal{P}}$ alone—which converges to a biased fixed point—IE-VI preserves $V^\dagger$ as an exact fixed point: both terms in Equation (10) multiply the current value error $\|V_k - V^\dagger\|_\infty$. The first term is the exact implicit-Euler contraction factor from Section 3.1, which improves monotonically

toward zero as h grows. The second term bounds the contraction loss from approximating the exact implicit step with m rounds of surrogate defect correction.

Increasing m suppresses this second term whenever $\rho(h) < 1$, or equivalently when

$$h\big(E_\infty - (1-\gamma)\big) < 1. \tag{11}$$

Inequality (11) compares the surrogate error E_∞ against the slow decay rate $1-\gamma$ that sets the effective horizon of the Bellman residual flow. Their ratio defines the *normalized model error*,

$$R_E := \frac{E_\infty}{1-\gamma}. \tag{12}$$

Because $\lim_{h\to\infty} \rho(h) = R_E$, the parameter design splits into two regimes:

Low-error regime ($R_E < 1$). When $E_\infty < 1-\gamma$, Inequality (11) holds for all $h > 0$. Sending $h \to \infty$ drives the exact implicit contraction term to zero while keeping $\lim_{h\to\infty} \rho(h) = R_E < 1$, so a single correction round ($m = 1$) contracts by at most R_E per step. This $(h \to \infty, m = 1)$ limit recovers OS-VI [Rakhsha et al., 2022] (Proposition 13).

High-error regime ($R_E \geq 1$). When $R_E \geq 1$, taking $h \to \infty$ gives $\rho(h) \to R_E \geq 1$, so the infinite-step limit no longer guarantees contraction. Instead, choosing a finite step size h to cap $\rho(h) \leq 1/2$ and setting m to cancel the $O(1/(1-\gamma))$ prefactor in Equation (10) yields the schedule

$$h = \frac{1}{(1-\gamma)(2R_E - 1)}, \qquad m = \left\lceil \log_2\left(\frac{8}{1-\gamma}\right)\right\rceil. \tag{13}$$

Schedule (13) guarantees an outer contraction factor of at most $1 - \frac{1}{4\bar{R}_E}$ per step (Lemma 19) at a cost of m true-model queries, and extends uniformly to all $R_E \geq 0$ upon replacing R_E with $\bar{R}_E := \max\{1, R_E\}$. Multiplying the resulting outer iteration count by m yields the query bounds in Theorem 2 (proved in Appendix D, including general $\epsilon \in (0,1)$ conventions and greedy policy extraction for Control).

Theorem 2 (Iteration complexity and polylogarithmic acceleration)**.** *For target precision $\epsilon \in (0, 1/e)$, IE-VI reaches relative value error $\|V_N - V^\dagger\|_\infty \leq \epsilon \|V_0 - V^\dagger\|_\infty$ with the following true-model query complexity $Q(\epsilon)$:*

Low-error regime ($R_E < 1$). *Under $h \to \infty$ and $m = 1$,*

$$Q(\epsilon) = O\left(\frac{1}{1-R_E} \log \frac{1}{\epsilon}\right). \tag{14}$$

High-error regime ($R_E \geq 1$). *Under Schedule (13),*

$$Q(\epsilon) = O\left(R_E \log \frac{1}{1-\gamma} \log \frac{1}{\epsilon}\right). \tag{15}$$

The uniform finite-step version of Equation (15), for all $R_E \geq 0$, is

$$Q(\epsilon) = O\left(\bar{R}_E \log \frac{8}{1-\gamma} \log \frac{1}{\epsilon}\right).$$

Consequence: polylogarithmic horizon dependence. *If $E_\infty \leq C(1-\gamma)\log\frac{1}{1-\gamma}$ for a constant $C > 0$ independent of γ, then $R_E = O(\log\frac{1}{1-\gamma})$ and the uniform finite-step schedule achieves*

$$Q(\epsilon) = O\left(\log^2 \frac{1}{1-\gamma} \log \frac{1}{\epsilon}\right). \tag{16}$$

Theorem 2 establishes both worst-case safety and long-horizon acceleration. Because $E_\infty \leq 2\gamma$ implies $R_E = O(\frac{1}{1-\gamma})$ for any Markov kernel $\hat{\mathcal{P}}$, under the uniform finite-step schedule, IE-VI's worst-case query complexity never exceeds VI's $O(\frac{1}{1-\gamma}\log\frac{1}{\epsilon})$ bound by more than a $\log\frac{1}{1-\gamma}$ factor. Conversely, whenever R_E grows at most logarithmically with the horizon $H := \frac{1}{1-\gamma}$, IE-VI reduces VI's linear dependence on H to polylogarithmic—extending provable acceleration well beyond the $R_E < 1$ regime certified by OS-VI. Moreover, when R_E is known, jointly tuning h and m in Equation (10) further tightens the horizon dependence in Equation (16) from $\log^2 H$ to $\log H(1 + \log\log H)$ (Appendix E.2). Per step, $1 - 1/(4\bar{R}_E)$ beats VI's γ exactly when $\bar{R}_E < H/4$.

4.3 Adaptive IE-VI

In Schedule (13), the correction depth m depends only on the known discount factor γ, whereas the step size h depends on the normalized model error R_E, which is rarely known in advance. Adaptive IE-VI selects h online using the true Bellman residual $\mathrm{BR}(V) \in \{\mathrm{BR}^\pi(V), \mathrm{BR}^*(V)\}$, which costs one true-model query and sandwiches the unknown value error (Lemmas 3 and 8):

$$\frac{\|\mathrm{BR}(V)\|_\infty}{1+\gamma} \le \|V - V^\dagger\|_\infty \le \frac{\|\mathrm{BR}(V)\|_\infty}{1-\gamma}. \tag{17}$$

By Equation (17), reducing $\|\mathrm{BR}(V)\|_\infty$ to at most $\frac{\epsilon(1-\gamma)}{1+\gamma}\|\mathrm{BR}(V_0)\|_\infty$ certifies relative value error $\|V - V^\dagger\|_\infty \le \epsilon\|V_0 - V^\dagger\|_\infty$.

Doubling schedule with convergence guarantees. Our theoretical adaptive procedure (Algorithms 2 and 3 in Appendices F–G) searches over hypotheses $\Lambda_i = 2^i$ for $\bar{R}_E$, starting at $\Lambda_0 = 1$. Epoch i sets $h_i = [(1-\gamma)(2\Lambda_i - 1)]^{-1}$ and runs a prescribed number of m-round IE-VI updates (restarting from V_0 or from the point with the smallest residual among V_0 and previous epoch endpoints; Appendix H). If the resulting residual exceeds the target threshold, the algorithm doubles $\Lambda_{i+1} = 2\Lambda_i$, shrinks h, and retries. Theorems 29 and 33 prove that this search terminates no later than the first epoch with $\Lambda_i \ge \bar{R}_E$, matching the uniform finite-step query bound of Theorem 2 up to logarithmic overhead from residual certification.

Practical residual-stall schedule. To avoid epoch restarts and multi-round overhead in practice, we also use a one-round ($m = 1$) trajectory-wise variant (Algorithm 4 in Appendix J). Starting from a large step size $h_{\mathrm{init}} \gg 1$ (approximating the OS-VI limit), it carries iterates forward along a single trajectory and shrinks $h \leftarrow \max\{h_{\min}, \beta h\}$ by a factor $\beta \in (0,1)$ whenever $\|\mathrm{BR}(V_k)\|_\infty$ fails to improve on the best residual recorded so far for S consecutive steps. Section 5 evaluates this practical schedule, while its formal termination bound remains open.

4.4 IE-DYNA: Implicit-Euler planning in reinforcement learning

In tabular reinforcement learning, the true transition operator $\mathcal{P}$ is accessed through sampled environment transitions (x_t, a_t, r_t, x_t') rather than exact expectations. IE-DYNA brings implicit-Euler integration to the Dyna architecture [Sutton, 1990], where an agent simultaneously learns a surrogate model $\hat{\mathcal{P}}_t$ and plans value updates from interaction.

IE-DYNA preserves the implicit-Euler structure of the Control update in Equations (7)–(8)—anchoring each step to the outer iterate V_k with step size $h_k > 0$ and planning under the damped discount $\gamma_{h_k} = h_k\gamma/(1+h_k)$—while replacing exact action-wise defect evaluations with a stochastic-approximation tracker $D_{k,j}(x,a)$.

During outer step k, V_k remains fixed and $U_{k,0} := V_k$. The defect tracker is carried over from the preceding outer step, with $D_0 = 0$. At inner stage j, after observing (x_t, a_t, r_t, x_t'), let $\bar{\mathcal{P}}_{k,j} := \hat{\mathcal{P}}_{t+1}$ denote the surrogate updated using this sample. IE-DYNA forms the single-sample defect target

$$Y_{k,j} := r_t + \gamma U_{k,j-1}(x_t') - \gamma\big(\bar{\mathcal{P}}_{k,j}^{a_t} U_{k,j-1}\big)(x_t)$$

and updates the visited coordinate:

$$D_{k,j}(x_t, a_t) \leftarrow (1-\alpha_{n_t})D_{k,j-1}(x_t,a_t) + \alpha_{n_t} Y_{k,j},$$

with step size $\alpha_{n_t} \in (0,1]$ indexed by the visit count n_t, leaving unvisited coordinates unchanged. The tracker combines sampled rewards and successor values with the surrogate correction, avoiding a full evaluation of the true transition operator. Each inner planning step then solves the implicit-step surrogate optimality equation

$$U_{k,j}(x) = \max_{a\in\mathcal{A}} \left\{ \frac{1}{1+h_k} V_k(x) + \frac{h_k}{1+h_k} D_{k,j}(x,a) + \gamma_{h_k}\big(\bar{\mathcal{P}}_{k,j}^a U_{k,j}\big)(x) \right\}.$$

After J inner stages, the algorithm sets $V_{k+1} := U_{k,J}$ and carries the updated tracker into the next outer step.

Taking $h_k \to \infty$ removes the anchor V_k and gives $\gamma_{h_k} \to \gamma$, recovering OS-Dyna [Rakhsha et al., 2022]. Keeping h_k finite lets IE-DYNA damp surrogate and sampling errors, as observed in Section 5. Appendix K specifies the sampling model and gives the complete procedure in Algorithm 5. While our theoretical guarantees cover exact-operator IE-VI, Section 5 evaluates IE-DYNA empirically.

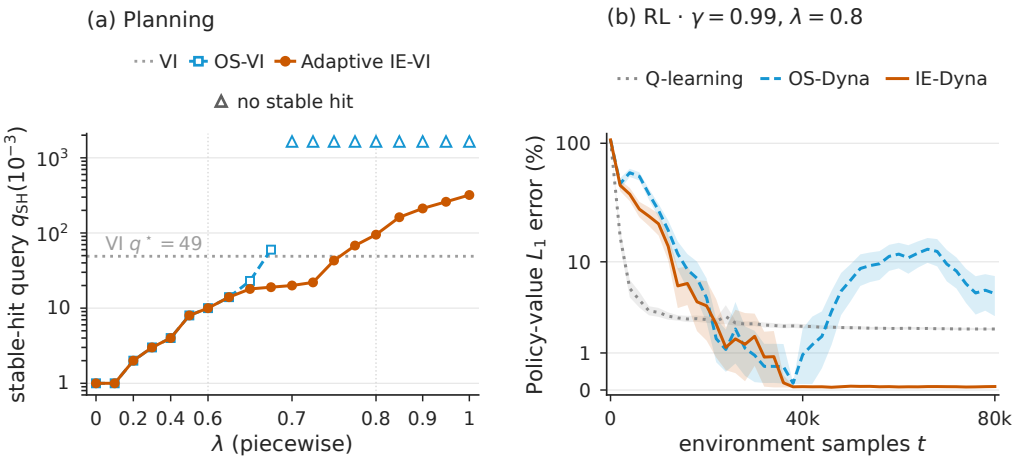


Figure 1: Adaptive IE-VI reaches a stable hit across the full displayed Cliffwalk model-error grid, including settings where OS-VI does not. (a) Planning at $\gamma = 0.99$ with support-uniform model error: stable-hit query indices $q_{\mathrm{SH}}(10^{-3})$ for the practical one-round stall schedule, OS-VI, and VI. The budget is 1000 recorded outer attempts; Section N.1 gives the counting convention. Upward triangles mark no stable hit within budget. The piecewise horizontal scale expands the high-error region. (b) RL at $\gamma = 0.99$, $\lambda = 0.8$ (support-uniform corruption): normalized all-state policy-value L_1 error over 80,000 nominal samples. Curves show means and 95% pointwise bootstrap intervals over 60 held-out seeds. The vertical scale is logarithmic in one plus percentage error; ticks show percent.

5 Empirical results

We evaluate IE-VI and IE-DYNA on (i) stability under large surrogate errors, (ii) query scaling with the effective horizon $H = 1/(1-\gamma)$, (iii) total compute and wall-clock runtime once surrogate work is charged, and (iv) sample efficiency in Dyna-style RL. Figure 1 summarizes our main Control and RL results on a 6×6 stochastic Cliffwalk ($\gamma = 0.99$), where the surrogate mixes its base model with a uniform distribution over each state-action support at corruption weight $\lambda \in [0, 1]$. Corruption is applied to the true transition kernel in planning and to the learned empirical model in RL. Full protocols, sensitivity checks, and additional benchmarks on Maze, FrozenLake, and Four Rooms appear in Appendix N.

Robustness to surrogate error in planning. Figure 1(a) reports the stable-hit index $q_{\mathrm{SH}}(10^{-3})$, the earliest recorded output from which the policy's normalized value error remains below 10^{-3} for the rest of the run. For low corruption ($\lambda \leq 0.65$), Adaptive IE-VI (using the practical stall schedule of Section 4.3) tracks OS-VI closely and reaches the threshold substantially earlier than VI (stable-hit index 49). At the tested levels $\lambda \geq 0.70$, OS-VI's $h = \infty$ updates oscillate and fail to stabilize within the 1,000-attempt budget, whereas Adaptive IE-VI reduces h upon detecting residual stalls and reaches a stable hit at every tested corruption level through $\lambda = 1$. Additional benchmarks show environment-dependent robustness gains (Figures N.3–N.13).

Horizon scaling. Evaluating a fixed Cliffwalk policy across horizons $H \in [10, 640]$ (Appendix N.13.2, Figure N.16) illustrates the horizon–accuracy trade-off of Section 4.2. When empirical surrogates are estimated with $16H$ samples per state so that model error shrinks with H, IE-VI (using the refined (h, m) schedule of Appendix E.2) exhibits sublinear query growth with log–log slope 0.66 versus 1.01 for VI (1,251 vs. 4,418 median queries at $H = 640$). Under a fixed budget of 80 samples per state, R_E grows with H and both tested IE-VI schedules use more queries than VI at long horizons. OS-VI remains more query-efficient than IE-VI in this horizon study.

Surrogate computation. Charging inner surrogate sweeps via the weighted cost $C_\kappa = N_{\mathcal{P}} + \kappa N_{\hat{\mathcal{P}}}$ with relative surrogate cost $\kappa > 0$ (Appendix N.13.1, Figure N.14) shows that query savings reduce total compute when κ lies below a break-even threshold, approximately 1% at $\lambda = 0.70$. For $\lambda \geq 0.775$, Adaptive IE-VI already uses more true-model queries than VI. Relaxing the inner solver's

value-error tolerance from 10^{-8} to 10^{-2} cuts surrogate sweeps by a median factor of $2.6\times$ without changing true-model query counts across all 19 corruption levels (Figure N.15).

Wall-clock runtime. On a 10,000-state advection–diffusion simulator ($\gamma = 0.995$; Appendix N.13.3, Figure N.17) where $\mathcal{P}$ integrates 128 fine microsteps and $\hat{\mathcal{P}}$ uses 4 to 32 coarse steps, fixed-step IE-VI achieves paired mean solver speedups of $8.25\times$ (cheapest surrogate) down to $1.18\times$ (most expensive) over VI, and $6.04\times$ to $1.17\times$ when operator-preparation time is included.

Dyna-style reinforcement learning. On Cliffwalk ($\gamma = 0.99$, $\lambda = 0.8$, support-uniform corruption; Figure 1(b)), Q-Learning has lower integrated L_1 policy-value error over the first 40,000 samples, while fixed-step IE-DYNA reaches lower late-stage error. At the final evaluation, mean errors are 0.07%, 2.1%, and 5.1% for IE-DYNA, Q-Learning, and OS-Dyna, respectively. Appendix K.4 details tuning and evaluation.

6 Related work

Operator splitting and model correction. The closest planners, OS-VI [Rakhsha et al., 2022] and MoCoVI [Rakhsha et al., 2024], preserve $V^\dagger$ using surrogate models (rate comparisons in Appendix M). OS-VI is the $h \to \infty$, $m = 1$ limit of IE-VI (Proposition 13); its supremum-norm bound certifies contraction for $R_E < 1$ and is otherwise inconclusive. MoCoVI corrects $\hat{\mathcal{P}}$ along past iterates under basis-approximation conditions. IE-VI applies iterated defect correction [Stetter, 1978, Böhmer and Stetter, 1984] to the finite-h resolvent [Rockafellar, 1976], choosing h, m to guarantee convergence for arbitrary Markov surrogates. Its edge over OS-VI is robustness, not speed.

Model-based RL and surrogate bias. Classical DYNA [Sutton, 1990] and modern model-based RL [Deisenroth and Rasmussen, 2011, Hafner et al., 2020, Schrittwieser et al., 2020] use learned models for planning; model errors can bias values and decisions. Short model rollouts [Janner et al., 2019] limit error propagation. IE-VI preserves the true Bellman fixed point through true-model defect corrections; IE-DYNA extends this construction with sampled corrections.

VI acceleration and continuous-time flows. Exact-model accelerations—Anderson VI [Geist and Scherrer, 2018], momentum VI [Goyal and Grand-Clément, 2021], PID-VI [Farahmand and Ghavamzadeh, 2021], Anchored VI [Lee and Ryu, 2023], and spectral deflation [Bertsekas, 1995, Lee et al., 2025]—accelerate explicit Bellman updates through extrapolation, feedback, anchoring, or spectral information. Continuous-time optimization views [Su et al., 2016, Wibisono et al., 2016] motivate algorithms through ODE discretizations. IE-VI instead approximates implicit steps through surrogate planning, with acceleration guarantees governed by surrogate accuracy.

7 Conclusion

Viewing Value Iteration as the unit-step explicit-Euler discretization of the Bellman residual flow traces its long-horizon bottleneck to numerical stiffness and motivates implicit integration. Approximating each implicit step via iterated defect correction reduces to solving a sequence of surrogate MDPs at a damped discount $\gamma_h < \gamma$ with true-model reward shifts. Rather than using $\hat{\mathcal{P}}$ as a stand-in for the environment—and inheriting its bias—IE-VI embeds $\hat{\mathcal{P}}$ into the update operator itself, reshaping how true-model residuals propagate across states while keeping the target fixed point intact. Unlike prior fixed-point-preserving surrogate planners, suitable choices of the step size h and depth m guarantee convergence across all Markov surrogates $\hat{\mathcal{P}}$ and widen the model-error regime with provable acceleration over VI, reducing query complexity from linear to polylogarithmic in $1/(1-\gamma)$. Empirically, this translates into stable planning and RL (IE-DYNA) under large model errors and faster runtimes when $\hat{\mathcal{P}}$ is cheap.

This bridge between numerical analysis and surrogate planning opens several directions: designing higher-order integrators that preserve the surrogate-MDP structure, developing state-visitation weighted-norm guarantees, and establishing stochastic convergence theory for IE-DYNA.

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Appendix roadmap

Appendix Roadmap: Appendix A establishes operator conventions and residual-error bounds. Appendix B derives exact stiffness ratios. Appendix C details the shifted-reward algebra used by Algorithm 1, and Appendix D gives the exact-operator convergence proofs for Policy Evaluation and Control. Appendix E justifies the parameter schedules. Appendices F, G and J give the adaptive search variants. Appendix K specifies the tabular IE-DYNA algorithm. Appendix L contains limitations and compute details. Appendix M provides extended related-work rate comparisons, and Appendix N details the empirical environments and sensitivity checks.

A Notation and residual bounds

This appendix gathers the measure-theoretic conventions deferred from Section 2. The MDP $(\mathcal{X}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$ takes $\mathcal{X}$ and $\mathcal{A}$ to be measurable spaces. Value functions belong to the Banach space $\mathcal{B}(\mathcal{X})$ of bounded measurable functions on $\mathcal{X}$, endowed with

$$\|V\|_\infty := \sup_{x\in\mathcal{X}} |V(x)|, \qquad \|S\|_\infty := \sup_{\|V\|_\infty \le 1} \|SV\|_\infty$$

for bounded linear operators $S : \mathcal{B}(\mathcal{X}) \to \mathcal{B}(\mathcal{X})$; let $\mathbf{1} \in \mathcal{B}(\mathcal{X})$ denote the constant function taking value 1 everywhere. For each action $a \in \mathcal{A}$, the reward kernel $\mathcal{R}(\cdot \mid x, a)$ has bounded mean

$$R^a(x) := \int_{\mathbb{R}} z\, \mathcal{R}(dz \mid x, a), \qquad \sup_{a\in\mathcal{A}} \|R^a\|_\infty < \infty,$$

and we also write $R(x, a) := R^a(x)$. The transition kernel $\mathcal{P}(\cdot \mid \cdot, a)$ acts on $\mathcal{B}(\mathcal{X})$ by

$$(\mathcal{P}^a V)(x) := \int_{\mathcal{X}} V(y)\, \mathcal{P}(dy \mid x, a),$$

and $\hat{\mathcal{P}}^a$ is defined analogously for the surrogate model. We assume that $\mathcal{X}$ and $\mathcal{A}$ are nonempty, and the standard measurability and selection regularity needed for the Bellman suprema in Section 2 to define elements of $\mathcal{B}(\mathcal{X})$. To cover also the suprema in the Control inner solves (Appendix D.2), we assume that for all $V_0, U, W \in \mathcal{B}(\mathcal{X})$ and $c \ge 0$ the function $x \mapsto \sup_{a\in\mathcal{A}}\{V_0(x) + cR^a(x) + (\mathcal{P}^a U)(x) + (\hat{\mathcal{P}}^a W)(x)\}$ is measurable; this holds automatically when $\mathcal{A}$ is countable. A stationary policy π induces the Markov operator and expected reward function

$$(\mathcal{P}^\pi V)(x) := \int_{\mathcal{A}} \int_{\mathcal{X}} V(y)\, \mathcal{P}(dy \mid x, a)\, \pi(da \mid x), \qquad r^\pi(x) := \int_{\mathcal{A}} R^a(x)\, \pi(da \mid x),$$

and $\hat{\mathcal{P}}^\pi$ is defined analogously. The induced operators $\mathcal{P}^\pi$ and $\hat{\mathcal{P}}^\pi$ are bounded linear Markov operators on $\mathcal{B}(\mathcal{X})$, so

$$\|\mathcal{P}^\pi\|_\infty = \|\hat{\mathcal{P}}^\pi\|_\infty = 1.$$

If the supremum in $(T^\star V)(x) = \sup_{a\in\mathcal{A}}\{R^a(x) + \gamma(\mathcal{P}^a V)(x)\}$ is attained, $T^\star V = T^{\pi'} V$ for every deterministic greedy selector π' of V (with ties broken arbitrarily but fixed); otherwise the Control analysis uses near-optimal selectors with vanishing slack.

Lemma 3 (Residual-error bounds)**.** *For every $V \in \mathcal{B}(\mathcal{X})$,*

$$\frac{1}{1+\gamma}\|\mathrm{BR}(V)\|_\infty \le \|V - V^\pi\|_\infty \le \frac{1}{(1-\gamma)}\|\mathrm{BR}(V)\|_\infty.$$

Proof. Both bounds follow from operator-norm estimates on $L^\pi = I - \gamma\mathcal{P}^\pi$. For the upper bound, write $\mathrm{BR}(V) = L^\pi(V^\pi - V)$, so $V^\pi - V = (L^\pi)^{-1}\,\mathrm{BR}(V)$. Since $\mathcal{P}^\pi$ is a Markov operator on $\mathcal{B}(\mathcal{X})$, $\|\mathcal{P}^\pi\|_\infty = 1$ and $\gamma < 1$, so the discounted Neumann series

$$\big(L^\pi\big)^{-1} = \big(I - \gamma\mathcal{P}^\pi\big)^{-1} = \sum_{t=0}^{\infty} \gamma^t (\mathcal{P}^\pi)^t$$

converges in operator norm. Therefore

$$\|(L^\pi)^{-1}\|_\infty \le \sum_{t=0}^{\infty} \gamma^t = \frac{1}{(1-\gamma)},$$
$$\|V^\pi - V\|_\infty \le (1-\gamma)^{-1}\|\mathrm{BR}(V)\|_\infty.$$

For the lower bound, the same identity gives

$$\|\mathrm{BR}(V)\|_\infty \le \|L^\pi\|_\infty \|V^\pi - V\|_\infty,$$

and the triangle inequality together with the Markov-operator bound yields

$$\|L^\pi\|_\infty = \|I - \gamma \mathcal{P}^\pi\|_\infty \le \|I\|_\infty + \gamma \|\mathcal{P}^\pi\|_\infty = 1 + \gamma.$$

□

The Policy Evaluation Bellman residual flow $\dot{V}(t) = \mathrm{BR}(V(t)) = r^\pi - L^\pi V(t)$ is a linear ODE on $\mathcal{B}(\mathcal{X})$, with unique fixed point V^π and homogeneous dynamics $V(t) - V^\pi = e^{-tL^\pi}(V(0) - V^\pi)$ governed by the Bellman semigroup $\{e^{-tL^\pi}\}_{t\ge 0}$; the implicit-Euler resolvent $(I + hL^\pi)^{-1}$ is its discrete-time counterpart at horizon h, with natural timescale $1/(1-\gamma)$. The next two results state the semigroup norm bound and the iterated-resolvent representation underlying Proposition 6; IE-VI (Section 4) realizes a finite truncation of the iterated-resolvent representation.

Lemma 4 (Bellman-residual-flow semigroup norm)**.** *Let L be either $L^\pi = I - \gamma \mathcal{P}^\pi$ or $\hat{L}^\pi = I - \gamma \hat{\mathcal{P}}^\pi$. For every $t \ge 0$,*

$$\|\exp(-tL)\|_\infty = e^{-(1-\gamma)t}.$$

Proof. Write $L = I - \gamma \mathcal{P}_L$, where $\mathcal{P}_L$ is either $\mathcal{P}^\pi$ or $\hat{\mathcal{P}}^\pi$. Since $-tI$ is a scalar multiple of the identity, it commutes with $t\gamma\mathcal{P}_L$, so

$$\exp(-tL) = \exp(-tI + t\gamma\mathcal{P}_L) = e^{-t}\exp(t\gamma\mathcal{P}_L).$$

The exponential series converges in operator norm. Since $\mathcal{P}_L$ is a Markov operator on $\mathcal{B}(\mathcal{X})$, $\|\mathcal{P}_L\|_\infty = 1$, and submultiplicativity gives the upper bound

$$\|\exp(t\gamma\mathcal{P}_L)\|_\infty \le \sum_{k=0}^{\infty} \frac{(t\gamma)^k}{k!} \|\mathcal{P}_L\|_\infty^k = e^{t\gamma}.$$

The reverse inequality follows from the action on the constant function $\mathbf{1}$. Since $\mathcal{P}_L \mathbf{1} = \mathbf{1}$, every power $\mathcal{P}_L^k$ fixes $\mathbf{1}$, and therefore

$$\exp(t\gamma\mathcal{P}_L)\mathbf{1} = e^{t\gamma}\mathbf{1},$$
$$\|\exp(t\gamma\mathcal{P}_L)\|_\infty \ge \frac{\|\exp(t\gamma\mathcal{P}_L)\mathbf{1}\|_\infty}{\|\mathbf{1}\|_\infty} = e^{t\gamma}.$$

Combining the upper and lower bounds gives

$$\|\exp(t\gamma\mathcal{P}_L)\|_\infty = e^{t\gamma}.$$

Multiplying by e^{-t} gives the claimed identity. □

Remark 5 (Bellman semigroup as iterated resolvent). By the exponential formula [Pazy, 1983, Theorem 1.8.3],

$$e^{-tL^\pi} = \lim_{n\to\infty} \left(I + \frac{t}{n} L^\pi\right)^{-n},$$

so each backward-Euler step $(I + hL^\pi)^{-1}$ is one term of an iterated-resolvent discretization of the Bellman residual flow.

Proposition 6 (Resolvent stability)**.** *For every $h > 0$,*

$$K(h)^{-1} = \frac{1}{h}\int_0^\infty e^{-t/h} e^{-tL^\pi}\, dt,$$
$$\hat{K}(h)^{-1} = \frac{1}{h}\int_0^\infty e^{-t/h} e^{-t\hat{L}^\pi}\, dt,$$

where both integrals converge in operator norm on $\mathcal{B}(\mathcal{X})$. Each resolvent is an exponentially weighted average of its Bellman semigroup with mean horizon h. Consequently,

$$\|K(h)^{-1}\|_\infty \le \frac{1}{1 + h(1-\gamma)}, \qquad \|\hat{K}(h)^{-1}\|_\infty \le \frac{1}{1 + h(1-\gamma)}.$$

Proof. Let $L \in \{L^\pi, \hat{L}^\pi\}$. The operator $A := h^{-1}I + L$ is bounded, so $-A$ generates the uniformly continuous semigroup $\{e^{-tA}\}_{t\geq 0}$. Since scalar multiples of the identity commute with L, $e^{-tA} = e^{-t/h}e^{-tL}$. By Lemma 4,

$$\|e^{-tA}\|_\infty \leq e^{-(h^{-1}+(1-\gamma))t}.$$

Thus the integral below converges in operator norm. Because A is bounded, $t \mapsto e^{-tA}$ is norm-differentiable in $\mathcal{L}(\mathcal{B}(\mathcal{X}))$, with derivative $-Ae^{-tA} = -e^{-tA}A$. Integrating $-\frac{d}{dt}e^{-tA} = Ae^{-tA} = e^{-tA}A$ over $[0, T]$ gives

$$A\int_0^T e^{-tA}\,dt = \left(\int_0^T e^{-tA}\,dt\right)A = I - e^{-TA}.$$

The boundary term vanishes in operator norm because

$$\|e^{-TA}\|_\infty \leq e^{-(h^{-1}+(1-\gamma))T} \to 0.$$

Hence $A^{-1} = \int_0^\infty e^{-tA}\,dt$, and therefore

$$\left(h^{-1}I + L\right)^{-1} = \int_0^\infty e^{-t/h}e^{-tL}\,dt.$$

Multiplying both sides by h^{-1} and using $K(h) = I + hL^\pi$ and $\hat{K}(h) = I + h\hat{L}^\pi$ yields the two claimed representations. The norm bounds follow by taking $\|\cdot\|_\infty$ inside the integral and applying $\|e^{-tL}\|_\infty = e^{-(1-\gamma)t}$. □

The same bound is also derived in Lemma 16 via Neumann series. The proposition supplies the stability bound used below: both resolvents contract uniformly in supremum norm, and the surrogate resolvent has the same unconditional bound as the true one.

For Control, the model-error scalar used in the body is the worst action-wise transition discrepancy

$$E_\infty := \gamma \sup_{a\in\mathcal{A}} \|\hat{\mathcal{P}}^a - \mathcal{P}^a\|_\infty.$$

This quantity is the supremum of the induced policy-evaluation model errors over stationary policies; the supremum need not be attained.

Proposition 7 (Action-wise model-error equivalence)**.** *Let* Π *be the set of stationary stochastic policies and define*

$$E_\Pi := \sup_{\pi\in\Pi} \|\gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi)\|_\infty.$$

Then $E_\Pi = E_\infty$.

Proof. For every stationary policy π and every $V \in \mathcal{B}(\mathcal{X})$ with $\|V\|_\infty \leq 1$,

$$\big(\gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi)V\big)(x) = \gamma\int_{\mathcal{A}} \big((\hat{\mathcal{P}}^a - \mathcal{P}^a)V\big)(x)\,\pi(da \mid x).$$

Taking absolute values gives

$$\begin{aligned}\Big|\big(\gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi)V\big)(x)\Big| &\leq \gamma\int_{\mathcal{A}} \Big|\big((\hat{\mathcal{P}}^a - \mathcal{P}^a)V\big)(x)\Big|\,\pi(da \mid x)\\ &\leq \gamma\int_{\mathcal{A}} \sup_{b\in\mathcal{A}}\|\hat{\mathcal{P}}^b - \mathcal{P}^b\|_\infty\,\pi(da \mid x) = E_\infty.\end{aligned}$$

Taking the supremum over $x \in \mathcal{X}$ and $\|V\|_\infty \leq 1$ yields

$$\|\gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi)\|_\infty \leq E_\infty,$$

and then $E_\Pi \leq E_\infty$.

Conversely, fix $\eta > 0$. By the definition of the supremum over $\mathcal{A}$, choose $a_\eta \in \mathcal{A}$ such that

$$\gamma\|\hat{\mathcal{P}}^{a_\eta} - \mathcal{P}^{a_\eta}\|_\infty \geq E_\infty - \eta.$$

Let π_{a_η} be the deterministic policy that selects a_η at every state. Then

$$E_\Pi \geq \|\gamma(\hat{\mathcal{P}}^{\pi_{a_\eta}} - \mathcal{P}^{\pi_{a_\eta}})\|_\infty = \gamma\|\hat{\mathcal{P}}^{a_\eta} - \mathcal{P}^{a_\eta}\|_\infty \geq E_\infty - \eta.$$

Since $\eta > 0$ is arbitrary, $E_\Pi \geq E_\infty$. Thus $E_\Pi = E_\infty$. □

Recall from Section 2 the Bellman optimality residual $\mathrm{BR}^\star(V) := T^\star V - V$.

Lemma 8 (Optimality residual-error bounds)**.** *For every $V \in \mathcal{B}(\mathcal{X})$,*

$$\frac{1}{1+\gamma}\|\mathrm{BR}^\star(V)\|_\infty \le \|V - V^\star\|_\infty \le \frac{1}{(1-\gamma)}\|\mathrm{BR}^\star(V)\|_\infty.$$

Proof. Using $V^\star = T^\star V^\star$, decompose

$$\mathrm{BR}^\star(V) = T^\star V - T^\star V^\star + V^\star - V.$$

For the upper error bound, the reverse triangle inequality and the γ-contraction of $T^\star$ give

$$\begin{aligned}\|\mathrm{BR}^\star(V)\|_\infty &\ge \|V^\star - V\|_\infty - \|T^\star V - T^\star V^\star\|_\infty \\ &\ge \|V^\star - V\|_\infty - \gamma\|V - V^\star\|_\infty \\ &= (1-\gamma)\|V - V^\star\|_\infty.\end{aligned}$$

For the lower error bound, the triangle inequality gives

$$\|\mathrm{BR}^\star(V)\|_\infty \le \|T^\star V - T^\star V^\star\|_\infty + \|V^\star - V\|_\infty \le (1+\gamma)\|V - V^\star\|_\infty.$$

□

Lemma 8 is the residual certificate used by the adaptive Control analogue in Appendix G.

B Numerical stiffness of the Bellman residual flow

B.1 Exact stiffness ratios for general state spaces

This subsection gives the L_∞ stiffness-ratio calculation used in Section 3. The state space $(\mathcal{X}, \Sigma)$ is measurable and all singletons are measurable.

Proposition 9 (Exact stiffness ratio for Policy Evaluation)**.** *Fix an arbitrary policy π. Let L^π act on $\mathcal{B}(\mathcal{X})$ and define*

$$L^\pi = I - \gamma\mathcal{P}^\pi, \qquad p_{\inf} := \inf_{x\in\mathcal{X}} \mathcal{P}^\pi(\{x\} \mid x).$$

Then

$$\rho_{\mathrm{stiff}}(L^\pi) := \|L^\pi\|_\infty \|(L^\pi)^{-1}\|_\infty = \frac{1+\gamma-2\gamma p_{\inf}}{1-\gamma}.$$

Proof. The inverse norm is exact. Since $\mathcal{P}^\pi$ is a Markov operator and $\gamma \in [0,1)$, the Neumann series gives

$$(L^\pi)^{-1} = \sum_{t=0}^{\infty} \gamma^t (\mathcal{P}^\pi)^t, \qquad \|(L^\pi)^{-1}\|_\infty \le \frac{1}{1-\gamma}.$$

The constant-one function $\mathbf{1}$ satisfies

$$(L^\pi)^{-1}\mathbf{1} = \frac{1}{1-\gamma}\mathbf{1}, \qquad \|\mathbf{1}\|_\infty = 1, \qquad \|(L^\pi)^{-1}\|_\infty = \frac{1}{1-\gamma}.$$

It remains to compute $\|L^\pi\|_\infty$. Fix $\|V\|_\infty \le 1$ and $x \in \mathcal{X}$, and write

$$p(x) := \mathcal{P}^\pi(\{x\} \mid x).$$

Splitting the transition integral at $\{x\}$ gives

$$(L^\pi V)(x) = V(x)(1 - \gamma p(x)) - \gamma \int_{\mathcal{X}\setminus\{x\}} V(y)\, \mathcal{P}^\pi(dy \mid x).$$

Since $1 - \gamma p(x) > 0$, the bound $|V(y)| \le 1$ gives

$$|(L^\pi V)(x)| \le 1 - \gamma p(x) + \gamma(1 - p(x)) = 1 + \gamma - 2\gamma p(x).$$

Taking the supremum over x yields

$$\|L^\pi\|_\infty \le 1 + \gamma - 2\gamma p_{\inf}.$$

For the reverse inequality, fix $\epsilon > 0$. By the definition of the infimum, choose $x_\epsilon \in \mathcal{X}$ with $p(x_\epsilon) < p_{\inf} + \epsilon$. The test function

$$V_\epsilon(y) := \begin{cases} 1, & y = x_\epsilon, \\ -1, & y \neq x_\epsilon \end{cases}$$

belongs to $\mathcal{B}(\mathcal{X})$, because singletons are measurable, and $\|V_\epsilon\|_\infty = 1$. At x_ϵ,

$$\begin{aligned}(L^\pi V_\epsilon)(x_\epsilon) &= 1 - \gamma p(x_\epsilon) - \gamma \int_{\mathcal{X}\setminus\{x_\epsilon\}} (-1)\, \mathcal{P}^\pi(dy \mid x_\epsilon) \\ &= 1 + \gamma - 2\gamma p(x_\epsilon).\end{aligned}$$

This gives

$$\|L^\pi\|_\infty \geq 1 + \gamma - 2\gamma p(x_\epsilon) \geq 1 + \gamma - 2\gamma(p_{\inf} + \epsilon).$$

Letting $\epsilon \downarrow 0$ gives $\|L^\pi\|_\infty \geq 1 + \gamma - 2\gamma p_{\inf}$. Multiplying the two operator norms proves the stiffness-ratio identity. If $p_{\inf} < 1$ and $\delta_\pi := 1 - p_{\inf}$, then

$$\rho_{\text{stiff}}(L^\pi) = 1 + \frac{2\gamma\delta_\pi}{1-\gamma} \to \infty \qquad \text{as } \gamma \uparrow 1.$$

□

Proposition 10 (Exact stiffness ratio for Control)**.** *Let*

$$F(V) := V - T^\star V = -\operatorname{BR}^\star(V)$$

on $\mathcal{B}(\mathcal{X})$. Let L_F and M_F be the optimal constants satisfying

$$\begin{aligned}\|F(V) - F(U)\|_\infty &\leq L_F \|V - U\|_\infty, \\ \|V - U\|_\infty &\leq M_F \|F(V) - F(U)\|_\infty.\end{aligned}$$

Set $\rho_{\text{stiff}}(F) := L_F M_F$ and $p^\star_{\inf} := \inf_{x\in\mathcal{X},\, a\in\mathcal{A}} \mathcal{P}^a(\{x\} \mid x)$. Then

$$\rho_{\text{stiff}}(F) = \frac{1 + \gamma - 2\gamma p^\star_{\inf}}{1-\gamma}.$$

Proof. For the inverse constant, write $D := V - U$. The optimality operator is a γ-contraction in $\|\cdot\|_\infty$, so

$$\begin{aligned}\|F(V) - F(U)\|_\infty &= \|D - (T^\star V - T^\star U)\|_\infty \\ &\geq \|D\|_\infty - \|T^\star V - T^\star U\|_\infty \\ &\geq (1-\gamma)\|D\|_\infty.\end{aligned}$$

This gives $M_F \leq (1-\gamma)^{-1}$. Equality follows by taking $U = 0$ and $V = c\mathbf{1}$ with $c > 0$. Since

$$\begin{aligned}T^\star(c\mathbf{1}) &= T^\star 0 + \gamma c\mathbf{1}, \\ F(c\mathbf{1}) - F(0) &= c(1-\gamma)\mathbf{1},\end{aligned}$$

and $M_F = (1-\gamma)^{-1}$.

For the forward constant, fix $x \in \mathcal{X}$, $\eta > 0$, and $D := V - U$. Choose $a_V \in \mathcal{A}$ with

$$(T^\star V)(x) \leq R^{a_V}(x) + \gamma(\mathcal{P}^{a_V} V)(x) + \eta.$$

Evaluating $T^\star U$ at the same action gives

$$(T^\star V)(x) - (T^\star U)(x) \leq \gamma(\mathcal{P}^{a_V} D)(x) + \eta.$$

The same argument with U and V interchanged gives an action a_U such that

$$\begin{aligned}(T^\star U)(x) - (T^\star V)(x) &\leq \gamma(\mathcal{P}^{a_U}(-D))(x) + \eta, \\ F(V)(x) - F(U)(x) &= D(x) - ((T^\star V)(x) - (T^\star U)(x)),\end{aligned}$$

so the two one-sided bounds imply

$$|F(V)(x) - F(U)(x)| \leq \sup_{a\in\mathcal{A}} |D(x) - \gamma(\mathcal{P}^a D)(x)| + \eta.$$

Letting $\eta \downarrow 0$, applying the singleton-isolation estimate from Proposition 9 to each action kernel, and taking the supremum over x gives

$$L_F \le 1 + \gamma - 2\gamma p^\star_{\inf}.$$

For the reverse inequality, fix $\epsilon > 0$ and choose (x_ϵ, a_ϵ) such that

$$\mathcal{P}^{a_\epsilon}(\{x_\epsilon\} \mid x_\epsilon) < p^\star_{\inf} + \epsilon.$$

Let $U = 0$ and, for $c > 0$, define

$$V_c(y) := \begin{cases} -c, & y = x_\epsilon, \\ c, & y \neq x_\epsilon. \end{cases}$$

Then $\|V_c - U\|_\infty = c$. The action a_ϵ gives

$$(T^\star V_c)(x_\epsilon) \ge R^{a_\epsilon}(x_\epsilon) + \gamma c\,(1 - 2\mathcal{P}^{a_\epsilon}(\{x_\epsilon\} \mid x_\epsilon))\,.$$

Therefore

$$\begin{aligned} F(V_c)(x_\epsilon) - F(U)(x_\epsilon) &= -c - (T^\star V_c)(x_\epsilon) + (T^\star 0)(x_\epsilon) \\ &\le -c\,(1 + \gamma - 2\gamma\mathcal{P}^{a_\epsilon}(\{x_\epsilon\} \mid x_\epsilon)) + \Delta_R, \end{aligned}$$

where

$$\Delta_R := (T^\star 0)(x_\epsilon) - R^{a_\epsilon}(x_\epsilon)$$

is independent of c. For large enough c, the right side is negative. Taking absolute values and dividing by c gives

$$\frac{\|F(V_c) - F(U)\|_\infty}{\|V_c - U\|_\infty} \ge 1 + \gamma - 2\gamma\mathcal{P}^{a_\epsilon}(\{x_\epsilon\} \mid x_\epsilon) - \frac{\Delta_R}{c}.$$

Let $c \to \infty$ and then $\epsilon \downarrow 0$. This gives

$$L_F \ge 1 + \gamma - 2\gamma p^\star_{\inf}.$$

Multiplying L_F by M_F proves the claimed stiffness ratio. If $p^\star_{\inf} < 1$ and $\delta_\star := 1 - p^\star_{\inf}$, then

$$\rho_{\text{stiff}}(F) = 1 + \frac{2\gamma\delta_\star}{1-\gamma} \to \infty \qquad \text{as } \gamma \uparrow 1.$$

□

B.2 A two-state stiffness example

We give the two-state PE example referenced in Section 3.1, on which explicit and implicit Euler admit a closed-form supremum-norm analysis. Consider an MDP with the state space $\mathcal{X} = \{x_1, x_2\}$, a single action (so PE and Control coincide), deterministic transitions that swap states ($x_1 \to x_2$ and $x_2 \to x_1$ with probability 1), and rewards $r(x_1) = 1$, $r(x_2) = 0$; that is, $\mathcal{P}^\pi = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ and $r^\pi = (1, 0)^\top$. The linear operator $L^\pi = I - \gamma\mathcal{P}^\pi$ is diagonal in the modal basis $e_\pm = (1, \pm 1)^\top$: $L^\pi e_+ = (1-\gamma)\,e_+$ and $L^\pi e_- = (1+\gamma)\,e_-$. The reward decomposes evenly across the two modes, $r^\pi = \frac{1}{2}e_+ + \frac{1}{2}e_-$, so starting from $V_0 = 0$ the initial error excites both.

Explicit Euler. The explicit-Euler step $V_{k+1} = V_k + h\,\mathrm{BR}(V_k)$ yields the error recursion $V_{k+1} - V^\pi = (I - hL^\pi)(V_k - V^\pi)$. For this MDP, $\|I - hL^\pi\|_\infty = |1-h| + h\gamma$. The recursion is a strict contraction precisely for $0 < h < 2/(1+\gamma)$; inside that stable range the modulus is minimized at $h = 1$ with value γ, which is the conventional VI step. Every horizon-scale choice $h \asymp 1/(1-\gamma)$ leaves the stable range as $\gamma \uparrow 1$.

Implicit Euler. The implicit-Euler step yields the error recursion $V_{k+1} - V^\pi = (I + hL^\pi)^{-1}(V_k - V^\pi)$. A direct row-sum calculation (Appendix B.2.1) gives

$$\|(I + hL^\pi)^{-1}\|_\infty = \frac{1}{1 + h(1-\gamma)} \qquad \text{for every } h > 0, \tag{18}$$

so the horizon-scale choice $h = 1/(1-\gamma)$ yields modulus $1/2$ regardless of γ.

What the example shows. The two eigenvalues of L^π sit at separated scales: the slow mode e_+ at $(1-\gamma)$, which sets the horizon scale, and the fast mode e_- at $1+\gamma$, which is bounded away from zero. Their ratio $(1+\gamma)/(1-\gamma)$ is the *stiffness ratio* of L^π and diverges as $\gamma \uparrow 1$. Explicit Euler propagates errors by the single multiplier $I - hL^\pi$, so the fast eigenvalue caps the admissible step size and no stable step achieves a supremum-norm modulus below γ. Implicit Euler instead inverts $I + hL^\pi$; the inverse acts on the constant function (a slow-mode eigenvector) as multiplication by $1/(1+h(1-\gamma))$, and that same factor controls the entire supremum norm. Inverting the operator rather than multiplying by it removes the fast mode's cap on the step size, the standard numerical-analysis remedy for stiffness [Hairer and Wanner, 1996, Ch. IV].

B.2.1 Calculation of the two-state Bellman-residual-flow stiffness

We verify the two-state calculation used above. Consider the single-action two-state MDP with

$$\mathcal{P}^\pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad r^\pi = (1,0)^\top, \qquad L^\pi = I - \gamma\mathcal{P}^\pi = \begin{bmatrix} 1 & -\gamma \\ -\gamma & 1 \end{bmatrix}.$$

Policy Evaluation and Control coincide because there is only one action.

Mode decomposition and fixed point. Let

$$e_+ = (1,1)^\top, \qquad e_- = (1,-1)^\top.$$

Then $\mathcal{P}^\pi e_+ = e_+$ and $\mathcal{P}^\pi e_- = -e_-$. It follows that

$$L^\pi e_+ = (1-\gamma)e_+, \qquad L^\pi e_- = (1+\gamma)e_-.$$

For any reward vector, write the value error as $e(t) := V(t) - V^\pi = (e_1(t), e_2(t))^\top$. It decomposes into a uniform component and a difference component:

$$e(t) = \underbrace{\frac{e_1(t)+e_2(t)}{2}}_{a(t)} e_+ + \underbrace{\frac{e_1(t)-e_2(t)}{2}}_{b(t)} e_-.$$

The transition operator swaps the two entries, preserving the uniform component and reversing the difference component. Consequently, $\dot{e} = -L^\pi e$ gives

$$\dot{a} = -(1-\gamma)a, \qquad \dot{b} = -(1+\gamma)b.$$

Equivalently, $e(t) = e^{-tL^\pi}e(0)$. Thus, the e_+ mode decays at rate $(1-\gamma)$, while the e_- mode decays at rate $1+\gamma$. Reducing the uniform component by a fixed factor requires continuous time of order $1/(1-\gamma)$, while the difference component decays on a timescale that remains bounded as $\gamma \to 1$. The ratio between the fast and slow rates is

$$\frac{1+\gamma}{(1-\gamma)},$$

which diverges as $\gamma \uparrow 1$.

The reward decomposes as $r^\pi = (1,0)^\top = \frac{1}{2}e_+ + \frac{1}{2}e_-$. Therefore

$$V^\star = V^\pi = (L^\pi)^{-1}r^\pi = \frac{1}{2(1-\gamma)}e_+ + \frac{1}{2(1+\gamma)}e_-.$$

From $V_0 = 0$, the initial error $-V^\pi$ has both a slow horizon-scale component and a fast alternating component.

The same separation appears in the exact L_∞ stiffness ratio of L^π. Since the induced L_∞ matrix norm is the maximum absolute row sum, $\|L^\pi\|_\infty = 1+\gamma$. The inverse is

$$(L^\pi)^{-1} = \frac{1}{1-\gamma^2}\begin{bmatrix} 1 & \gamma \\ \gamma & 1 \end{bmatrix},$$

so $\|(L^\pi)^{-1}\|_\infty = (1+\gamma)/(1-\gamma^2) = 1/(1-\gamma)$, and

$$\rho_{\text{stiff}}(L^\pi) = \|L^\pi\|_\infty \|(L^\pi)^{-1}\|_\infty = \frac{1+\gamma}{(1-\gamma)}.$$

Explicit Euler. For the Bellman residual flow $\dot{V} = r^\pi - L^\pi V$, the explicit-Euler step with step size h is

$$V_{k+1} = V_k + h(r^\pi - L^\pi V_k).$$

Subtracting the fixed point V^π, which satisfies $r^\pi - L^\pi V^\pi = 0$, gives the error recursion and its matrix form:

$$V_{k+1} - V^\pi = (I - hL^\pi)(V_k - V^\pi),$$
$$I - hL^\pi = \begin{bmatrix} 1-h & h\gamma \\ h\gamma & 1-h \end{bmatrix}.$$

The induced norm is $\|I - hL^\pi\|_\infty = |1-h| + h\gamma$. The step is a strict contraction in supremum norm if and only if this quantity is less than 1.

If $0 < h \leq 1$, then

$$|1-h| + h\gamma = 1 - h + h\gamma = 1 - h(1-\gamma).$$

This is less than 1 for every $h \in (0, 1]$, and it is decreasing on this interval. Its minimum over $0 < h \leq 1$ occurs at $h = 1$, where the modulus is

$$1 - (1-\gamma) = \gamma.$$

If $h > 1$, then

$$|1-h| + h\gamma = h - 1 + h\gamma = h(1+\gamma) - 1.$$

The strict-contraction condition is $h(1+\gamma) - 1 < 1$, or $h < 2/(1+\gamma)$. On the interval $h > 1$, the function $h(1+\gamma) - 1$ is increasing, and its limit as $h \downarrow 1$ is γ. No $h > 1$ gives a contraction modulus smaller than γ.

Combining the two cases, explicit Euler is a strict supremum-norm contraction if and only if

$$0 < h < \frac{2}{1+\gamma},$$

and the minimizer of the contraction modulus over all such steps is $h = 1$, with modulus γ. At $h = 1$, the update is $V_{k+1} = V_k + \mathrm{BR}(V_k) = T^\pi V_k$, which is ordinary VI.

Implicit Euler. A component with decay rate λ is multiplied by $1/(1+h\lambda)$ under implicit Euler, whereas explicit Euler multiplies it by $1 - h\lambda$. Both error components are therefore damped by implicit Euler for every $h > 0$. The implicit-Euler step and its linear system form are

$$V_{k+1} = V_k + h(r^\pi - L^\pi V_{k+1}), \qquad (I + hL^\pi)V_{k+1} = V_k + hr^\pi.$$

Subtracting the fixed point gives the error recursion and the matrices needed below:

$$V_{k+1} - V^\pi = (I + hL^\pi)^{-1}(V_k - V^\pi),$$
$$I + hL^\pi = \begin{bmatrix} 1+h & -h\gamma \\ -h\gamma & 1+h \end{bmatrix},$$
$$(I + hL^\pi)^{-1} = \frac{1}{(1+h)^2 - (h\gamma)^2} \begin{bmatrix} 1+h & h\gamma \\ h\gamma & 1+h \end{bmatrix}.$$

For $h > 0$ and $\gamma \in (0, 1)$, all entries of this inverse are nonnegative. Its induced supremum norm is its row sum:

$$\begin{aligned} \|(I + hL^\pi)^{-1}\|_\infty &= \frac{1 + h + h\gamma}{(1+h)^2 - (h\gamma)^2} \\ &= \frac{1 + h(1+\gamma)}{(1 + h - h\gamma)(1 + h + h\gamma)} \\ &= \frac{1 + h(1+\gamma)}{(1 + h(1-\gamma))(1 + h(1+\gamma))} \\ &= \frac{1}{1 + h(1-\gamma)}. \end{aligned}$$

The implicit-Euler step is a strict supremum-norm contraction for every $h > 0$. With $h = 1/(1-\gamma)$, its contraction modulus is

$$\frac{1}{1 + (1/(1-\gamma))(1-\gamma)} = \frac{1}{2}.$$

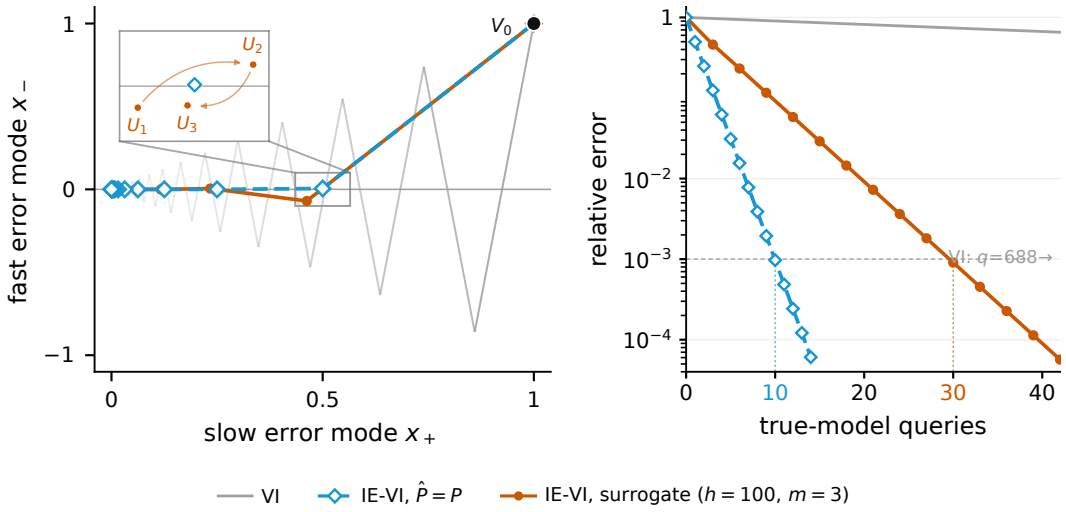


Figure B.1: Two-state swap MDP at $\gamma = 0.99$, $h = 100$. **(a)** Modal coordinates (x_+, x_-): VI (grey, rendered at contraction 0.86 for legibility) zigzags in $|x_-| \le x_+$; exact IE-VI with $\hat{\mathcal{P}}^\pi = \mathcal{P}^\pi$ (open blue diamonds, dashed) lands at $x_- = 1/200$, next to the slow eigenvector $x_- = 0$, in one outer step; surrogate IE-VI with the asymmetric self-loop $a = 0.6, b = 0$ and $m = 3$ (orange filled circles) reaches a nearby point in three inner defect-correction rounds. **(b)** Relative L_∞ value-function error against charged true-model queries; the 10^{-3} threshold is crossed at $q = 10$ for exact IE-VI, $q = 30$ for surrogate IE-VI, and $q = 688$ for VI. The rendering choice, inset details, and closed-form derivation of the three crossings are described below and in Equation (19).

Asymmetric self-loop surrogate (figure construction). Figure B.1 renders VI, exact IE-VI ($\hat{\mathcal{P}}^\pi = \mathcal{P}^\pi$, $m = 1$), and IE-VI with an asymmetric self-loop surrogate on this MDP, all in modal coordinates (x_+, x_-) defined by the basis $e_+ = (1, 1)^\top, e_- = (1, -1)^\top$. The surrogate transition operator is

$$\hat{\mathcal{P}}^\pi = \begin{bmatrix} a & 1-a \\ 1-b & b \end{bmatrix}, \qquad a = 0.6, \quad b = 0,$$

so state x_1 self-loops with probability $a = 0.6$ under $\hat{\mathcal{P}}^\pi$ while state x_2 keeps the true swap dynamics. Because $\hat{\mathcal{P}}^\pi \neq \mathcal{P}^\pi$, the surrogate resolvent $\hat{K}(h)^{-1}$ is not the true resolvent; the algorithm closes the gap through the m inner defect-correction rounds of Algorithm 1. With $\hat{\mathcal{P}}^\pi = \mathcal{P}^\pi$, $m = 1$ recovers the exact implicit-Euler step.

The figure uses $\gamma = 0.99$ (so $(1 - \gamma) = 10^{-2}$), $h = 1/(1 - \gamma) = 100$, and $m = 3$. In the phase portrait, the grey VI trajectory is rendered with contraction 0.86 for legibility: step n is drawn at $(0.86^n, (-1)^n 0.86^n)$ in these coordinates, preserving the cone shape $|x_-| \le x_+$. The true $\gamma = 0.99$ trajectory would need ~ 300 zigzags to fill the same viewport. The floating inset zooms into the box of the left panel and shows the three inner iterates U_1, U_2, U_3 and the exact-resolvent target. The inset's vertical axis is restricted to $[-0.3, 0.3]$ so the $x_- = 0$ reference line crosses its geometric midline; within this symmetric viewport the inner-iterate vertical positions are placed illustratively, and the algorithm and parameter choices are unchanged.

The right panel plots the relative L_∞ value-function error $\|V_k - V^\star\|_\infty / \|V_0 - V^\star\|_\infty$ against charged true-model queries. Once IE-VI's first outer step kills the fast mode, the slow eigenvector dominates this norm, so the curves track γ^q for VI and 2^{-q} for exact IE-VI with $h = 1/(1 - \gamma)$; surrogate IE-VI uses the same $m = 3$ recurrence as the left panel, with outer iterate k plotted at query count $q = mk$. At the 10^{-3} threshold, exact IE-VI crosses at $q = 10$, surrogate IE-VI crosses at $q = 30$, and VI crosses at $q = 688$.

Closed-form slow-mode contraction with the asymmetric surrogate. This paragraph gives a closed form for the IE-VI error operator on the two-state swap MDP with the surrogate above and recovers Figure B.1's three threshold-crossings analytically.

Modal action of $\hat{\mathcal{P}}^\pi$. Every two-state stochastic matrix fixes e_+ with eigenvalue 1, so $\hat{\mathcal{P}}^\pi$ and $\mathcal{P}^\pi$ share the slow eigenpair. In the modal basis,

$$\hat{\mathcal{P}}^\pi = \begin{bmatrix} 1 & a-b \\ 0 & a+b-1 \end{bmatrix}, \qquad \mathcal{P}^\pi - \hat{\mathcal{P}}^\pi = \begin{bmatrix} 0 & -(a-b) \\ 0 & -(a+b) \end{bmatrix}.$$

Both are upper-triangular; the surrogate's fast-mode eigenvalue is $a+b-1$, and its modal coupling is $a-b$.

Modal action of $\hat{K}(h)^{-1}$. Set $\hat{\beta} := 1+\gamma(1-a-b)$, the fast-mode eigenvalue of $\hat{L}^\pi := I - \gamma\hat{\mathcal{P}}^\pi$. Then $\hat{K}(h)^{-1}$ is also upper-triangular in the modal basis,

$$\hat{K}(h)^{-1} = \begin{bmatrix} \dfrac{1}{1+h(1-\gamma)} & \dfrac{h\gamma(a-b)}{(1+h(1-\gamma))(1+h\hat{\beta})} \\ 0 & \dfrac{1}{1+h\hat{\beta}} \end{bmatrix},$$

and its $(+,+)$ entry equals the exact-resolvent slow-mode rate $1/(1+h(1-\gamma))$ of Proposition 6, recovered without appeal to any surrogate-quality assumption.

Linear error operator. The IE-VI error $e_k := V_k - V^\pi$ satisfies $e_{k+1} = \mathsf{M}_m\, e_k$. Define

$$A := \hat{K}(h)^{-1}, \qquad B := h\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi).$$

The polynomial comes from the inner error recurrence. Put $\tilde{e}_j := U_j - V^\pi$. Subtracting the fixed-point identity for V^π from the shifted-reward equation for U_j gives

$$\tilde{e}_j = Ae_k + AB\tilde{e}_{j-1}, \qquad \tilde{e}_0 = e_k.$$

Unrolling this recurrence through m inner rounds yields

$$\mathsf{M}_m = (AB)^m + \sum_{j=0}^{m-1} (AB)^j A.$$

In modal coordinates both A and B are upper-triangular and the $(+,+)$ entry of B is zero, so for every $m \geq 1$,

$$[\mathsf{M}_m]_{++} = \frac{1}{1+h(1-\gamma)}. \tag{19}$$

The slow mode contracts at the exact-implicit rate $1/(1+h(1-\gamma))$ per outer step, irrespective of the surrogate's fast-mode eigenvalue $a+b-1$ or modal coupling $a-b$; only the fast-mode entry and the fast–slow off-diagonal of M_m are surrogate-dependent.

Specialization at the figure's parameters. For $a = 0.6, b = 0, \gamma = 0.99, h = 100, m = 3$,

$$\mathsf{M}_3 = \begin{bmatrix} 1/2 & -7.50 \\ 0 & -7.0 \times 10^{-2} \end{bmatrix} \quad \text{(modal basis)}.$$

The figure normalizes by the initial modal coordinates of $V_0 - V^\pi$; under this normalization M_3 becomes $\begin{bmatrix} 1/2 & -3.77\times10^{-2} \\ 0 & -7.0\times10^{-2} \end{bmatrix}$, and iterating from $(x_+(0), x_-(0)) = (1,1)$,

$$|x_+(k)| = 1,\ 0.462,\ 0.234,\ 0.117,\ 0.058,\ \ldots,$$

the surrogate sequence in panel (b).

Threshold-crossings at 10^{-3}. The relative L_∞ error tracks $|x_+|$ once the fast component is small. VI contracts at rate γ per query, so $q = \lceil \log(10^{-3})/\log\gamma \rceil = 688$; exact IE-VI ($\hat{\mathcal{P}}^\pi = \mathcal{P}^\pi, m = 1$) contracts at $1/2$ per query, giving $q = \lceil \log_2(10^3) \rceil = 10$; surrogate IE-VI contracts at $1/2$ per outer step at $m = 3$ charged queries per outer step, giving $q = 30$.

C Exact shifted-reward derivation

This appendix expands the algebra used in Algorithm 1. The exact implicit-Euler policy-evaluation step solves

$$K(h)Z_h^\pi(V_k) = V_k + hr^\pi, \qquad K(h) := I + hL^\pi.$$

Write $b_k := V_k + hr^\pi$. Since

$$\begin{aligned}\hat{K}(h) &:= I + h\hat{L}^\pi = (1+h)I - h\gamma\hat{\mathcal{P}}^\pi,\\ D^\pi &:= L^\pi - \hat{L}^\pi = \gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi),\\ K(h) &= \hat{K}(h) + hD^\pi,\end{aligned}$$

the corresponding defect-correction step solves

$$\begin{aligned}\hat{K}(h)U_{j+1} &= b_k - hD^\pi U_j\\ &= V_k + h\big(r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_j\big)\\ &= V_k + h\Delta^\pi(U_j).\end{aligned}$$

Dividing by $1+h$ yields the surrogate Bellman equation as the operator identity on $\mathcal{B}(\mathcal{X})$, with $\gamma_h := h\gamma/(1+h)$:

$$\begin{aligned}\left(I - \gamma_h\hat{\mathcal{P}}^\pi\right)U_{j+1} &= \tilde{r}[V_k, U_j],\\ \tilde{r}[V,U] &= \frac{1}{1+h}V + \frac{h}{1+h}\Delta^\pi(U),\end{aligned}$$

where $\Delta^\pi(U) := r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U$. Thus each inner round computes the value function of a surrogate MDP on $\mathcal{B}(\mathcal{X})$, with a shifted reward and effective discount γ_h.

Proposition 11 (Truncated computation of the implicit step)**.** *Let $Y^\pi := -h\hat{K}(h)^{-1}D^\pi$. After m inner rounds,*

$$V_{k+1} = V_k + h\left(\sum_{j=0}^{m-1}(Y^\pi)^j\right)\hat{K}(h)^{-1}\,\mathrm{BR}(V_k).$$

Equivalently, the inner loop computes a truncated Neumann expansion of the true resolvent based at the surrogate resolvent. Let $Z_h^\pi(V_k)$ be the exact implicit-Euler target. The error recursion

$$U_m - Z_h^\pi(V_k) = (Y^\pi)^m\big(V_k - Z_h^\pi(V_k)\big)$$

and the factorization $K(h) = \hat{K}(h)(I - Y^\pi)$ give

$$\big(I - (Y^\pi)^m\big)K(h)^{-1} = \left(\sum_{j=0}^{m-1}(Y^\pi)^j\right)\hat{K}(h)^{-1}.$$

The m inner rounds compute the first m terms of the Neumann expansion of $K(h)^{-1}$, with truncation operator $(Y^\pi)^m K(h)^{-1}$. This identity is the one used in Theorem 1 to decompose the iteration's error operator.

Proof. The inner loop can be written

$$U_{j+1} = \hat{K}(h)^{-1}b_k + Y^\pi U_j.$$

Let $Z_h^\pi(V_k)$ solve the exact implicit system $K(h)Z_h^\pi(V_k) = b_k$. Then $Z_h^\pi(V_k) = \hat{K}(h)^{-1}b_k + Y^\pi Z_h^\pi(V_k)$, so

$$\begin{aligned}U_{j+1} - Z_h^\pi(V_k) &= Y^\pi(U_j - Z_h^\pi(V_k)).\\ U_m &= Z_h^\pi(V_k) + (Y^\pi)^m(V_k - Z_h^\pi(V_k)).\end{aligned}$$

The exact implicit displacement is $Z_h^\pi(V_k) - V_k = hK(h)^{-1}\,\mathrm{BR}(V_k)$. It remains to identify the truncated resolvent inside the inner loop. Since $K(h) = \hat{K}(h) + hD^\pi$ and $Y^\pi = -h\hat{K}(h)^{-1}D^\pi$,

$$K(h) = \hat{K}(h)\left(I + h\hat{K}(h)^{-1}D^\pi\right) = \hat{K}(h)(I - Y^\pi).$$

The inverses used here are well defined because

$$\hat{K}(h) = (1+h)\left(I - \gamma_h\hat{\mathcal{P}}^\pi\right),$$

and $\hat{\mathcal{P}}^\pi$ is a Markov operator, so

$$\|\gamma_h \hat{\mathcal{P}}^\pi\|_\infty = \gamma_h < 1.$$

Thus the right-hand factor is invertible by the Neumann series. The same argument with $\mathcal{P}^\pi$ in place of $\hat{\mathcal{P}}^\pi$ proves that $K(h)$ is invertible. Since $K(h) = \hat{K}(h)(I - Y^\pi)$, we have $I - Y^\pi = \hat{K}(h)^{-1}K(h)$, so $I - Y^\pi$ is invertible and

$$K(h)^{-1} = (I - Y^\pi)^{-1}\hat{K}(h)^{-1}.$$

The finite geometric identity

$$I - (Y^\pi)^m = \left(\sum_{j=0}^{m-1} (Y^\pi)^j\right)(I - Y^\pi)$$

is an ordinary operator identity. Right-multiplying both sides by $(I - Y^\pi)^{-1}$ and using associativity gives

$$\big(I - (Y^\pi)^m\big)(I - Y^\pi)^{-1} = \left(\sum_{j=0}^{m-1} (Y^\pi)^j\right)(I - Y^\pi)(I - Y^\pi)^{-1} = \sum_{j=0}^{m-1} (Y^\pi)^j.$$

Right-multiplying this identity by $\hat{K}(h)^{-1}$ and using $K(h)^{-1} = (I - Y^\pi)^{-1}\hat{K}(h)^{-1}$ gives

$$(I - (Y^\pi)^m)K(h)^{-1} = \left(\sum_{j=0}^{m-1} (Y^\pi)^j\right)\hat{K}(h)^{-1},$$

which is the truncated-resolvent identity. Combining it with the inner-error relation and the displacement identity gives

$$\begin{aligned} U_m - V_k &= \big(I - (Y^\pi)^m\big)(Z_h^\pi(V_k) - V_k) \\ &= h\big(I - (Y^\pi)^m\big)K(h)^{-1}\,\mathrm{BR}(V_k) \\ &= h\left(\sum_{j=0}^{m-1} (Y^\pi)^j\right)\hat{K}(h)^{-1}\,\mathrm{BR}(V_k). \end{aligned}$$

Since $V_{k+1} = U_m$, adding V_k to both sides proves the claim. □

Proposition 12 (Arbitrary precision of the iterated defect correction for Policy Evaluation and Control)**.** *Consider the exact implicit-Euler target $Z_h^\dagger(V_k)$ evaluated at outer iterate V_k. For Policy Evaluation, $Z_h^\pi(V_k) = K(h)^{-1}(V_k + hr^\pi)$; for Control, $Z_h^\star(V_k)$ is the unique solution to the exact implicit optimality equation. Let the corresponding Bellman residual be defined as $\mathrm{BR}^\dagger(V_k)$, representing either $\mathrm{BR}(V_k)$ or $\mathrm{BR}^\star(V_k)$, and let $R_E := \frac{E_\infty}{1-\gamma}$ denote the normalized model error.*

The algorithm branches its parameter choices and behavior based on the error ratio:

1. ***The Low-Error Regime*** *($R_E < 1$). The algorithm takes the limit $h \to \infty$ and sets the inner loop depth $m = 1$. In this limit, the exact resolvent contraction term vanishes, and the defect contraction base reduces to $\rho(h) \to R_E < 1$.*

2. ***The High-Error Regime*** *($R_E \geq 1$). The defect-correction contraction base $\rho(h) := \frac{hE_\infty}{1+h(1-\gamma)}$ is strictly less than 1 if and only if the step size $h > 0$ is chosen within the admissible domain*

$$h \in \left(0, \frac{1}{(1-\gamma)(R_E - 1)}\right),$$

where $1/0$ is interpreted as ∞. To ensure this, the algorithm sets

$$h = \frac{1}{(1-\gamma)(2R_E - 1)},$$

which gives $\rho(h) \leq 1/2$.

When h is chosen within its valid regime, the iterated defect correction procedure refines the approximation to the exact true implicit step to arbitrary precision. Defining the explicit initial displacement bound as

$$E_{\text{init}} := \frac{h}{1+h(1-\gamma)} \|\text{BR}^\dagger(V_k)\|_\infty,$$

the approximation error after $m \geq 0$ inner rounds initialized at $U_0 = V_k$ satisfies

$$\|U_m - Z_h^\dagger(V_k)\|_\infty \leq \rho(h)^m E_{\text{init}}.$$

For every desired precision $\epsilon > 0$, guaranteeing $\|U_m - Z_h^\dagger(V_k)\|_\infty \leq \epsilon$ requires at most $m(\epsilon)$ inner rounds. The depth also depends on the fixed values of $\rho(h)$ and E_{init} and is defined as

$$m(\epsilon) = \begin{cases} 0, & \text{if } E_{\text{init}} \leq \epsilon, \\ 1, & \text{if } E_{\text{init}} > \epsilon \text{ and } \rho(h) = 0, \\ \left\lceil \dfrac{\ln(E_{\text{init}}/\epsilon)}{\ln(1/\rho(h))} \right\rceil, & \text{if } E_{\text{init}} > \epsilon \text{ and } 0 < \rho(h) < 1. \end{cases}$$

Furthermore, under the High-Error schedule ($R_E \geq 1$, $h = [(1-\gamma)(2R_E - 1)]^{-1}$), the required inner depth is bounded by the base-2 envelope

$$m(\epsilon) \leq \max\{0, \lceil \log_2(E_{\text{init}}/\epsilon) \rceil\}.$$

Proof. We first derive the exact analytical domain for finite h that guarantees $\rho(h) < 1$. The strict contraction condition requires

$$\frac{hE_\infty}{1+h(1-\gamma)} < 1 \iff hE_\infty < 1 + h(1-\gamma) \iff h\big(E_\infty - (1-\gamma)\big) < 1.$$

Dividing by the strictly positive scalar $1-\gamma$ exposes the dependency on the normalized error ratio $R_E = \frac{E_\infty}{1-\gamma}$:

$$h(1-\gamma)(R_E - 1) < 1.$$

If $R_E < 1$, then $R_E - 1 < 0$. Since $h(1-\gamma) > 0$, the left-hand side is strictly negative, so the inequality holds for arbitrarily large step sizes $h \in (0, \infty)$. Conversely, if $R_E \geq 1$, isolating h mandates

$$h < \frac{1}{(1-\gamma)(R_E - 1)},$$

where $1/0$ is interpreted as ∞.

Under the branched algorithmic choices, the Low-Error Regime $R_E < 1$ takes the limit $h \to \infty$ with $m = 1$, meeting the condition and evaluating an OS-VI step. As $h \to \infty$, $1/(1+h(1-\gamma)) \to 0$ and $\rho(h) = E_\infty/(h^{-1} + 1 - \gamma) \to R_E$. In the High-Error Regime $R_E \geq 1$, the schedule sets

$$h = [(1-\gamma)(2R_E - 1)]^{-1}.$$

Because $R_E \geq 1$, we have $2R_E - 1 \geq R_E > 0$, so the chosen h lies in the strictly contractive domain. Proposition 18 proves that this choice gives $\rho(h) \leq 1/2$. Assuming h is drawn from the valid domain, we prove the geometric contraction bound.

Part 1: Policy Evaluation (Linear Defect Correction). Following Algorithm 1, the j-th inner round computes the exact solution U_j to the surrogate equation $(I - \gamma_h \hat{\mathcal{P}}^\pi)U_j = \tilde{r}_j^\pi$. We explicitly expand the shifted reward from the algorithm definition:

$$\tilde{r}_j^\pi = \frac{1}{1+h} V_k + \frac{h}{1+h}\Big(r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_{j-1}\Big).$$

Multiplying the surrogate equation by $1+h$ and substituting $\gamma_h = \frac{h\gamma}{1+h}$ yields

$$\big((1+h)I - h\gamma\hat{\mathcal{P}}^\pi\big)U_j = V_k + hr^\pi + h\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)U_{j-1}.$$

We recognize the left-hand operator as $I + h(I - \gamma\hat{\mathcal{P}}^\pi) = I + h\hat{L}^\pi = \hat{K}(h)$. On the right side, the defect operator is

$$D^\pi := L^\pi - \hat{L}^\pi = \gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi),$$

meaning $h\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi) = -hD^\pi = \hat{K}(h) - K(h)$. Substituting these identities yields

$$\hat{K}(h)U_j = V_k + hr^\pi + \big(\hat{K}(h) - K(h)\big)U_{j-1}.$$

Subtracting $\hat{K}(h)U_{j-1}$ from both sides and left-multiplying by the inverse operator $\hat{K}(h)^{-1}$ recovers the standard linear defect correction update:

$$U_j = U_{j-1} + \hat{K}(h)^{-1}\big(V_k + hr^\pi - K(h)U_{j-1}\big).$$

Since the exact target satisfies $K(h)Z_h^\pi(V_k) = V_k + hr^\pi$, substituting this into the recurrence and subtracting $Z_h^\pi(V_k)$ yields the error dynamics:

$$U_j - Z_h^\pi(V_k) = \Big(I - \hat{K}(h)^{-1}K(h)\Big)\big(U_{j-1} - Z_h^\pi(V_k)\big).$$

The error-propagation operator is

$$Y^\pi := I - \hat{K}(h)^{-1}K(h) = \hat{K}(h)^{-1}\big(\hat{K}(h) - K(h)\big) = -h\hat{K}(h)^{-1}D^\pi.$$

Unrolling over m rounds yields

$$\|U_m - Z_h^\pi(V_k)\|_\infty \le \|Y^\pi\|_\infty^m \|V_k - Z_h^\pi(V_k)\|_\infty.$$

By Lemma 16, $\|\hat{K}(h)^{-1}\|_\infty \le \frac{1}{1+h(1-\gamma)}$. Submultiplicativity bounds the modulus as $\|Y^\pi\|_\infty \le \frac{hE_\infty}{1+h(1-\gamma)} = \rho(h)$. The initial displacement is derived by subtracting $K(h)V_k = (I + hL^\pi)V_k = V_k + hL^\pi V_k$ from the exact system $K(h)Z_h^\pi(V_k) = V_k + hr^\pi$, yielding

$$K(h)\big(Z_h^\pi(V_k) - V_k\big) = h(r^\pi - L^\pi V_k) = h\,\mathrm{BR}(V_k).$$

Inverting $K(h)$ provides the explicit bound

$$\|Z_h^\pi(V_k) - V_k\|_\infty \le \frac{h}{1+h(1-\gamma)}\|\mathrm{BR}(V_k)\|_\infty = E_{\text{init}}.$$

Part 2: Control (Nonlinear Defect Correction). For Control, the extension in Section 4.1 uses the same outer-inner structure, solving a surrogate MDP exactly in each inner round. Lemma 23 establishes that this is equivalent to finding the unique fixed point of the strictly contractive operator $\hat{T}_{h,V_k,U_{j-1}}$. By the definition of Ψ_{h,V_k} (Appendix D.2), this solution is the output of the action-parameterized operator $\Psi_{h,V_k}(U_{j-1})$. Thus, the algorithmic step maps flawlessly to

$$U_j = \Psi_{h,V_k}(U_{j-1}).$$

Because the space of bounded measurable functions $\mathcal{B}(\mathcal{X})$ equipped with the supremum norm is a complete Banach space, and Lemma 25 establishes that Ψ_{h,V_k} is strictly $\rho(h)$-Lipschitz, the conditions for the Banach fixed-point theorem are fully satisfied. Lemma 24 establishes that the exact implicit step $Z_h^\star(V_k)$ is precisely the unique fixed point of Ψ_{h,V_k}. Therefore, the metric distance to the target strictly decays:

$$\begin{aligned}\|U_j - Z_h^\star(V_k)\|_\infty &= \|\Psi_{h,V_k}(U_{j-1}) - \Psi_{h,V_k}(Z_h^\star(V_k))\|_\infty \\ &\le \rho(h)\|U_{j-1} - Z_h^\star(V_k)\|_\infty.\end{aligned}$$

Unrolling identically yields

$$\|U_m - Z_h^\star(V_k)\|_\infty \le \rho(h)^m \|V_k - Z_h^\star(V_k)\|_\infty.$$

To bound the initial displacement tightly, multiply the exact implicit optimality equation (uniquely solvable by Lemma 22)

$$Z_h^\star(V_k) = \frac{1}{1+h}V_k + \frac{h}{1+h}T^\star Z_h^\star(V_k)$$

by $1+h$ and subtract $(1+h)V_k$:

$$(1+h)\big(Z_h^\star(V_k) - V_k\big) = h\big(T^\star Z_h^\star(V_k) - V_k\big).$$

Adding and subtracting $hT^\star V_k$ inside the right-hand parentheses gives

$$(1+h)\big(Z_h^\star(V_k) - V_k\big) = h\big(T^\star Z_h^\star(V_k) - T^\star V_k\big) + h\big(T^\star V_k - V_k\big).$$

Recognizing $T^\star V_k - V_k = \mathrm{BR}^\star(V_k)$, we take the supremum norm and apply the triangle inequality alongside the rigorous γ-contraction property of the Bellman optimality operator $T^\star$ (Remark 21):

$$(1+h)\|Z_h^\star(V_k) - V_k\|_\infty \le h\gamma\|Z_h^\star(V_k) - V_k\|_\infty + h\|\mathrm{BR}^\star(V_k)\|_\infty,$$
$$\big(1+h(1-\gamma)\big)\|Z_h^\star(V_k) - V_k\|_\infty \le h\|\mathrm{BR}^\star(V_k)\|_\infty.$$

Dividing by the scalar strictly isolates the identical initial error bound:

$$\|Z_h^\star(V_k) - V_k\|_\infty \le \frac{h}{1+h(1-\gamma)}\|\mathrm{BR}^\star(V_k)\|_\infty = E_{\text{init}}.$$

Part 3: Unified finite-step formula. We solve for m to guarantee $\rho(h)^m E_{\text{init}} \le \epsilon$. By choice of h, $0 \le \rho(h) < 1$. If $E_{\text{init}} \le \epsilon$, the condition is met at $m = 0$. If $E_{\text{init}} > \epsilon$ and the surrogate is exact ($\rho(h) = 0$), then exactly $m = 1$ round yields an error of $0 \le \epsilon$. If $E_{\text{init}} > \epsilon$ and $0 < \rho(h) < 1$, isolating m yields $\rho(h)^m \le \epsilon / E_{\text{init}}$. Because $\ln(\rho(h))$ is strictly negative, dividing by it reverses the inequality, yielding

$$m \ge \frac{\ln(E_{\text{init}}/\epsilon)}{\ln(1/\rho(h))}.$$

Taking the ceiling gives the needed integer round count. Under the High-Error schedule, $\rho(h) \le 1/2$, so $\ln(1/\rho(h)) \ge \ln(2)$. Substituting the smaller denominator gives

$$\left\lceil \frac{\ln(E_{\text{init}}/\epsilon)}{\ln(1/\rho(h))} \right\rceil \le \lceil \log_2(E_{\text{init}}/\epsilon) \rceil .$$

The cases $m(\epsilon) = 0$ and $m(\epsilon) = 1$ also obey the envelope, since $E_{\text{init}} > \epsilon$ gives $\lceil \log_2(E_{\text{init}}/\epsilon) \rceil \ge 1$. □

Proposition 13 (IE-VI subsumes OS-VI in the $h \to \infty$, $m = 1$ limit)**.** *Recall the defect-corrected reward $\Delta^\pi(V) := r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)V$. Let the OS-VI policy-evaluation update be the standard operator-splitting equation for target U, given V:*

$$\hat{L}^\pi U = \Delta^\pi(V)$$

[Rakhsha et al., 2022]. Equivalently, its residual form is

$$T_{\mathrm{OS}}(V) := V + (\hat{L}^\pi)^{-1}\,\mathrm{BR}(V).$$

The one-round IE-VI *update is*

$$T_h^{\text{IE-VI}}(V) := V + h\hat{K}(h)^{-1}\,\mathrm{BR}(V).$$

For every fixed V, $T_h^{\text{IE-VI}}(V) \to T_{\mathrm{OS}}(V)$ as $h \to \infty$. The convergence is uniform over every bounded supremum-norm ball.

Remark 14 (OS-VI as a shifted-reward limit). The identity $\hat{L}^\pi U = \Delta^\pi(V)$ is equivalent to $U = \Delta^\pi(V) + \gamma\hat{\mathcal{P}}^\pi U$. Thus OS-VI solves a surrogate policy-evaluation problem whose reward is the defect-corrected reward $\Delta^\pi(V)$ under discount γ. One-round IE-VI solves the corresponding surrogate problem with reward $(1+h)^{-1}V + h(1+h)^{-1}\Delta^\pi(V)$ and discount γ_h; sending $h \to \infty$ gives the OS-VI equation.

Proof. We first verify the residual form. From $L^\pi = I - \gamma\mathcal{P}^\pi$ and $\hat{L}^\pi = I - \gamma\hat{\mathcal{P}}^\pi$,

$$\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi) = \hat{L}^\pi - L^\pi.$$

Substituting this identity into $\hat{L}^\pi U = \Delta^\pi(V)$ gives

$$\hat{L}^\pi U = r^\pi + \hat{L}^\pi V - L^\pi V.$$
$$\hat{L}^\pi(U - V) = r^\pi - L^\pi V = \mathrm{BR}(V).$$

Since $\hat{\mathcal{P}}^\pi$ is a Markov operator and $\gamma < 1$, $\hat{L}^\pi = I - \gamma\hat{\mathcal{P}}^\pi$ is invertible by the discounted Neumann series. Thus $U = V + (\hat{L}^\pi)^{-1}\,\mathrm{BR}(V) = T_{\mathrm{OS}}(V)$. Subtracting V^π from the residual form and using $r^\pi = L^\pi V^\pi$ gives

$$\begin{aligned} T_{\mathrm{OS}}(V) - V^\pi &= V - V^\pi + (\hat{L}^\pi)^{-1}\,\mathrm{BR}(V) \\ &= V - V^\pi - (\hat{L}^\pi)^{-1} L^\pi (V - V^\pi) \\ &= (\hat{L}^\pi)^{-1}(\hat{L}^\pi - L^\pi)(V - V^\pi). \end{aligned}$$

Since $\hat{L}^\pi - L^\pi = \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)$, the OS-VI error operator around the true fixed point is

$$T_{\mathrm{OS}}(V) - V^\pi = (\hat{L}^\pi)^{-1}\gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)(V - V^\pi),$$

and hence the standard supremum-norm bound

$$\|T_{\mathrm{OS}}(V) - V^\pi\|_\infty \le \frac{\gamma\|\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi\|_\infty}{1-\gamma}\|V - V^\pi\|_\infty = R_E\|V - V^\pi\|_\infty.$$

This is the OS-VI comparison factor used in Section 4.2.

The implicit-step operator in the one-round IE-VI update can be rewritten as

$$h\hat{K}(h)^{-1} = \left(\frac{1}{h}I + \hat{L}^\pi\right)^{-1}.$$

Define the resolvent difference used below by

$$\Xi_h := \left(\frac{1}{h}I + \hat{L}^\pi\right)^{-1} - (\hat{L}^\pi)^{-1}.$$

By the resolvent identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$, with $A = h^{-1}I + \hat{L}^\pi$ and $B = \hat{L}^\pi$,

$$\Xi_h = -\frac{1}{h}\left(\frac{1}{h}I + \hat{L}^\pi\right)^{-1}(\hat{L}^\pi)^{-1}.$$

The discounted Neumann-series bound gives $\|(\hat{L}^\pi)^{-1}\|_\infty \le 1/(1-\gamma)$ for the surrogate Bellman generator. For the resolvent of $-\hat{L}^\pi$ at parameter $1/h$, recall that $\gamma_h = h\gamma/(1+h) = \gamma/(1+1/h)$ and factor

$$\frac{1}{h}I + \hat{L}^\pi = \frac{1+h}{h}\left(I - \gamma_h\hat{\mathcal{P}}^\pi\right).$$

Since $\gamma_h < 1$ and $\|\hat{\mathcal{P}}^\pi\|_\infty = 1$,

$$\begin{aligned} \left\|\left(I - \gamma_h\hat{\mathcal{P}}^\pi\right)^{-1}\right\|_\infty &\le \sum_{t=0}^{\infty}\gamma_h^t = \frac{1}{1-\gamma_h} = \frac{1+h}{1+h(1-\gamma)}, \\ \left\|\left(\frac{1}{h}I + \hat{L}^\pi\right)^{-1}\right\|_\infty &\le \frac{h}{1+h}\frac{1+h}{1+h(1-\gamma)} = \frac{h}{1+h(1-\gamma)}, \\ \|\Xi_h\|_\infty &\le \frac{1}{h}\frac{h}{1+h(1-\gamma)}\frac{1}{(1-\gamma)} = \frac{1}{(1-\gamma)(1+h(1-\gamma))} \xrightarrow[h\to\infty]{} 0. \end{aligned}$$

For a fixed V,

$$\|T_h^{\text{IE-VI}}(V) - T_{\mathrm{OS}}(V)\|_\infty \le \|\Xi_h\|_\infty\|\mathrm{BR}(V)\|_\infty,$$

which proves pointwise convergence. If $\mathcal{B}_R := \{V : \|V\|_\infty \le R\}$, then

$$\begin{aligned} \sup_{V\in\mathcal{B}_R}\|\mathrm{BR}(V)\|_\infty &\le \|r^\pi\|_\infty + (1+\gamma)R, \\ \sup_{V\in\mathcal{B}_R}\|T_h^{\text{IE-VI}}(V) - T_{\mathrm{OS}}(V)\|_\infty &\le \|\Xi_h\|_\infty\left(\|r^\pi\|_\infty + (1+\gamma)R\right). \end{aligned}$$

Writing $C_R := \|r^\pi\|_\infty + (1+\gamma)R$, the last display and the bound on Ξ_h give

$$\sup_{V\in\mathcal{B}_R}\|T_h^{\text{IE-VI}}(V) - T_{\mathrm{OS}}(V)\|_\infty \le \frac{C_R}{(1-\gamma)(1+h(1-\gamma))} \xrightarrow[h\to\infty]{} 0.$$

This proves uniform convergence on $\mathcal{B}_R$. □

D Convergence proofs

This appendix supplies the algebra deferred from Section 4.2. The policy-evaluation proof first identifies the exact error-propagation operator for Algorithm 1. The control proof then repeats the same parameter-choice calculation for the nonlinear optimality map, using the action-wise shifted-reward realization in Equations (7) and (8).

D.1 Policy-evaluation proofs

Throughout this subsection we use the defect and inner-loop operators introduced in Appendix C:

$$D^\pi \;:=\; L^\pi - \hat{L}^\pi \;=\; \gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi), \qquad Y^\pi \;:=\; -h\hat{K}(h)^{-1}D^\pi.$$

The proof of Theorem 1 derives the defect-correction residual $R_{h,m} = h(Y^\pi)^m K(h)^{-1}L^\pi$ used in the contraction bound.

Proposition 15 (Fidelity preservation)**.** *For every $h > 0$, every integer $m \geq 1$, and every Markov surrogate $\hat{\mathcal{P}}$, the* IE-VI *iteration map fixes V^π: $\Phi_h^m(V^\pi) = V^\pi$.*

Proof. Set $V_k = U_0 = V^\pi$. Using $r^\pi = L^\pi V^\pi$, the defect-corrected reward at V^π satisfies $\Delta^\pi(V^\pi) = r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)V^\pi = V^\pi - \gamma\hat{\mathcal{P}}^\pi V^\pi$, so the shifted reward is $\tilde{r}[V^\pi, V^\pi] = (I - \gamma_h\hat{\mathcal{P}}^\pi)V^\pi$. The inner equation Algorithm 1 then reads $(I - \gamma_h\hat{\mathcal{P}}^\pi)U_1 = (I - \gamma_h\hat{\mathcal{P}}^\pi)V^\pi$, giving $U_1 = V^\pi$ by uniqueness; induction gives $U_j = V^\pi$ for all $j \leq m$. □

Proposition 15 establishes $\Phi_h^m(V^\pi) = V^\pi$; we use it freely below.

Lemma 16 (Resolvent and defect bounds)**.** *For every $h > 0$, the operators $K(h) = I + hL^\pi$ and $\hat{K}(h) = I + h\hat{L}^\pi$ are invertible. Let $\rho(h) := hE_\infty/(1 + h(1-\gamma))$. Then*

$$\|K(h)^{-1}\|_\infty \leq \frac{1}{1+h(1-\gamma)}, \qquad \|\hat{K}(h)^{-1}\|_\infty \leq \frac{1}{1+h(1-\gamma)},$$
$$\|Y^\pi\|_\infty \leq \rho(h), \qquad \|K(h)^{-1}L^\pi\|_\infty \leq \frac{1+\gamma+h(1-\gamma)}{(1+h)(1+h(1-\gamma))}.$$

Proof. Write $\gamma_h := h\gamma/(1+h)$. The true resolvent factors as

$$K(h) = (1+h)\,(I - \gamma_h\mathcal{P}^\pi)\,.$$

Since $\mathcal{P}^\pi$ is a Markov operator,

$$\|\gamma_h\mathcal{P}^\pi\|_\infty = \gamma_h < 1.$$

Therefore the inverse exists by the Neumann series, and

$$\|K(h)^{-1}\|_\infty \leq \frac{1}{1+h}\sum_{t=0}^{\infty}\gamma_h^t = \frac{1}{1+h(1-\gamma)}.$$

The same factorization with $\hat{\mathcal{P}}^\pi$ gives the same bound for $\hat{K}(h)^{-1}$. Since $Y^\pi = -h\hat{K}(h)^{-1}D^\pi$, the bound on $\|Y^\pi\|_\infty$ follows from submultiplicativity and $\|D^\pi\|_\infty = E_\infty$.

It remains to bound $K(h)^{-1}L^\pi$. Fix $V \in \mathcal{B}(\mathcal{X})$ and set $U := K(h)^{-1}V$. The equation $K(h)U = V$ is equivalent to

$$U = \frac{1}{1+h}V + \gamma_h\mathcal{P}^\pi U.$$

Subtracting V and using the Markov-operator bound gives

$$\begin{aligned}\|U - V\|_\infty &\leq \frac{h}{1+h}\,(\gamma\|U\|_\infty + \|V\|_\infty)\\ &\leq \frac{h}{1+h}\left(\frac{\gamma}{1+h(1-\gamma)} + 1\right)\|V\|_\infty\\ &= \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}\|V\|_\infty,\end{aligned}$$

where the second line uses $\|K(h)^{-1}\|_\infty \le (1+h(1-\gamma))^{-1}$. Since $K(h) = I + hL^\pi$, we have $K(h)^{-1} - I = -hK(h)^{-1}L^\pi$. Thus $U - V = -hK(h)^{-1}L^\pi V$, so taking the supremum over $\|V\|_\infty \le 1$ gives

$$\|hK(h)^{-1}L^\pi\|_\infty \le \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}.$$

Dividing by $h > 0$ proves the stated bound. □

The following result is helpful.

Proposition 17 (Constant-vector eigenidentity)**.** *For every $h > 0$ and $m \ge 1$, $\hat{M}_h^m \mathbf{1} = (1+h(1-\gamma))^{-1}\mathbf{1}$, where $\mathbf{1} \in \mathcal{B}(\mathcal{X})$ is the constant-one function.*

Proof. $\mathcal{P}^\pi \mathbf{1} = \hat{\mathcal{P}}^\pi \mathbf{1} = \mathbf{1}$, so $D^\pi \mathbf{1} = 0$ and $Y^\pi \mathbf{1} = -h\hat{K}(h)^{-1}D^\pi \mathbf{1} = 0$; the residual $h(Y^\pi)^m K(h)^{-1}L^\pi \mathbf{1}$ vanishes on $\mathbf{1}$. Since $L^\pi \mathbf{1} = (1-\gamma)\mathbf{1}$, $K(h)^{-1}\mathbf{1} = (1+h(1-\gamma))^{-1}\mathbf{1}$. □

Proof of Theorem 1. By Proposition 11,

$$V_{k+1} = V_k + h\left(\sum_{j=0}^{m-1}(Y^\pi)^j\right)\hat{K}(h)^{-1}\,\mathrm{BR}(V_k).$$

Since $r^\pi = L^\pi V^\pi$,

$$\mathrm{BR}(V_k) = r^\pi - L^\pi V_k = -L^\pi(V_k - V^\pi).$$

Subtracting V^π from the update therefore gives

$$V_{k+1} - V^\pi = \left[I - h\left(\sum_{j=0}^{m-1}(Y^\pi)^j\right)\hat{K}(h)^{-1}L^\pi\right](V_k - V^\pi).$$

The geometric identity from Proposition 11 gives

$$\left(\sum_{j=0}^{m-1}(Y^\pi)^j\right)\hat{K}(h)^{-1} = (I - (Y^\pi)^m)K(h)^{-1}.$$

Substituting this identity into the error recursion gives the unsimplified error-propagation operator

$$\hat{M}_h^m = I - h(I-(Y^\pi)^m)K(h)^{-1}L^\pi = I - hK(h)^{-1}L^\pi + h(Y^\pi)^m K(h)^{-1}L^\pi.$$

Because $K(h) = I + hL^\pi$, we have $hL^\pi = K(h) - I$, and multiplying by $K(h)^{-1}$ gives

$$hK(h)^{-1}L^\pi = K(h)^{-1}(K(h) - I) = I - K(h)^{-1}.$$
$$I - hK(h)^{-1}L^\pi = K(h)^{-1},$$

and the error-propagation operator admits the resolvent-defect decomposition:

$$\hat{M}_h^m = \underbrace{K(h)^{-1}}_{\text{resolvent contraction}} + \underbrace{h(Y^\pi)^m K(h)^{-1}L^\pi}_{\text{defect-correction residual } R_{h,m}}.$$

Taking induced norms and applying Lemma 16,

$$\|\hat{M}_h^m\|_\infty \le \frac{1}{1+h(1-\gamma)} + \rho(h)^m \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}.$$

For Control, $V_{k+1} = U_m$ with $U_0 = V_k$ and $U_{j+1} = \Psi_{h,V_k}(U_j)$ (Appendix D.2), so Equation (10) with $V^\dagger = V^\star$ is Lemma 26. □

The constant-vector eigenidentity $\hat{M}_h^m \mathbf{1} = (1+h(1-\gamma))^{-1}\mathbf{1}$ of Proposition 17 combined with $\|\mathbf{1}\|_\infty = 1$ yield the operator-norm lower bound $\|\hat{M}_h^m\|_\infty \ge 1/(1+h(1-\gamma))$ used in the discussion of Theorem 1.

Proposition 18 (Step-size choice)**.** *Let* $\Lambda \geq 1$ *and* $\Lambda \geq R_E$*. Set* $h_\Lambda := [(1-\gamma)(2\Lambda-1)]^{-1}$*. Then*

$$\rho(h_\Lambda) \leq \frac{1}{2}, \qquad \frac{1}{1+h_\Lambda(1-\gamma)} = 1 - \frac{1}{2\Lambda}, \qquad \frac{h_\Lambda}{1+h_\Lambda(1-\gamma)} = \frac{1}{2(1-\gamma)\Lambda}.$$

Proof. Since $h_\Lambda(1-\gamma) = (2\Lambda-1)^{-1}$, we have $1 + h_\Lambda(1-\gamma) = 2\Lambda/(2\Lambda-1)$. This gives $1/(1+h_\Lambda(1-\gamma)) = 1 - 1/(2\Lambda)$ and $h_\Lambda/(1+h_\Lambda(1-\gamma)) = 1/(2(1-\gamma)\Lambda)$. The defect-correction base is

$$\rho(h_\Lambda) = \frac{h_\Lambda E_\infty}{1+h_\Lambda(1-\gamma)} = \frac{E_\infty/(1-\gamma)}{2\Lambda} = \frac{R_E}{2\Lambda} \leq \frac{1}{2}.$$

□

Lemma 19 (Per-step modulus under the parameter choice)**.** *Let* $\Lambda \geq 1$ *and* $\Lambda \geq R_E$*, let* $h = h_\Lambda = [(1-\gamma)(2\Lambda-1)]^{-1}$*, and let* $m = \lceil \log_2(8/(1-\gamma)) \rceil$*. Then*

$$\frac{1}{1+h(1-\gamma)} + \rho(h)^m \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))} \leq 1 - \frac{1}{2\Lambda} + \frac{1}{8\Lambda} = 1 - \frac{3}{8\Lambda} \leq 1 - \frac{1}{4\Lambda}.$$

Proof. By Proposition 18, $\rho(h) \leq 1/2$, so $\rho(h)^m \leq (1-\gamma)/8$; also $1/(1+h(1-\gamma)) = 1 - 1/(2\Lambda)$ and $h/(1+h(1-\gamma)) = 1/(2(1-\gamma)\Lambda)$. The defect-correction prefactor factors as

$$\frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))} = \frac{h}{1+h(1-\gamma)} \cdot \frac{1+\gamma+h(1-\gamma)}{1+h}.$$

Using $(1-\gamma) = 1-\gamma$,

$$\frac{1+\gamma+h(1-\gamma)}{1+h} = 1 + \gamma\frac{1-h}{1+h} \leq 1+\gamma \leq 2,$$

so the rational coefficient is at most $1/((1-\gamma)\Lambda)$ and the defect-correction residual is at most $((1-\gamma)/8) \cdot 1/((1-\gamma)\Lambda) = 1/(8\Lambda)$. Combining the resolvent and defect bounds gives the displayed chain. □

Proof of Theorem 2. We split the proof according to the two algorithmic regimes.

Part 1: Low-Error Regime $(R_E < 1)$**.** Set $m = 1$ and take the limit $h \to \infty$. By Propositions 11 and 13, this update reduces to OS-VI. In this limit, the exact resolvent contraction term $1/(1+h(1-\gamma))$ goes to zero, and the per-step contraction modulus converges to

$$\lim_{h\to\infty} \rho(h) = R_E.$$

Therefore

$$\|V_N - V^\pi\|_\infty \leq R_E^N \|V_0 - V^\pi\|_\infty.$$

Because $m = 1$, each outer step costs one true-model query. The relative error ϵ is reached when $R_E^N \leq \epsilon$, which for $R_E > 0$ requires

$$N \geq \frac{\ln(1/\epsilon)}{\ln(1/R_E)};$$

for $R_E = 0$, one step suffices. Using $\ln(1/R_E) \geq 1 - R_E$, the iteration complexity is

$$O\left(\frac{1}{1-R_E} \log \frac{1}{\epsilon}\right).$$

Part 2: High-Error Regime $(R_E \geq 1)$**.** Use the finite step-size schedule

$$h = \frac{1}{(1-\gamma)(2R_E-1)}, \qquad m = \lceil \log_2(8/(1-\gamma)) \rceil.$$

By Theorem 1 and Lemma 19 applied with $\Lambda = R_E$, $h = h_{R_E}$, and $m = \lceil \log_2(8/(1-\gamma)) \rceil$,

$$\|V_{k+1} - V^\pi\|_\infty \leq \left(1 - \frac{1}{4R_E}\right) \|V_k - V^\pi\|_\infty.$$

Therefore,

$$\|V_N - V^\pi\|_\infty \le \left(1 - \frac{1}{4R_E}\right)^N \|V_0 - V^\pi\|_\infty.$$

Using $(1-x)^N \le e^{-xN}$ for $x \in (0,1)$, the right side is at most $\epsilon\|V_0 - V^\pi\|_\infty$ whenever

$$N \ge 4R_E \ln(1/\epsilon).$$

The choice $N = \lceil 4R_E \ln(1/\epsilon)\rceil$ suffices. Each outer iteration uses $m = \lceil \log_2(8/(1-\gamma))\rceil$ true-model correction rounds. Since $E_\infty \le 2\gamma$, $R_E \ge 1$ forces $\gamma \ge 1/3$, so $\log_2(8/(1-\gamma)) \le 7\log_2(1/(1-\gamma))$, and the total query complexity is

$$N \cdot m = O\left(R_E \log \frac{1}{(1-\gamma)} \log \frac{1}{\epsilon}\right).$$

Sublinear model-error acceleration. Assume $E_\infty \le C_0(1-\gamma)\log(1/(1-\gamma))$ for some constant $C_0 \ge 0$, and let C_1 be C_0 times the conversion factor to base 2. This implies

$$R_E \le C_1 \log_2(1/(1-\gamma)).$$

Use the uniform finite-step schedule, that is, Part 2 with R_E replaced by $\bar{R}_E := \max\{1, R_E\}$ (Lemma 19 allows every $\Lambda \ge \max\{1, R_E\}$); this also covers $R_E < 1$. Since $\bar{R}_E \le 1 + C_1 \log_2(1/(1-\gamma))$, substituting into the Part 2 step count gives

$$\lceil 4\bar{R}_E \ln(1/\epsilon)\rceil \le 4\big(1 + C_1 \log_2(1/(1-\gamma))\big)\ln(1/\epsilon) + 1,$$

and the inner depth satisfies

$$\lceil \log_2(8/(1-\gamma))\rceil \le \log_2(8/(1-\gamma)) + 1.$$

Multiplying the two bounds gives

$$Q(\epsilon) \le \Big[4\big(1 + C_1 \log_2(1/(1-\gamma))\big)\ln(1/\epsilon) + 1\Big]\Big[\log_2(8/(1-\gamma)) + 1\Big].$$

It remains only to bound the displayed product by a constant multiple of $x^2 L_\epsilon$. Set

$$x := \log_2(8/(1-\gamma)) = 3 + \log_2(1/(1-\gamma)) \ge 3, \qquad L_\epsilon := \max\{1, \ln(1/\epsilon)\}.$$

Then $1 + C_1 \log_2(1/(1-\gamma)) \le 1 + C_1 x$ and $L_\epsilon \ge 1$ give

$$4(1 + C_1 x)\ln(1/\epsilon) + 1 \le (5 + 4C_1 x)L_\epsilon.$$

The lower bound $x \ge 3$ also gives the auxiliary estimates

$$\log_2(8/(1-\gamma)) + 1 \le \frac{4}{3}x, \qquad (5 + 4C_1 x)L_\epsilon \le \left(\frac{5}{3} + 4C_1\right) x L_\epsilon.$$

Combining these estimates yields

$$Q(\epsilon) \le C_2 x^2 L_\epsilon$$

for a constant C_2 depending only on the constant in the hypothesis. Substituting back $x = \log_2(8/(1-\gamma))$ gives

$$Q(\epsilon) = O\big(\log^2(8/(1-\gamma)) \max\{1, \log(1/\epsilon)\}\big).$$

For $\epsilon \le e^{-1}$, this reduces to the usual $\log^2(8/(1-\gamma))\log(1/\epsilon)$ scaling. □

D.2 Control proof

For Control, write $\gamma_h := h\gamma/(1+h)$, and define the action-wise defect-corrected reward at inner argument U by

$$\Delta^a(U) := R^a + \gamma(\mathcal{P}^a - \hat{\mathcal{P}}^a)U.$$

The action-parameterized shifted reward $\tilde{r}^a[V, U] \in \mathcal{B}(\mathcal{X})$ is

$$\tilde{r}^a[V, U] := \frac{1}{1+h}V + \frac{h}{1+h}\Delta^a(U).$$

For fixed V, let $\Psi_{h,V}(U)$ be the optimal value function of the surrogate MDP with transition operators $\hat{\mathcal{P}}^a$, rewards $\tilde{r}^a[V,U]$, and effective discount γ_h:

$$\Psi_{h,V}(U)(x) = \sup_{a\in\mathcal{A}} \left\{ \tilde{r}^a[V,U](x) + \gamma_h(\hat{\mathcal{P}}^a \Psi_{h,V}(U))(x) \right\}.$$

Starting from $U_0 = V$ and setting $U_{j+1} = \Psi_{h,V}(U_j)$ gives the action-parameterized shifted-reward realization of $\Phi_h^{\star,m}$ given by Equations (7) and (8).

IE-VI for Control replaces the linear generator L^π with the nonlinear operator $I - T^\star$. The mechanism differs from the linear case: in place of a literal operator identity, fidelity rests on a nonlinear fixed-point equivalence and the inner loop is bounded through a Lipschitz-to-target argument. The same contraction-modulus bound still appears.

Lemma 20 (Accretivity of $I - T^\star$)**.** *$T^\star$ is a γ-contraction on $(\mathcal{B}(\mathcal{X}), \|\cdot\|_\infty)$, so $I - T^\star$ is accretive: for every $U, W \in \mathcal{B}(\mathcal{X})$ and $\lambda > 0$,*

$$\|U - W + \lambda\big((I - T^\star)U - (I - T^\star)W\big)\|_\infty \geq \|U - W\|_\infty.$$

Proof. Set $D = U - W$ and $G = T^\star U - T^\star W$. The γ-contraction of $T^\star$ gives $\|G\|_\infty \leq \gamma\|D\|_\infty$; the triangle inequality then yields $\|(1+\lambda)D - \lambda G\|_\infty \geq (1+\lambda)\|D\|_\infty - \lambda\|G\|_\infty \geq (1 + \lambda(1-\gamma))\|D\|_\infty \geq \|D\|_\infty$. □

Accretivity is the nonlinear analogue of the linear stability bound $\|(I + \lambda L^\pi)^{-1}\|_\infty \leq 1$; together with the range condition (Lemma 22), it gives the Crandall–Liggett interpretation [Crandall and Liggett, 1971] of the exact implicit-Euler step as a term in an iterated-resolvent discretization of the Control Bellman residual flow. The exact implicit target $Z_h^\star(V_k)$ is the unique fixed point of $Z \mapsto (1+h)^{-1}V_k + h(1+h)^{-1}T^\star Z$, a strict γ_h-contraction (Lemma 22).

Remark 21 (γ-contraction of $T^\star$ under ϵ-suboptimal selectors). Fix arbitrary $U, W \in \mathcal{B}(\mathcal{X})$, $x \in \mathcal{X}$, and $\eta > 0$. Choose $a_U \in \mathcal{A}$ with $(T^\star U)(x) \leq R^{a_U}(x) + \gamma(\mathcal{P}^{a_U}U)(x) + \eta$; evaluating $T^\star W$ at the same action gives $(T^\star W)(x) \geq R^{a_U}(x) + \gamma(\mathcal{P}^{a_U}W)(x)$. Subtracting yields $(T^\star U)(x) - (T^\star W)(x) \leq \gamma\|U - W\|_\infty + \eta$; swapping U, W, sending $\eta \downarrow 0$, and taking the supremum over x gives $\|T^\star U - T^\star W\|_\infty \leq \gamma\|U - W\|_\infty$.

Lemma 22 (Exact nonlinear implicit-Euler contraction)**.** *For every $h > 0$ and every $V \in \mathcal{B}(\mathcal{X})$, the exact implicit-Euler optimality equation*

$$Z = \frac{1}{1+h}V + \frac{h}{1+h}T^\star Z$$

has a unique solution $Z_h^\star(V)$ and

$$\|Z_h^\star(V) - V^\star\|_\infty \leq \frac{1}{1 + h(1-\gamma)}\|V - V^\star\|_\infty.$$

Proof. Define $F_V(Z) := (1+h)^{-1}V + h(1+h)^{-1}T^\star Z$. Remark 21 gives that $T^\star$ is a γ-contraction on the complete metric space $(\mathcal{B}(\mathcal{X}), \|\cdot\|_\infty)$. Therefore, for every $Z, W \in \mathcal{B}(\mathcal{X})$,

$$\|F_V(Z) - F_V(W)\|_\infty \leq \gamma_h\|Z - W\|_\infty$$

and $\gamma_h < 1$. Thus F_V is a strict contraction, so $Z_h^\star(V)$ exists and is unique by the contraction mapping theorem. Subtracting the fixed-point identity $V^\star = (1+h)^{-1}V^\star + h(1+h)^{-1}T^\star V^\star$ from the defining equation for $Z_h^\star(V)$ yields

$$(1+h)(Z_h^\star(V) - V^\star) = (V - V^\star) + h(T^\star Z_h^\star(V) - T^\star V^\star).$$

Taking supremum norms and using the γ-contraction of $T^\star$,

$$(1+h)\|Z_h^\star(V) - V^\star\|_\infty \leq \|V - V^\star\|_\infty + h\gamma\|Z_h^\star(V) - V^\star\|_\infty.$$

Rearranging gives the stated bound. □

Lemma 23 (Fixed shifted-reward surrogate contraction)**.** *For fixed $h > 0$, $V \in \mathcal{B}(\mathcal{X})$, and $U \in \mathcal{B}(\mathcal{X})$, define*

$$(\hat{T}_{h,V,U}Z)(x) := \sup_{a\in\mathcal{A}} \left\{ \tilde{r}^a[V,U](x) + \gamma_h(\hat{\mathcal{P}}^a Z)(x) \right\}.$$

Then $\hat{T}_{h,V,U}$ is a strict supremum-norm contraction on $\mathcal{B}(\mathcal{X})$ with modulus γ_h, and it has a unique fixed point.

Proof. Fix $Z_1, Z_2 \in \mathcal{B}(\mathcal{X})$, $x \in \mathcal{X}$, and $\eta > 0$. Choose a_η so that

$$(\hat{T}_{h,V,U} Z_1)(x) \le \tilde{r}^{a_\eta}[V,U](x) + \gamma_h(\hat{\mathcal{P}}^{a_\eta} Z_1)(x) + \eta.$$

Evaluating $\hat{T}_{h,V,U} Z_2$ at the same action gives

$$(\hat{T}_{h,V,U} Z_2)(x) \ge \tilde{r}^{a_\eta}[V,U](x) + \gamma_h(\hat{\mathcal{P}}^{a_\eta} Z_2)(x).$$

The shifted reward cancels, so

$$(\hat{T}_{h,V,U} Z_1)(x) - (\hat{T}_{h,V,U} Z_2)(x) \le \gamma_h(\hat{\mathcal{P}}^{a_\eta}(Z_1 - Z_2))(x) + \eta \le \gamma_h \|Z_1 - Z_2\|_\infty + \eta.$$

Exchanging Z_1 and Z_2, sending $\eta \downarrow 0$, and taking the supremum over x gives

$$\|\hat{T}_{h,V,U} Z_1 - \hat{T}_{h,V,U} Z_2\|_\infty \le \gamma_h \|Z_1 - Z_2\|_\infty.$$

Since $\gamma_h < 1$, the Banach fixed-point theorem gives the unique fixed point. □

Lemma 24 (Equivalence to the action-wise shifted-reward surrogate MDP)**.** *The exact implicit value $Z_h^\star(V)$ is the unique fixed point of $U \mapsto \Psi_{h,V}(U)$.*

Proof. Let $Z = Z_h^\star(V)$. Pointwise, the exact implicit equation is

$$Z(x) = \frac{1}{1+h} V(x) + \frac{h}{1+h} \sup_{a \in \mathcal{A}} \{R^a(x) + \gamma(\mathcal{P}^a Z)(x)\}.$$

The term $V(x)/(1+h)$ is independent of a, so it may be moved inside the supremum. Substituting $\mathcal{P}^a Z = \hat{\mathcal{P}}^a Z + (\mathcal{P}^a - \hat{\mathcal{P}}^a) Z$ inside the action value gives

$$\begin{aligned} Z(x) &= \sup_{a \in \mathcal{A}} \left\{ \frac{1}{1+h} V(x) + \frac{h}{1+h} \big(R^a + \gamma(\mathcal{P}^a - \hat{\mathcal{P}}^a) Z\big)(x) + \gamma_h(\hat{\mathcal{P}}^a Z)(x) \right\} \\ &= \sup_{a \in \mathcal{A}} \left\{ \frac{1}{1+h} V(x) + \frac{h}{1+h} \Delta^a(Z)(x) + \gamma_h(\hat{\mathcal{P}}^a Z)(x) \right\} \\ &= \sup_{a \in \mathcal{A}} \left\{ \tilde{r}^a[V,Z](x) + \gamma_h(\hat{\mathcal{P}}^a Z)(x) \right\}. \end{aligned}$$

This is exactly the surrogate Bellman equation defining $\Psi_{h,V}(Z)$, so Z is a fixed point by the uniqueness in Lemma 23. To prove uniqueness for the outer fixed-point equation, let $W = \Psi_{h,V}(W)$. Then W obeys

$$W(x) = \sup_{a \in \mathcal{A}} \left\{ \tilde{r}^a[V,W](x) + \gamma_h(\hat{\mathcal{P}}^a W)(x) \right\}.$$

Substituting the definition of $\tilde{r}^a[V,W]$ cancels the surrogate-model terms inside each action value and yields

$$W(x) = \frac{1}{1+h} V(x) + \frac{h}{1+h} \sup_{a \in \mathcal{A}} \{R^a(x) + \gamma(\mathcal{P}^a W)(x)\}.$$

Equivalently, $W = (1+h)^{-1} V + h(1+h)^{-1} T^\star W$. By Lemma 22, this equation has the unique solution $Z_h^\star(V)$. Hence every fixed point of $U \mapsto \Psi_{h,V}(U)$ equals $Z_h^\star(V)$. □

Lemma 25 (Control inner-map Lipschitz bound)**.** *For fixed V, the map $U \mapsto \Psi_{h,V}(U)$ is $\rho(h)$-Lipschitz in supremum norm:*

$$\|\Psi_{h,V}(U) - \Psi_{h,V}(W)\|_\infty \le \rho(h) \|U - W\|_\infty.$$

Proof. Let $Z_U = \Psi_{h,V}(U)$ and $Z_W = \Psi_{h,V}(W)$. Fix $x \in \mathcal{X}$ and $\eta > 0$. By the definition of the supremum in the surrogate Bellman equation for $Z_U(x)$, choose $a_U \in \mathcal{A}$ such that

$$Z_U(x) \le \tilde{r}^{a_U}[V,U](x) + \gamma_h(\hat{\mathcal{P}}^{a_U} Z_U)(x) + \eta.$$

Evaluating the surrogate Bellman equation for $Z_W(x)$ at the same action and subtracting gives

$$\begin{aligned} Z_W(x) &\ge \tilde{r}^{a_U}[V,W](x) + \gamma_h(\hat{\mathcal{P}}^{a_U} Z_W)(x), \\ Z_U(x) - Z_W(x) &\le \tilde{r}^{a_U}[V,U](x) - \tilde{r}^{a_U}[V,W](x) + \gamma_h(\hat{\mathcal{P}}^{a_U}(Z_U - Z_W))(x) + \eta. \end{aligned}$$

Bounding the right-hand side yields

$$Z_U(x) - Z_W(x) \le \sup_{a\in\mathcal{A}} \|\tilde{r}^a[V,U] - \tilde{r}^a[V,W]\|_\infty + \gamma_h \|Z_U - Z_W\|_\infty + \eta.$$

Reversing the roles of U and W gives the same upper bound for $Z_W(x) - Z_U(x)$. Sending $\eta \downarrow 0$, taking the supremum over x, and rearranging gives

$$\|Z_U - Z_W\|_\infty \le \frac{1}{1-\gamma_h} \sup_{a\in\mathcal{A}} \|\tilde{r}^a[V,U] - \tilde{r}^a[V,W]\|_\infty.$$

The shifted-reward difference isolates the defect-corrected reward:

$$\begin{aligned}
\tilde{r}^a[V,U] - \tilde{r}^a[V,W] &= \frac{h}{1+h}\big(\Delta^a(U) - \Delta^a(W)\big) \\
&= \gamma_h(\mathcal{P}^a - \hat{\mathcal{P}}^a)(U - W), \\
&\Rightarrow \sup_{a\in\mathcal{A}} \|\tilde{r}^a[V,U] - \tilde{r}^a[V,W]\|_\infty \le \frac{h}{1+h} E_\infty \|U - W\|_\infty.
\end{aligned}$$

Since $(1-\gamma_h)^{-1} = (1+h)/(1+h(1-\gamma))$, the claim follows. □

Lemma 26 (Finite-inner-loop control error)**.** *Let $U_0 = V$ and $U_{j+1} = \Psi_{h,V}(U_j)$. Then*

$$\|U_m - V^\star\|_\infty \le \left(\frac{1}{1+h(1-\gamma)} + \rho(h)^m \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}\right) \|V - V^\star\|_\infty.$$

Proof. By Lemma 24, $Z_h^\star(V)$ is the unique fixed point of $\Psi_{h,V}$. Applying Lemma 25 for m inner rounds gives

$$\|U_m - Z_h^\star(V)\|_\infty \le \rho(h)^m \|V - Z_h^\star(V)\|_\infty.$$

To bound the displacement, subtract V from the exact implicit equation:

$$Z_h^\star(V) - V = \frac{h}{1+h}\big(T^\star Z_h^\star(V) - V\big) = \frac{h}{1+h}\big(T^\star Z_h^\star(V) - T^\star V^\star + V^\star - V\big),$$

where the second equality adds and subtracts $V^\star = T^\star V^\star$. Taking supremum norms and applying the triangle inequality, the γ-contraction of $T^\star$, and Lemma 22,

$$\begin{aligned}
\|Z_h^\star(V) - V\|_\infty &\le \frac{h}{1+h}\big(\|T^\star Z_h^\star(V) - T^\star V^\star\|_\infty + \|V^\star - V\|_\infty\big) \\
&\le \frac{h}{1+h}\big(\gamma\|Z_h^\star(V) - V^\star\|_\infty + \|V - V^\star\|_\infty\big) \\
&\le \frac{h}{1+h}\left(\frac{\gamma}{1+h(1-\gamma)} + 1\right)\|V - V^\star\|_\infty \\
&= \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}\|V - V^\star\|_\infty.
\end{aligned}$$

Combining this displacement bound with the triangle inequality gives

$$\begin{aligned}
\|U_m - V^\star\|_\infty &\le \|Z_h^\star(V) - V^\star\|_\infty + \rho(h)^m \|V - Z_h^\star(V)\|_\infty \\
&\le \left(\frac{1}{1+h(1-\gamma)} + \rho(h)^m \frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))}\right)\|V - V^\star\|_\infty.
\end{aligned}$$

□

Proof of Theorem 2. We split the proof by the model-error ratio.

Part 1: Low-Error Regime $(R_E < 1)$**.** Set $m = 1$ and take the limit $h \to \infty$. Solving the shifted-reward surrogate problem gives the OS-VI update $Z_{\mathrm{OS}}(U)$, the optimal value function of the surrogate MDP with transition operators $\hat{\mathcal{P}}^a$, rewards $\Delta^a(U)$, and discount γ. Indeed, $\tilde{r}^a[V,U] + \gamma_h \hat{\mathcal{P}}^a Z - (\Delta^a(U) + \gamma\hat{\mathcal{P}}^a Z) = (1+h)^{-1}(V - \Delta^a(U) - \gamma\hat{\mathcal{P}}^a Z)$ for every $Z \in \mathcal{B}(\mathcal{X})$; taking $Z = Z_{\mathrm{OS}}(U)$ and using the γ_h-contraction of $\hat{T}_{h,V,U}$ (Lemma 23) gives $\|\Psi_{h,V}(U) - Z_{\mathrm{OS}}(U)\|_\infty \le c/((1+h)(1-\gamma))$

with $c := \sup_{a\in\mathcal{A}}\|V - \Delta^a(U) - \gamma\hat{\mathcal{P}}^a Z_{\mathrm{OS}}(U)\|_\infty$ independent of h, so the one-round update $\Psi_{h,V}(V)$ converges to $Z_{\mathrm{OS}}(V)$. In this limit, the exact resolvent contraction term $(1+h(1-\gamma))^{-1}$ goes to zero, and $\rho(h) \to R_E$. The combined per-step contraction modulus is R_E. Since $m = 1$, each outer iteration uses one action-wise true-model query. The value-iterate complexity is

$$O\left(\frac{1}{1-R_E}\log\frac{1}{\epsilon}\right).$$

The extraction of an ϵ-optimal greedy policy tightens the value target by a factor of $(1-\gamma)$, giving the policy-extraction complexity

$$O\left(\frac{1}{1-R_E}\log\frac{1}{\epsilon(1-\gamma)}\right).$$

Part 2: High-Error Regime ($R_E \geq 1$)**.** Set $\Lambda = R_E$, $h = h_{R_E} = 1/((1-\gamma)(2R_E-1))$, and $m = \lceil \log_2(8/(1-\gamma))\rceil$. By Lemma 26 and Lemma 19, one control update contracts the value iterate by at most $1 - \frac{1}{4R_E}$. After $N = \lceil 4R_E \ln(1/\epsilon)\rceil$ outer control updates,

$$\|V_N - V^\star\|_\infty \leq \left(1 - \frac{1}{4R_E}\right)^N \|V_0 - V^\star\|_\infty \leq \epsilon\|V_0 - V^\star\|_\infty.$$

Each inner round requires one action-wise true-model application $V \mapsto \{\mathcal{P}^a V\}_{a\in\mathcal{A}}$, while solving the surrogate MDP is uncharged. Since $E_\infty \leq 2\gamma$, $R_E \geq 1$ forces $\gamma \geq 1/3$, so $\log_2(8/(1-\gamma)) \leq 7\log_2(1/(1-\gamma))$, and the query count $N \cdot m$ is

$$O\left(R_E \log\frac{1}{(1-\gamma)}\log\frac{1}{\epsilon}\right).$$

Policy extraction. Let π_k satisfy

$$T^{\pi_k} V_k \geq T^\star V_k - \xi_k \mathbf{1},$$

and write $\mathcal{P}^{\pi_k}$ for its Markov operator. Since $V^{\pi_k} = T^{\pi_k}V^{\pi_k}$ and $V^\star = T^\star V^\star$, the fixed-point expansion gives $V^{\pi_k} - V^\star = (I - \gamma\mathcal{P}^{\pi_k})^{-1}(T^{\pi_k}V^\star - T^\star V^\star)$, and the Neumann series gives $\|(I-\gamma\mathcal{P}^{\pi_k})^{-1}\|_\infty \leq 1/(1-\gamma)$. Since $T^{\pi_k} \leq T^\star$, $\|T^{\pi_k}V_k - T^\star V_k\|_\infty \leq \xi_k$, so the γ-contraction of T^{π_k} and of $T^\star$ gives $\|T^{\pi_k}V^\star - T^\star V^\star\|_\infty \leq 2\gamma\|V_k - V^\star\|_\infty + \xi_k$. Together,

$$\|V^{\pi_k} - V^\star\|_\infty \leq \frac{2\gamma}{(1-\gamma)}\|V_k - V^\star\|_\infty + \frac{\xi_k}{(1-\gamma)}.$$

If $V_0 = V^\star$, the initial policy is bounded by the ξ_0 term. Assume $V_0 \neq V^\star$. In the High-Error Regime, set

$$N_{\mathrm{pol}} = \lceil 4R_E \max\left\{0, \ln\left(\frac{4\gamma}{\epsilon(1-\gamma)}\right)\right\}\rceil.$$

The value-iterate contraction gives

$$\|V_{N_{\mathrm{pol}}} - V^\star\|_\infty \leq \min\left\{1, \frac{\epsilon(1-\gamma)}{4\gamma}\right\}\|V_0 - V^\star\|_\infty.$$

By Lemma 8,

$$\frac{1}{1+\gamma}\|\mathrm{BR}^\star(V_0)\|_\infty \leq \|V_0 - V^\star\|_\infty.$$

If the extracted policy satisfies

$$\xi_{N_{\mathrm{pol}}} \leq \frac{\epsilon(1-\gamma)}{2(1+\gamma)}\|\mathrm{BR}^\star(V_0)\|_\infty,$$

then

$$\xi_{N_{\mathrm{pol}}} \leq \frac{\epsilon(1-\gamma)}{2}\|V_0 - V^\star\|_\infty, \qquad \|V^{\pi_{N_{\mathrm{pol}}}} - V^\star\|_\infty \leq \epsilon\|V_0 - V^\star\|_\infty.$$

Multiplying N_{pol} by $m = \lceil \log_2(8/(1-\gamma))\rceil$ counts the inner-round true-model applications, and one final action-wise query extracts the terminal greedy policy:

$$Q_{\mathrm{pol}}(\epsilon) \leq N_{\mathrm{pol}}\lceil \log_2(8/(1-\gamma))\rceil + 1.$$

Sublinear model-error acceleration for Control. Assume $E_\infty \le C_0(1-\gamma)\log(1/(1-\gamma))$, and put

$$x := \log_2(8/(1-\gamma)), \qquad L_\epsilon := \max\{1, \ln(1/\epsilon)\}.$$

Use the uniform finite-step schedule, that is, Part 2 with R_E replaced by $\bar{R}_E := \max\{1, R_E\}$; this also covers $R_E < 1$. The assumption bounds $R_E \le C_0 x$, so $\bar{R}_E \le 1 + C_0 x$. By Part 2,

$$Q(\epsilon) \le \lceil 4\bar{R}_E \ln(1/\epsilon)\rceil \lceil \log_2(8/(1-\gamma))\rceil.$$

The first factor is at most

$$\lceil 4\bar{R}_E \ln(1/\epsilon)\rceil \le 4(1 + C_0 x)L_\epsilon + 1 \le \left(\frac{5}{3} + 4C_0\right) x L_\epsilon,$$

and the second is $\lceil x\rceil \le x + 1 \le 4x/3$. This proves the value-iterate rate.

For the policy-value statement, set

$$L_{\mathrm{pol}} := \max\left\{1, \ln\left(\frac{4\gamma}{\epsilon(1-\gamma)}\right)\right\}.$$

The same ceiling bounds give

$$N_{\mathrm{pol}} \le C_1 x L_{\mathrm{pol}}, \qquad C_1 := \frac{5}{3} + 4C_0.$$

Since $\lceil x\rceil \le 4x/3$, $x \ge 3$, and $L_{\mathrm{pol}} \ge 1$,

$$\begin{aligned} Q_{\mathrm{pol}}(\epsilon) &\le N_{\mathrm{pol}} \lceil x\rceil + 1 \\ &\le \left(\frac{4}{3}C_1 + \frac{1}{9}\right) x^2 L_{\mathrm{pol}}. \end{aligned}$$

Since $L_{\mathrm{pol}} = O\big(\max\{1, \log(1/(\epsilon(1-\gamma)))\}\big)$, the stated policy-value bound follows. □

E Parameter choices and optimality discussion

E.1 Known-parameter choices

In the High-Error Regime $R_E \ge 1$, the known-parameter algorithm uses $h = h_{R_E}$, where

$$h_\Lambda := [(1-\gamma)(2\Lambda - 1)]^{-1}.$$

Proposition 18 shows the more general fact needed below: every choice of $\Lambda \ge R_E \ge 1$ gives $\rho(h_\Lambda) \le 1/2$ and the same modulus calculation as Theorem 2, with R_E replaced by Λ.

E.2 Simultaneous tuning of the contraction-modulus bound

In the High-Error Regime, the deterministic bounds use the parameter choices $h = h_{R_E}$ and $m = \lceil \log_2(8/(1-\gamma))\rceil$. These constants make the proof uniform over surrogate accuracies. This subsection optimizes the contraction-modulus bound over h and m. The bound is

$$\mu(h, m) := \frac{1}{1 + h(1-\gamma)} + \rho(h)^m \frac{h(1 + \gamma + h(1-\gamma))}{(1+h)(1 + h(1-\gamma))}, \qquad \rho(h) = \frac{hE_\infty}{1 + h(1-\gamma)},$$

where the first term is the exact resolvent contraction and the second term is the defect-correction residual after m rounds. Increasing h improves the exact resolvent factor, but it also increases the defect-correction base $\rho(h)$. As $h \to \infty$, $\rho(h) \to E_\infty/(1-\gamma) = R_E$, so the weak-surrogate case $R_E \ge 1$ requires a finite-step tradeoff.

Theorem 27 (Simultaneous tuning of the contraction-modulus bound)**.** *Let $\gamma \in (0,1)$, $(1-\gamma) = 1-\gamma$, $E_\infty \ge 0$, and $R_E = E_\infty/(1-\gamma)$. Assume the High-Error Regime $R_E \ge 1$. Let $\Lambda \ge R_E \ge 1$, and define*

$$m^\star_{\mathrm{env}} := \min\left\{m \in \{1, 2, \ldots\} : \exists h > 0 \text{ such that } \mu(h, m) \le 1 - \frac{1}{4\Lambda}\right\}$$

(the set is nonempty by part 2, since $(4\Lambda/R_E)^m \ge 4^m$ eventually exceeds $8e(m+1)\min\{1/(1-\gamma), \Lambda\}$). The following statements hold.

1.

$$\left(\frac{4\Lambda}{R_E}\right)^{m^\star_{\mathrm{env}}} > (m^\star_{\mathrm{env}}+1)\min\{1/(1-\gamma), R_E\}.$$

2. If an integer $m \geq 1$ satisfies

$$\left(\frac{4\Lambda}{R_E}\right)^{m} \geq 8e(m+1)\min\{1/(1-\gamma), \Lambda\},$$

then there exists $h > 0$ such that $\mu(h, m) \leq 1 - 1/(4\Lambda)$. In particular, $m^\star_{\mathrm{env}} \leq m$.

3. If $\Lambda = R_E \geq 1$, $Y := \min\{1/(1-\gamma), R_E\}$, and $X := 8eY$, then

$$\frac{1}{2}\log_2 Y + \frac{1}{2} < m^\star_{\mathrm{env}} \leq \lceil \frac{1}{2}\log_2 X + \log_2(1 + \log_2 X)\rceil.$$

Proof. Set

$$u := \frac{h(1-\gamma)}{1+h(1-\gamma)} \in (0,1).$$

Then $1-u = (1+h(1-\gamma))^{-1}$, $\rho(h) = uR_E$, and

$$\frac{h(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))} = uK(u), \qquad K(u) := \frac{1+\gamma(1-u)}{(1-\gamma)+\gamma u}.$$

The target inequality $\mu(h,m) \leq 1 - 1/(4\Lambda)$ is equivalent to

$$u(uR_E)^m K(u) \leq u - \frac{1}{4\Lambda}.$$

Dividing by u^{m+1} gives

$$R_E^m K(u) \leq u^{-m} - \frac{1}{4\Lambda}u^{-m-1}. \tag{20}$$

The numerator of $K(u)$ is larger than 1, and the denominator is $(1-\gamma)+\gamma u = 1-\gamma(1-u) < 1$, so $K(u) > 1$.

Let m and u satisfy (20). The right side of (20) must be positive, so $u > 1/(4\Lambda)$. Since $K(u) > 1$,

$$R_E^m < u^{-m}\left(1 - \frac{1}{4\Lambda u}\right) < u^{-m},$$

which implies $u < 1/R_E$. The denominator of $K(u)$ obeys $(1-\gamma)+\gamma u < (1-\gamma)+u < (1-\gamma)+1/R_E$, which gives

$$K(u) > \frac{1}{(1-\gamma)+1/R_E}.$$

Writing $c := 1/(4\Lambda)$ and

$$f_m(u) := u^{-m} - cu^{-m-1},$$

(20) gives $R_E^m/((1-\gamma)+1/R_E) < f_m(u)$. The derivative

$$f_m'(u) = u^{-m-2}\big((m+1)c - mu\big)$$

shows that f_m has its maximum at $u = c(m+1)/m$, with

$$\max_{u>0} f_m(u) = c^{-m}\left(\frac{m}{m+1}\right)^m \frac{1}{m+1} \leq \frac{(4\Lambda)^m}{2(m+1)}.$$

The last inequality uses $(1+1/m)^m \geq 2$. Since

$$\frac{1}{(1-\gamma)+1/R_E} \geq \frac{1}{2}\min\{1/(1-\gamma), R_E\},$$

every feasible m satisfies

$$(m+1)\min\{1/(1-\gamma), R_E\} < \left(\frac{4\Lambda}{R_E}\right)^m.$$

Applying this to $m = m^\star_{\mathrm{env}}$ proves the lower bound.

For the sufficient condition, fix an integer $m \geq 1$ and choose

$$u := \frac{m+1}{4\Lambda m}.$$

Since $\Lambda \geq 1$, this choice lies in $(0,1)$. It corresponds to the step size $h = u/((1-\gamma)(1-u))$. For this u,

$$f_m(u) = (4\Lambda)^m \left(\frac{m}{m+1}\right)^m \frac{1}{m+1} \geq \frac{(4\Lambda)^m}{e(m+1)},$$

because $(1+1/m)^{-m} \geq e^{-1}$. The numerator of $K(u)$ is at most $1+\gamma < 2$, while its denominator is at least $\max\{(1-\gamma), 1/(4\Lambda)\}$, since $(1-\gamma)+\gamma u \geq (1-\gamma)$ and $(1-\gamma)+\gamma u \geq u \geq 1/(4\Lambda)$. Therefore

$$K(u) \leq \frac{2}{\max\{(1-\gamma), 1/(4\Lambda)\}} \leq 8\min\{1/(1-\gamma), \Lambda\}.$$

The target (20) is met whenever

$$R_E^m \, 8\min\{1/(1-\gamma), \Lambda\} \leq \frac{(4\Lambda)^m}{e(m+1)},$$

which is the stated sufficient condition.

It remains to prove the displayed finite bounds for $\Lambda = R_E \geq 1$. The lower bound reduces to

$$4^{m^\star_{\mathrm{env}}} > (m^\star_{\mathrm{env}} + 1)Y.$$

Taking base-two logarithms and using $\log_2(m^\star_{\mathrm{env}} + 1) \geq 1$ gives

$$2m^\star_{\mathrm{env}} > 1 + \log_2 Y.$$

For the upper bound, let $L := \log_2 X$ and set

$$m_0 := \lceil \frac{1}{2}L + \log_2(1+L) \rceil.$$

Since $X \geq 8e$, we have $L > 4$. Then

$$4^{m_0} \geq X(1+L)^2,$$
$$m_0 + 1 \leq \frac{1}{2}L + \log_2(1+L) + 2 \leq (1+L)^2,$$

where the last inequality follows from $L > 4$. This gives

$$4^{m_0} \geq X(m_0+1),$$

so the sufficient condition holds with $m = m_0$. □

For $R_E \leq 1$, the same construction can use the error upper bound 1. It then gives $m = 5$ and $h = 3/(7(1-\gamma))$, with outer contraction factor at most $3/4$, since the bound is monotone in the model error. This provides a constant-depth finite-step alternative to the infinite-step limit, including near $R_E = 1$. Query expressions for general $0 < \epsilon < 1$ use $\max\{1, \log(1/\epsilon)\}$. The companion lower bound concerns optimization of the stated contraction bound, not all planning algorithms.

For the uniform conservative schedule, comparing its outer factor with γ establishes only a per-outer-step comparison. A sufficient comparison per true-model application is

$$1 - \frac{1}{4\bar{R}_E} < \gamma^m,$$

because each IE-VI step costs m applications. Neither comparison includes surrogate computation.

The preceding theorem fixes the contraction target $1 - 1/(4\Lambda)$. The next statement gives the corresponding lower bound for the cost factor that multiplies $\ln(1/\epsilon)$ in the contraction proof.

Proposition 28 (Cost lower bound for simultaneous tuning)**.** *Assume the High-Error Regime $R_E \geq 1$, set $Y := \min\{1/(1-\gamma), R_E\}$, and suppose $Y > 2$. For every $h > 0$ and integer $m \geq 1$ with $\mu(h,m) < 1$,*

$$\frac{m}{1-\mu(h,m)} \geq eR_E \ln(Y/2).$$

There are also choices of h and m, with $\Lambda = R_E$, such that

$$\frac{m}{1-\mu(h,m)} \leq 4R_E\lceil \frac{1}{2}\log_2(8eY) + \log_2(1+\log_2(8eY))\rceil.$$

Proof. Use the same fractional step $u = h(1-\gamma)/(1+h(1-\gamma))$ and write $s := uR_E$. If $\mu(h,m) < 1$, then

$$s^m K(u) < 1.$$

Since $K(u) > 1$, we have $s < 1$. Also

$$1-\mu(h,m) = u(1-s^m K(u)) \leq u = \frac{s}{R_E}.$$

The same lower bound on $K(u)$ used above gives

$$K(u) > \frac{1}{(1-\gamma)+1/R_E} \geq \frac{Y}{2}.$$

Therefore $s^m < 2/Y$. Since $Y > 2$,

$$m\ln(1/s) > \ln(Y/2).$$

Combining these inequalities yields

$$\frac{m}{1-\mu(h,m)} \geq R_E\frac{m}{s} > R_E\frac{\ln(Y/2)}{s\ln(1/s)} \geq eR_E\ln(Y/2),$$

because $\sup_{0<s<1} s\ln(1/s) = 1/e$.

For the upper bound, take the constructive choice from Theorem 27 with $\Lambda = R_E$ and

$$m = \lceil \frac{1}{2}\log_2(8eY) + \log_2(1+\log_2(8eY))\rceil.$$

Then $\mu(h,m) \leq 1 - 1/(4R_E)$, which gives

$$\frac{m}{1-\mu(h,m)} \leq 4R_E m.$$

□

When R_E scales as $1/(1-\gamma)$, Proposition 28 gives a lower bound of order $R_E\log(1/(1-\gamma))$ for the cost factor $m/(1-\mu(h,m))$ under simultaneous tuning of h and m. The parameter choice in the main theorem has the same order, since it gives $1-\mu(h_{R_E},m) \geq 1/(4R_E)$ and $m = \lceil\log_2(8/(1-\gamma))\rceil$. For smaller R_E, the result identifies the part of the parameter choice that is conservative: simultaneous tuning depends on $\log\min\{1/(1-\gamma), R_E\}$ rather than only on $\log(1/(1-\gamma))$. In particular, if $E_\infty \leq C(1-\gamma)\log H$ with $H = 1/(1-\gamma)$, then $\bar{R}_E = O(\log H)$ gives depth $m = O(1+\log\log H)$, and $\lceil 4\bar{R}_E\ln(1/\epsilon)\rceil$ outer steps give the bound $O(\log H(1+\log\log H)\log(1/\epsilon))$ quoted after Theorem 2.

F Adaptive policy evaluation

The parameter schedule requires knowledge of R_E, which is rarely available in advance. Adaptive IE-VI replaces this with a doubling search over hypotheses $\Lambda_i = 2^i$, each epoch tested against the residual certificate

$$\frac{1}{1+\gamma}\|\mathrm{BR}(V)\|_\infty \leq \|V - V^\pi\|_\infty \leq \frac{1}{(1-\gamma)}\|\mathrm{BR}(V)\|_\infty$$

(Lemma 3). The search starts at $\Lambda_0 = 1$ and finds the first epoch with $\Lambda_i \geq 1$ and $\Lambda_i \geq R_E$. If $R_E < 1$, the initial hypothesis $\Lambda_0 = 1$ is already a valid upper bound for the schedule.

Theorem 29 (Adaptive IE-VI complexity)**.** *With* $K_\epsilon := \max\{1, \ln(1/\epsilon) + 2\ln((1+\gamma)/(1-\gamma))\}$, Adaptive IE-VI *for Policy Evaluation reaches relative error* ϵ *using at most*

$$Q(\epsilon) \le \begin{cases} C\ln(8/(1-\gamma))(K_\epsilon + 1), & \text{if } R_E < 1, \\ C\ln(8/(1-\gamma))\big(R_E K_\epsilon + \log_2(2R_E)\big), & \text{if } R_E \ge 1 \end{cases}$$

true-model queries, for an explicit universal constant $C \le 32$.

Compared with the uniform finite-step schedule in Theorem 2, the adaptive bound retains the linear dependence on $\bar{R}_E$, replaces $\log(1/\epsilon)$ by K_ϵ up to additive constants, and adds the displayed search overhead.

Algorithm 2 Adaptive IE-VI for policy evaluation

Require: Initial value function V_0, target relative precision $\epsilon \in (0,1)$
  Compute $B_0 \leftarrow \|\mathrm{BR}(V_0)\|_\infty$
  if $B_0 = 0$ **then**
    return V_0
  end if
  Set $T_\epsilon \leftarrow \epsilon(1-\gamma)B_0/(1+\gamma)$
  Set $K_\epsilon \leftarrow \max\{1, \ln(1/\epsilon) + 2\ln((1+\gamma)/(1-\gamma))\}$
  Initialize $\Lambda_0 \leftarrow 1$
  for $i = 0, 1, 2, \ldots$ **do**
    Set $h_i \leftarrow \lfloor (1-\gamma)(2\Lambda_i - 1)\rfloor^{-1}$, $m_i \leftarrow \lceil \log_2(8/(1-\gamma))\rceil$, and $N_i \leftarrow \lceil 4\Lambda_i K_\epsilon\rceil$
    Restart $W \leftarrow V_0$
    for $k = 1, \ldots, N_i$ **do**
      Set $W \leftarrow \Phi_{h_i}^{m_i}(W)$
    end for
    if $\|\mathrm{BR}(W)\|_\infty \le T_\epsilon$ **then**
      return W
    end if
    Set $\Lambda_{i+1} \leftarrow 2\Lambda_i$
  end for

The restart in Algorithm 2 is part of the proof contract: the first successful epoch is analyzed from the common initial error $\|V_0 - V^\pi\|_\infty$, so failed earlier hypotheses do not enter that epoch's error term. The theorem here analyzes the restarted algorithm; Corollary 34 gives a version that retains the best-residual epoch endpoint.

Lemma 30 (Adaptive residual threshold)**.** *Let*

$$K_\epsilon := \max\left\{1, \ln(1/\epsilon) + 2\ln\left(\frac{1+\gamma}{(1-\gamma)}\right)\right\}.$$

Then

$$e^{-K_\epsilon} \le \epsilon\left(\frac{(1-\gamma)}{1+\gamma}\right)^2.$$

Proof. Let $X := \ln(1/\epsilon) + 2\ln((1+\gamma)/(1-\gamma))$. Since $K_\epsilon = \max\{1, X\} \ge X$ and $z \mapsto e^{-z}$ is decreasing,

$$e^{-K_\epsilon} \le e^{-X} = \epsilon\left(\frac{(1-\gamma)}{1+\gamma}\right)^2.$$

□

Lemma 31 (Successful-epoch contraction for Adaptive IE-VI)**.** *Let* $i^\star$ *be the first epoch such that* $\Lambda_{i^\star} \ge 1$ *and* $\Lambda_{i^\star} \ge R_E$. *During epoch* $i^\star$,

$$\|\hat{M}_{h_{i^\star}}^{m_{i^\star}}\|_\infty \le 1 - \frac{1}{4\Lambda_{i^\star}}.$$

Proof. Because the search gives $\Lambda_{i^\star} \geq 1$ and $\Lambda_{i^\star} \geq R_E$, Lemma 16, applied in induced norm to the decomposition $\hat{M}_h^m = K(h)^{-1} + h(Y^\pi)^m K(h)^{-1} L^\pi$ from the proof of Theorem 1, and Lemma 19 apply with $\Lambda = \Lambda_{i^\star}$, $h = h_{i^\star}$, and $m = m_{i^\star}$, giving

$$\|\hat{M}_{h_{i^\star}}^{m_{i^\star}}\|_\infty \leq 1 - \frac{1}{4\Lambda_{i^\star}}.$$

□

Lemma 32 (Restarted doubling query count)**.** *Let $K \geq 1$, $m = \lceil \log_2(8/(1-\gamma)) \rceil$, and $\Lambda_i = 2^i$. Let $i^\star$ be the first index with $\Lambda_{i^\star} \geq 1$ and $\Lambda_{i^\star} \geq R_E$. Suppose epochs $0, \ldots, i^\star$ use $N_i = \lceil 4\Lambda_i K \rceil$ outer updates, each outer update costs m charged correction rounds, each epoch ends with one residual check, and there is one initial residual evaluation. Then the total charged query count satisfies*

$$Q \leq \begin{cases} C \ln(8/(1-\gamma))(K+1), & \text{if } R_E < 1, \\ C \ln(8/(1-\gamma))\big(R_E K + \log_2(2R_E)\big), & \text{if } R_E \geq 1 \end{cases}$$

for a universal constant $C \leq 32$.

Proof. The accounting assumption gives

$$Q \leq 1 + \sum_{i=0}^{i^\star} \big(m\lceil 4\Lambda_i K \rceil + 1\big).$$

Using $\lceil x \rceil \leq x + 1$ and $\sum_{i=0}^{i^\star} \Lambda_i = 2\Lambda_{i^\star} - 1$,

$$Q \leq 1 + 4mK(2\Lambda_{i^\star} - 1) + (m+1)(i^\star + 1).$$

If $R_E < 1$, then $\Lambda_0 = 1$ is accepted, so $i^\star = 0$. Therefore

$$Q \leq 1 + 4mK + (m+1) \leq 4mK + m + 2.$$

Set $x := \log_2(8/(1-\gamma))$. Since $x \geq 3$ and $m \leq 4x/3$, this is bounded by $C \ln(8/(1-\gamma))(K+1)$ for a universal $C \leq 32$.

If $R_E = 1$, then $i^\star = 0$, $\Lambda_{i^\star} = 1 < 2R_E$, and $i^\star + 1 = 1 < 2 + \log_2 R_E$. If $R_E > 1$, then $i^\star \geq 1$ and the previous epoch satisfies $\Lambda_{i^\star - 1} < R_E$, so $\Lambda_{i^\star} < 2R_E$ and $i^\star + 1 < 2 + \log_2 R_E$. In both cases,

$$\Lambda_{i^\star} \leq 2R_E, \qquad i^\star + 1 \leq 2 + \log_2 R_E.$$

Hence

$$Q \leq 16mKR_E + (m+1)(2 + \log_2 R_E) + 1.$$

Set $x := \log_2(8/(1-\gamma))$ and $L := \log_2(2R_E)$. Since $x \geq 3$ and $L \geq 1$,

$$2 + \log_2 R_E \leq 2L, \qquad m \leq \frac{4}{3}x,$$
$$m + 1 \leq \frac{5}{3}x, \qquad 1 \leq \frac{1}{3}xL.$$

Substitution gives

$$\begin{aligned} Q &\leq \frac{64}{3}xR_EK + \frac{11}{3}xL \\ &\leq 22x(R_EK + L). \end{aligned}$$

Converting the outer logarithm from base 2 to the natural base gives the displayed bound with $C \leq 22/\ln 2 < 32$. □

Proof of Theorem 29. If $B_0 = 0$, then $\mathrm{BR}(V_0) = 0$, so $V_0 = V^\pi$ and the claim is immediate. Assume $B_0 > 0$.

Let $i^\star$ be the first epoch satisfying $\Lambda_{i^\star} \geq 1$ and $\Lambda_{i^\star} \geq R_E$. By Theorem 1 and Lemma 19 with $\Lambda = \Lambda_{i^\star}$, $h = h_{i^\star}$, and $m = m_{i^\star}$, after $N_{i^\star}$ restarted outer updates in that epoch,

$$\|W - V^\pi\|_\infty \leq \left(1 - \frac{1}{4\Lambda_{i^\star}}\right)^{N_{i^\star}} \|V_0 - V^\pi\|_\infty \leq e^{-K_\epsilon} \|V_0 - V^\pi\|_\infty.$$

By Lemma 30,

$$e^{-K_\epsilon} \le \epsilon \left(\frac{(1-\gamma)}{1+\gamma}\right)^2.$$

$$\|\mathrm{BR}(W)\|_\infty \le (1+\gamma)\|W - V^\pi\|_\infty \le \epsilon \frac{(1-\gamma)^2}{1+\gamma}\|V_0 - V^\pi\|_\infty.$$

By Lemma 3, $B_0 \ge (1-\gamma)\|V_0 - V^\pi\|_\infty$. Hence the last display is at most

$$\epsilon \frac{(1-\gamma)B_0}{1+\gamma} = T_\epsilon.$$

Algorithm 2 terminates no later than the first successful epoch.

Let W_{out} denote the value returned by the algorithm. Its stopping test and Lemma 3 give

$$\|W_{\mathrm{out}} - V^\pi\|_\infty \le \frac{1}{(1-\gamma)}\|\mathrm{BR}(W_{\mathrm{out}})\|_\infty \le \frac{T_\epsilon}{(1-\gamma)} = \frac{\epsilon B_0}{1+\gamma} \le \epsilon\|V_0 - V^\pi\|_\infty,$$

where the last inequality uses the lower bound on the value error at V_0 in Lemma 3.

Lemma 32 applies with $K = K_\epsilon$, where the charged queries are policy-evaluation true-model applications. This gives the stated dual-regime bounds. □

G Adaptive control variant

The adaptive Control analogue uses the same restarted doubling search, but replaces the policy-evaluation residual by the optimality residual

$$\mathrm{BR}^\star(V) = T^\star V - V.$$

Given $B_0^\star := \|\mathrm{BR}^\star(V_0)\|_\infty$, it sets

$$T_\epsilon^\star := \frac{\epsilon(1-\gamma)B_0^\star}{1+\gamma}, \qquad K_\epsilon := \max\{1, \ln(1/\epsilon) + 2\ln((1+\gamma)/(1-\gamma))\},$$

and in epoch i applies $N_i = \lceil 4\Lambda_i K_\epsilon \rceil$ updates of the control map $\Phi_{h_i}^{\star,m_i}$, with $h_i = [(1-\gamma)(2\Lambda_i - 1)]^{-1}$ and $m_i = \lceil \log_2(8/(1-\gamma)) \rceil$, restarted from V_0. The algorithm returns once $\|\mathrm{BR}^\star(W)\|_\infty \le T_\epsilon^\star$.

Algorithm 3 Adaptive IE-VI for control

Require: Initial value function V_0, target relative precision $\epsilon \in (0,1)$
  Compute $B_0^\star \leftarrow \|\mathrm{BR}^\star(V_0)\|_\infty$
  if $B_0^\star = 0$ **then**
    return V_0
  end if
  Set $T_\epsilon^\star \leftarrow \epsilon(1-\gamma)B_0^\star/(1+\gamma)$
  Set $K_\epsilon \leftarrow \max\{1, \ln(1/\epsilon) + 2\ln((1+\gamma)/(1-\gamma))\}$
  Initialize $\Lambda_0 \leftarrow 1$
  for $i = 0, 1, 2, \ldots$ **do**
    Set $h_i \leftarrow [(1-\gamma)(2\Lambda_i - 1)]^{-1}$, $m_i \leftarrow \lceil \log_2(8/(1-\gamma)) \rceil$, and $N_i \leftarrow \lceil 4\Lambda_i K_\epsilon \rceil$
    Restart $W \leftarrow V_0$
    for $k = 1, \ldots, N_i$ **do**
      Set $W \leftarrow \Phi_{h_i}^{\star,m_i}(W)$
    end for
    if $\|\mathrm{BR}^\star(W)\|_\infty \le T_\epsilon^\star$ **then**
      return W
    end if
    Set $\Lambda_{i+1} \leftarrow 2\Lambda_i$
  end for

Theorem 33 (Adaptive Control complexity)**.** *The restarted adaptive Control variant returns W satisfying*

$$\|W - V^\star\|_\infty \le \epsilon \|V_0 - V^\star\|_\infty$$

within the same dual-regime iteration-complexity limits as Theorem 29:

$$Q(\epsilon) \le \begin{cases} C \ln(8/(1-\gamma))(K_\epsilon + 1), & \text{if } R_E < 1, \\ C \ln(8/(1-\gamma))\big(R_E K_\epsilon + \log_2(2R_E)\big), & \text{if } R_E \ge 1, \end{cases}$$

with policy-evaluation true-model queries replaced by action-wise true-model applications $V \mapsto \{\mathcal{P}^a V\}_{a\in\mathcal{A}}$.

Proof. If $B_0^\star = 0$, then $\mathrm{BR}^\star(V_0) = 0$, so $V_0 = V^\star$. Otherwise, let $i^\star$ be the first epoch with $\Lambda_{i^\star} \ge 1$ and $\Lambda_{i^\star} \ge R_E$. By Lemma 26 and Lemma 19 applied with $\Lambda = \Lambda_{i^\star}$, $h = h_{i^\star}$, and $m = m_{i^\star}$, one control update in epoch $i^\star$ satisfies

$$\|\Phi_{h_{i^\star}}^{\star, m_{i^\star}}(V) - V^\star\|_\infty \le \left(1 - \frac{1}{4\Lambda_{i^\star}}\right) \|V - V^\star\|_\infty.$$

After $N_{i^\star}$ restarted updates,

$$\|W - V^\star\|_\infty \le e^{-K_\epsilon} \|V_0 - V^\star\|_\infty \le \epsilon \left(\frac{(1-\gamma)}{1+\gamma}\right)^2 \|V_0 - V^\star\|_\infty.$$

Here the second inequality uses Lemma 30. Using Lemma 8,

$$\|\mathrm{BR}^\star(W)\|_\infty \le (1+\gamma)\|W - V^\star\|_\infty \le \epsilon \frac{(1-\gamma)^2}{1+\gamma} \|V_0 - V^\star\|_\infty \le \frac{\epsilon(1-\gamma)B_0^\star}{1+\gamma} = T_\epsilon^\star.$$

The algorithm terminates no later than epoch $i^\star$. Let W be the returned value function. The upper bound on the value error in Lemma 8 gives

$$\|W - V^\star\|_\infty \le \frac{1}{(1-\gamma)} \|\mathrm{BR}^\star(W)\|_\infty.$$

The stopping test gives $\|\mathrm{BR}^\star(W)\|_\infty \le T_\epsilon^\star$, hence

$$\|W - V^\star\|_\infty \le \frac{T_\epsilon^\star}{(1-\gamma)} = \frac{\epsilon B_0^\star}{1+\gamma}.$$

Finally, the lower bound on the value error applied at V_0 gives

$$B_0^\star = \|\mathrm{BR}^\star(V_0)\|_\infty \le (1+\gamma)\|V_0 - V^\star\|_\infty,$$
$$\|W - V^\star\|_\infty \le \epsilon \|V_0 - V^\star\|_\infty.$$

Lemma 32 applies with $K = K_\epsilon$, where the charged queries are action-wise true-model applications $V \mapsto \{\mathcal{P}^a V\}_{a\in\mathcal{A}}$. This gives the stated dual-regime query bound. □

H Adaptive epochs initialized from a residual incumbent

The restarted analysis protects each successful epoch from errors accumulated under earlier, unsuccessful parameter choices. We can preserve this protection while keeping useful progress: retain the epoch endpoint with the smallest true Bellman residual and start the next epoch from it.

Corollary 34 (Epochs initialized from a residual incumbent)**.** *In the exact-operator setting of Theorems 29 and 33, let $0 < \epsilon < 1$ and $B(V) := \|TV - V\|_\infty$, using $T = T^\pi$ for Policy Evaluation and $T = T^\star$ for Control. Assume $B_0 := B(V_0) > 0$, and set*

$$K := \max\left\{1, \ln(1/\epsilon) + 2\ln\left(\frac{1+\gamma}{(1-\gamma)}\right)\right\}.$$

The epoch parameters are

$$\Lambda_i := 2^i, \qquad h_i := [(1-\gamma)(2\Lambda_i - 1)]^{-1},$$
$$m := \lceil \log_2(8/(1-\gamma)) \rceil, \qquad N_i := \lceil 4\Lambda_i K \rceil.$$

Initialize $Z = V_0$. At epoch i, apply N_i exact m-round updates with step size h_i to Z to obtain W, and retain whichever of Z and W has smaller residual, keeping Z on ties. Stop when the retained residual is at most $\epsilon(1-\gamma)B_0/(1+\gamma)$. With the incumbent residual cached, this procedure has the same true-model query bound and relative value-error guarantee as the restarted procedure, for both Policy Evaluation and Control.

Proof. The retained iterate always satisfies $B(Z) \le B_0$, hence $\|Z - V^\dagger\|_\infty \le B_0/(1-\gamma)$. At the first epoch with $\Lambda_i \ge \max\{1, R_E\}$, the contraction bound of Theorem 1 (Lemma 26 for Control) with Lemma 19 gives

$$\|W - V^\dagger\|_\infty \le \left(1 - \frac{1}{4\Lambda_i}\right)^{N_i} \frac{B_0}{(1-\gamma)} \le e^{-K} \frac{B_0}{(1-\gamma)}.$$

Consequently, by Lemma 30,

$$B(W) \le (1+\gamma)e^{-K} \frac{B_0}{(1-\gamma)} \le \frac{\epsilon(1-\gamma)B_0}{1+\gamma}.$$

The retained iterate meets the same threshold and therefore the relative-error target, by Lemmas 3 and 8. Comparing cached residuals adds no true-model query: the cost through epoch i remains at most

$$1 + \sum_{j=0}^{i} (mN_j + 1).$$

The original query-count argument in Lemma 32 applies. □

If $B_0 = 0$, return V_0 immediately. This corollary concerns prescribed epochs in which the inner planning problems are solved exactly; it does not establish a bound for the practical one-round stall schedule.

I Numerical one-round updates

The main analysis assumes that each surrogate planning problem is solved exactly. For the one-round update, numerical error enters as an additive perturbation, which allows us to quantify the accuracy needed from the inner solver.

Let $F_h(V)$ be the exact one-round update with $h > 0$, and write $E = E_\infty$, with the action-wise error used for Control. Suppose the numerical output satisfies

$$\|\widetilde{V}_{k+1} - F_{h_k}(\widetilde{V}_k)\|_\infty \le \delta_k.$$

The error $e_k := \|\widetilde{V}_k - V^\dagger\|_\infty$ then obeys

$$e_{k+1} \le \mu_1(h_k) e_k + \delta_k,$$

where substitution of $m = 1$ into Theorem 1 gives

$$\mu_1(h) = \frac{1}{1+h(1-\gamma)} + \frac{h^2 E(1+\gamma+h(1-\gamma))}{(1+h)(1+h(1-\gamma))^2}. \tag{21}$$

If $\mu_1(h_k) \le q < 1$ and $\delta_k \le \bar{\delta}$ uniformly, induction yields

$$e_k \le q^k e_0 + \bar{\delta} \frac{1-q^k}{1-q}.$$

Thus a fixed accuracy for the surrogate value computation gives a bound on the error floor. A shifted surrogate Bellman residual at most τ_k provides the value-error bound $\delta_k \le \tau_k/(1-\gamma_{h_k})$.

The residual monitored by the adaptive procedure admits a direct bound as well. With $B(V) := \|TV - V\|_\infty$,

$$B(F_h(V)) \le \frac{1+hE}{1+h(1-\gamma)} B(V). \tag{22}$$

For Control, define

$$S_V(W)(x) := \sup_{a\in\mathcal{A}} \left\{ R^a(x) + \gamma(\mathcal{P}^a V)(x) + \gamma\big(\hat{\mathcal{P}}^a(W-V)\big)(x) \right\};$$

for Policy Evaluation, use the fixed-policy counterpart. Then S_V is a γ-contraction, $S_V(V) = TV$, and

$$\|TW - S_V(W)\|_\infty \le E\|W - V\|_\infty.$$

The last inequality follows from the action-wise bound and the inequality for differences of suprema. The exact one-round equation $F = V/(1+h) + hS_V(F)/(1+h)$ gives

$$\|F - V\|_\infty \le \frac{h}{1+h(1-\gamma)} B(V),$$
$$S_V(F) - F = \frac{F-V}{h}.$$

Combining these relations proves Equation (22). For a numerical output $\widetilde{F}$ within δ of F, the $(1+\gamma)$-Lipschitz property of B gives

$$B(\widetilde{F}) \le \frac{1+hE}{1+h(1-\gamma)} B(V) + (1+\gamma)\delta.$$

When $E < (1-\gamma)$ and $B(V) > 0$, exact updates strictly decrease the residual. Numerical updates do so if

$$(1+\gamma)\delta < \left(1 - \frac{1+hE}{1+h(1-\gamma)}\right) B(V).$$

At $E = (1-\gamma)$ the exact bound gives only nonincrease; above it the bound does not certify decrease. These sufficient conditions do not give a termination theorem for the practical stall schedule. The true-residual stopping certificate still applies whenever its target is met and the true residual is evaluated exactly.

J Practical residual-driven adaptive schedule

The practical Control studies use a one-round Adaptive IE-VI schedule that reduces h when the Bellman residual stalls, carrying the current iterate forward instead of restarting from V_0 at a doubled error hypothesis. Algorithm 4 describes this schedule. Its reference parameters are $h_{\text{init}} = 10^6$, $h_{\min} = 1$, $\beta = 0.1$, and $S = 3$, with one defect-correction round per outer update. Figures N.6 and N.7 examine threshold and schedule sensitivity for this practical variant. The separate common-solver study specifies its numerical solver and stopping rule in Section N.13.1. Figure 1 (a) uses this practical schedule with the numerical implementation described in Section N.1.

The endpoint choices have direct numerical interpretations. The floor $h_{\min} = 1$ sets the lower end of the schedule at the unit Bellman-residual-flow time scale; under explicit Euler, this unit step is ordinary VI (Section 2). The initial value $h_{\text{init}} = 10^6$ represents the opposite end of the same scale. With $m = 1$, IE-VI tends to OS-VI as $h \to \infty$ (Proposition 13); for $\gamma = 0.99$, $h_{\text{init}}(1-\gamma) = 10^4$, so the first attempted step lies in this large-h regime while using ordinary floating-point arithmetic. The remaining parameters β and S are schedule choices; Figure N.7 reports their sensitivity at $q_{\text{SH}}(10^{-3})$.

Algorithm 4 shares the residual-feedback principle of Adaptive IE-VI: progress is judged against the best-so-far Bellman residual, and the step size h is contracted when the residual stalls. Where Algorithm 3 maintains an explicit hypothesis Λ_i for $\bar{R}_E$ and tests it through the residual certificate of Lemma 8, Algorithm 4 replaces the explicit hypothesis with the empirical signal of stalled residuals and contracts h by a fixed factor β along the trajectory rather than restarting at a doubled hypothesis. No analogue of Theorem 33's complexity bound is proved for this practical schedule. The planning panel in Figure 1 (a), the Control sensitivity studies, and the common-solver study use this schedule.

The counter in Algorithm 4 uses the common-solver study's convention of actual true-model applications, including initialization; a rejected surrogate candidate does not increase it. The floor guard returns the current value estimate when the rejection update leaves the step size at its minimum. Neither a budget exit nor a guard-failure exit certifies convergence. The original practical plots

Algorithm 4 Practical one-round residual-driven adaptive schedule for control

Require: Initial value V_0, initial step and floor $h_{\text{init}} \geq h_{\min} > 0$, decay factor $\beta \in (0,1)$, integer stall patience $S \geq 1$, integer query budget $Q \geq 1$
Compute $U^a \leftarrow \mathcal{P}^a V_0$ for all $a \in \mathcal{A}$; set $B^\star \leftarrow \|\mathrm{BR}^\star(V_0)\|_\infty$
Initialize $V \leftarrow V_0$, $h \leftarrow h_{\text{init}}$, $c \leftarrow 0$, $q \leftarrow 1$
while $q < Q$ **do**
  Form $\tilde{r}^a[V,V] \leftarrow \frac{1}{1+h} V + \frac{h}{1+h}\big(R^a + \gamma(U^a - \hat{\mathcal{P}}^a V)\big)$ for each $a \in \mathcal{A}$
  $U_{\text{new}} \leftarrow$ numerical approximation to the optimal value of the surrogate MDP with rewards $\tilde{r}^a[V,V]$, transitions $\hat{\mathcal{P}}^a$, and discount γ_h, using the solver and stopping rule specified for the study
  ▷ surrogate-only computation, uncharged
  if U_{new} is nonfinite or $\|U_{\text{new}}\|_\infty$ exceeds the divergence envelope **then**
    $h \leftarrow \max(h_{\min}, \beta h)$, $c \leftarrow 0$ ▷ reject; no true-model query
    if $h \leq h_{\min}$ **then**
      return V with guard-failure status
    end if
    continue ▷ V unchanged
  end if
  Compute $U^a_{\text{new}} \leftarrow \mathcal{P}^a U_{\text{new}}$ for all $a \in \mathcal{A}$; set $\eta \leftarrow \|T^\star U_{\text{new}} - U_{\text{new}}\|_\infty$ ▷ one charged query
  $V \leftarrow U_{\text{new}}$, $U^a \leftarrow U^a_{\text{new}}$, $q \leftarrow q+1$
  if $\eta < B^\star$ **then**
    $B^\star \leftarrow \eta$, $c \leftarrow 0$
  else
    $c \leftarrow c+1$
    if $c \geq S$ **then**
      $h \leftarrow \max(h_{\min}, \beta h)$, $c \leftarrow 0$
    end if
  end if
end while
return V

instead retain their outer-attempt indices, as detailed in Section N.1. Numerical iteration caps and additional stopping tests are specified by each study.

A variant that restores the smallest-residual estimate at step-size reductions is specified and evaluated in Section N.13.5.

K IE-DYNA sampling model and algorithm

IE-DYNA is the RL analogue of the Control construction in Equations (7) and (8). The deterministic inner loop evaluates a full action-wise defect term before each surrogate-planning round. IE-DYNA keeps the same surrogate-planning structure, but replaces each full defect sweep by coordinate-wise sampled updates.

K.1 Sampling model and surrogate update

We assume a finite discounted MDP and a generative sampling mechanism. When queried at (x,a), the environment returns a random reward $\mathsf{R}(x,a) \sim \mathcal{R}(\cdot \mid x,a)$ and successor $\mathsf{X}^+(x,a)$, with

$$\mathbb{E}\left[\mathsf{R}(x,a)\right] = R(x,a), \qquad \mathbb{P}\left\{\mathsf{X}^+(x,a) = y\right\} = \mathcal{P}(y \mid x,a).$$

The algorithm samples state-action coordinates uniformly,

$$(X_t, A_t) \sim \mathrm{Unif}(\mathcal{X} \times \mathcal{A}),$$

so each inner stochastic update uses one physical transition sample.

The local visit count indexes the stochastic-approximation learning rates. After t samples, define

$$N_t(x,a,y) := \sum_{\ell=0}^{t-1} \mathbf{1}\{(X_\ell, A_\ell, X_\ell^+) = (x,a,y)\}, \qquad N_t(x,a) := \sum_{y \in \mathcal{X}} N_t(x,a,y).$$

The IE-DYNA update does not prescribe how the surrogate is estimated. It takes as input any update rule that maps the sample history to a row-stochastic kernel $\hat{\mathcal{P}}_t$. Maximum-likelihood tabular rows, parametric model updates, and experiment-specific perturbations all fit this requirement. At an unvisited pair, $N_t(x,a) = 0$ and every probability row maximizes the likelihood; the implementation keeps the row of the initial surrogate $\hat{\mathcal{P}}_0$ there. The model-error parameter λ used in the plots belongs to the surrogate families in Appendix N; it is not an input of the IE-DYNA algorithm.

K.2 Stochastic defect tracker

Fix an outer step k, step size h_k, and inner sample budget J. During outer step k, V_k remains fixed. The algorithm initializes

$$U_{k,0} := V_k, \qquad D_{k,0} := D_k,$$

where $D_{k,0} \in \mathbb{R}^{|\mathcal{X}|\times|\mathcal{A}|}$ is a stochastic defect tracker. Define the effective discount

$$\gamma_{h_k} := \frac{h_k\gamma}{1+h_k}.$$

The limiting convention $h_k = \infty$ is interpreted as $(1+h_k)^{-1} = 0$, $h_k/(1+h_k) = 1$, and $\gamma_{h_k} = \gamma$.

At inner stage $j \in \{1,\dots,J\}$, let $t = kJ + j - 1$. The algorithm draws a sample, updates the transition counts to N_{t+1}, and applies the surrogate update rule to form $\bar{\mathcal{P}}_{k,j} := \hat{\mathcal{P}}_{t+1}$. The defect-corrected reward that deterministic IE-VI would evaluate at the current inner value and this surrogate is

$$\Delta_{k,j}(x,a) := R(x,a) + \gamma \sum_{y\in\mathcal{X}} \big(\mathcal{P}(y \mid x,a) - \bar{\mathcal{P}}_{k,j}(y \mid x,a)\big) U_{k,j-1}(y).$$

After drawing (X_t, A_t), let R_t and X_t^+ denote the observed reward and successor. IE-DYNA forms the single-sample defect-corrected reward target

$$Y_{k,j} := R_t + \gamma U_{k,j-1}(X_t^+) - \gamma \sum_{y\in\mathcal{X}} \bar{\mathcal{P}}_{k,j}(y \mid X_t, A_t) U_{k,j-1}(y). \tag{23}$$

Let $\{\alpha_n\}_{n\ge1} \subset (0,1]$ be a learning-rate sequence indexed by local state-action visits, and set

$$n_t := N_{t+1}(X_t, A_t).$$

The tracker is updated only at the sampled coordinate:

$$D_{k,j}(x,a) := \begin{cases} (1-\alpha_{n_t}) D_{k,j-1}(x,a) + \alpha_{n_t} Y_{k,j}, & (x,a) = (X_t, A_t), \\ D_{k,j-1}(x,a), & (x,a) \neq (X_t, A_t). \end{cases}$$

From the updated tracker, IE-DYNA constructs the shifted reward

$$\tilde{r}_{k,j}(x,a) := \frac{1}{1+h_k} V_k(x) + \frac{h_k}{1+h_k} D_{k,j}(x,a).$$

The next inner value $U_{k,j}$ is the optimal value function of the surrogate MDP with transition $\bar{\mathcal{P}}_{k,j}$, reward $\tilde{r}_{k,j}$, and effective discount γ_{h_k}:

$$U_{k,j}(x) = \max_{a\in\mathcal{A}} \left\{ \tilde{r}_{k,j}(x,a) + \gamma_{h_k} \sum_{y\in\mathcal{X}} \bar{\mathcal{P}}_{k,j}(y \mid x,a) U_{k,j}(y) \right\}.$$

The updated surrogate is retained as $\hat{\mathcal{P}}_{t+1}$ for the next stochastic correction.

K.3 IE-DYNA algorithm

The algorithm is zero-initialized in the defect tracker: $D_0(x,a) = 0$ for all (x,a). Thus unsampled state-action pairs receive no artificial reward value. Reward information enters through realized samples and then propagates through surrogate planning.

Algorithm 5 IE-DYNA

Require: Initial value V_0, step sizes $\{h_k\}$, inner sample budget J, local-count learning rates $\{\alpha_n\}_{n\geq 1}$, initial row-stochastic surrogate $\hat{\mathcal{P}}_0$, and a surrogate update rule
Initialize transition counts and tracker to $N_0 \leftarrow 0$ and $D_0 \leftarrow 0$
for $k = 0, 1, 2, \ldots$ **do**
  Set $U_{k,0} \leftarrow V_k$, $D_{k,0} \leftarrow D_k$, and effective discount γ_{h_k} as above
  for $j = 1, \ldots, J$ **do**
    Set $t \leftarrow kJ + j - 1$
    Sample $(X_t, A_t) \sim \mathrm{Unif}(\mathcal{X} \times \mathcal{A})$ and query the environment for R_t, X_t^+
    Update counts, set $n_t \leftarrow N_{t+1}(X_t, A_t)$, and fit $\bar{\mathcal{P}}_{k,j} \leftarrow \hat{\mathcal{P}}_{t+1}$
    Form the stochastic defect-corrected reward target $Y_{k,j}$ as in Equation (23)
    Update $D_{k,j}(X_t, A_t)$ toward $Y_{k,j}$ using step size α_{n_t}
    Set $\tilde{r}_{k,j}(x, a) \leftarrow \frac{1}{1+h_k} V_k(x) + \frac{h_k}{1+h_k} D_{k,j}(x, a)$
    Set $U_{k,j}$ to the optimal value of the surrogate MDP $(\tilde{r}_{k,j}, \bar{\mathcal{P}}_{k,j}, \gamma_{h_k})$
  end for
  Set $V_{k+1} \leftarrow U_{k,J}$ and $D_{k+1} \leftarrow D_{k,J}$
end for

K.4 Run variant used in the Cliffwalk RL panel

The Cliffwalk RL panel in Figure 1 (b) uses the coordinate update and carried defect tracker in Algorithm 5, with a fixed step size $h = 0.03$, one inner planning round ($J = 1$), and a batch of $B = 10$ sampled coordinate updates before that round. Its learning rate is $\alpha_n = 3/(3 + n)$, where n is the updated state-action pair's visit count. OS-Dyna uses $B = 30$ and constant learning rate 0.01; Q-Learning uses $\alpha_n = n^{-0.8}$.

These settings were selected on ten tuning seeds by normalized policy-value L_1 error integrated through $40{,}000$ nominal samples. Surrogate planning uses five policy-iteration sweeps per round.

L Limitations and compute details

Scope. The exact-operator IE-VI guarantees are stated for bounded value functions on measurable state and action spaces under the regularity assumptions in Section 2 and Appendix A. The policy-evaluation analysis is linear under a fixed policy. The Control analysis uses the Bellman optimality operator on $\mathcal{B}(\mathcal{X})$, near-optimal action selectors with vanishing slack when suprema are not attained, and the action-wise model-error scalar in Proposition 7. The one-round perturbation analysis in Appendix I covers bounded numerical errors in surrogate planning under a uniform contraction margin and gives a sufficient condition for residual decrease. The numerical surrogate planning computations in Figure 1 (a) are documented separately in Section N.1; the exact-operator adaptive complexity theorem is not a numerical-solver error guarantee. A termination bound for the practical stall schedule, function approximation, partial observability, stochastic IE-DYNA convergence, and wall-clock runtime guarantees remain outside the analysis. Weighted-L^p approximate-DP bounds use concentrability coefficients [Munos, 2007, Munos and Szepesvári, 2008, Scherrer et al., 2015]. OS-VI also provides a weighted-norm analysis using a χ^2-type model discrepancy as a sufficient construction [Rakhsha et al., 2022]. A weighted-norm theorem for the IE-VI resolvent–defect decomposition is left to follow-up work.

Higher-order implicit schemes. The paper does not rule out higher-order, multistep, or Runge–Kutta discretizations for Bellman residual flows. The results here cover the backward-Euler construction because the proof uses two facts preserved by its resolvent: unconditional contraction in $\|\cdot\|_\infty$ and positivity after the true kernel is replaced by $\hat{\mathcal{P}}$. Related contractivity results for numerical initial-value methods suggest care when combining higher order with maximum-norm nonexpansiveness [Spijker, 1983, Kraaijevanger, 1991]. Schemes with signed stages, history terms, or intermediate equations that do not correspond to MDPs require a proof that the induced Bellman update still contracts in the norm used here and that its surrogate subproblems remain valid planning problems.

Cost model. The iteration-complexity bounds count access to the true model. For Policy Evaluation, one query is an application $V \mapsto \mathcal{P}^\pi V$. For Control, one query is an action-wise application $V \mapsto \{\mathcal{P}^a V\}_{a\in\mathcal{A}}$. Surrogate-only planning, including solving the discounted surrogate MDPs inside IE-VI, is not charged. This convention isolates scarce true-model interaction; it is not a wall-clock runtime bound. The weighted-cost and timing experiments separately account for surrogate computation.

IE-DYNA boundary. Appendix K specifies IE-DYNA as a tabular RL algorithm, including the sampling model, learned surrogate, defect tracker, shifted reward, and pseudocode. The manuscript does not include a proof that its iterates fix the true value function, nor does it state unbiasedness or convergence theorems for this algorithm.

Polylogarithmic-horizon regime. The polylogarithmic-in-$1/(1-\gamma)$ true-model query complexity claim of Theorem 2 is established under the surrogate-accuracy condition $E_\infty \le C(1-\gamma)\log(1/(1-\gamma))$ for a constant $C \ge 0$ independent of γ. For arbitrary Markov surrogates, the uniform finite-step schedule stays within a $\log(1/(1-\gamma))$ factor of VI's standard true-model query bound in the worst case. This comparison is between true-model query bounds. It does not bound every practical run's query count or running time, and the finite-range horizon experiment does not establish an empirical polylogarithmic law.

Reproducibility scope. Appendix N specifies the finite-MDP environments, perturbation families, metrics, and target-construction conventions used to generate Figure 1 and the additional appendix panels. Reproduction materials and a figure-provenance register are archived with this revision at the restricted-access location in Section N.13.4. The register distinguishes the accepted fresh executions underlying the displayed results from retained historical outputs and records the extent of execution verification. Public artifact access remains pending.

Related work. See Section 6 for the operator-splitting, VI-acceleration, and model-based-RL comparisons.

M Extended related work

The appendix gives the full related-work comparison behind Section 6: operator-splitting and model-correction planners, model-based RL with surrogate bias, value-iteration acceleration, optimization-as-ODE methods, and the numerical-analysis sources used in the exact-operator proofs. Section M.1 compares contraction bounds, asymptotic rates, and query complexity for representative methods. Sections M.2 to M.5 extend each comparison with the technical background and the precise difference from IE-VI. Section M.6 states the resolvent, defect-correction, and contraction ingredients used in the body construction and lists the scope boundaries.

M.1 Convergence-rate comparison

Table M.1 compares the rate statements for the methods discussed in Section 6. A supremum-norm contraction bound and an asymptotic spectral factor are different guarantees; the table identifies which is available. Query counts omit the common $\log(1/\epsilon)$ factor except in the adaptive row. The IE-VI entries count all true-model queries in the inner correction rounds. The OS-VI row is the exact Policy-Evaluation specialization of Rakhsha et al. [2022, Thm. 1]. The MoCoVI row uses the exact-query Policy-Evaluation bound in Rakhsha et al. [2024, Prop. 1 and § 4], retaining its KL-based error measure rather than replacing it by R_E.

For MoCoVI, let

$$\eta_{\mathrm{KL}} := \frac{\gamma\sqrt{2}}{(1-\gamma)} \sup_{x,a} \sqrt{D_{\mathrm{KL}}\Big(\mathcal{P}(\cdot \mid x,a) \,\|\, \hat{\mathcal{P}}(\cdot \mid x,a)\Big)}, \qquad q_{\mathrm{MC}} := \eta_{\mathrm{KL}}\alpha_d.$$

Here $\alpha_d \le 1$ uniformly bounds the ratio of the best supremum-norm approximation error of V^π in the span of the current d measurement functions to $\|V^\pi - V_{k-1}\|_\infty$, before convergence. The ratio is at most one because the measurement set includes V_{k-1}. The bound assumes exact queries and a well-defined finite-KL correction. Pinsker's inequality gives $R_E \le \eta_{\mathrm{KL}}$; it does not give the reverse

Table M.1: Rate bounds and asymptotic factors for representative methods. Query counts omit a common $\log(1/\epsilon)$ factor except for Adaptive IE-VI, whose K_ϵ is defined in Theorem 29. Spectral factors describe asymptotic decay, not uniform one-step supremum-norm contraction or a finite-time query bound. The last column identifies possible model-substitution bias, under the conditions stated. Costs of constructing a deflation basis and MoCoVI's initial measurement queries are not included in the iteration comparison.

Method	Rate factor	Query count / rate scaling	Conditions	Biased target
VI	γ	$O(1/(1-\gamma))$	exact model	no
VI in a fixed surrogate	γ	converges to surrogate value	fixed Markov surrogate	possible
OS-VI (PE)	R_E	$O(1/(1-R_E))$	$R_E < 1$	no
MoCoVI (exact PE)	q_{MC}	$O(1/(1-q_{\mathrm{MC}}))$	$q_{\mathrm{MC}} < 1$; see text	no
DDVI (PE)	$\gamma\lvert\lambda_{s+1}\rvert$	asymptotic spectral factor	exact rank-s deflation	no
Anderson VI	no general bound	empirical acceleration	mixing coefficients	no
PID-VI	no general bound	special-case spectral results	stable gains	no
A-VC / M-VC	$1-q$; $\frac{1-q}{1+q}$	$1-\Theta(\sqrt{(1-\gamma)})$ asymptotic factors	reversible PE; prescribed gains	no
IE-VI at $m=1$, $h\to\infty$	R_E	$O(1/(1-R_E))$	$R_E < 1$	no
IE-VI unified	$1-1/(4\bar{R}_E)$	$O((1+R_E)\log(1/(1-\gamma)))$	every Markov surrogate	no
IE-VI polylog regime	$1-1/(4\bar{R}_E)$	$O(\log^2(1/(1-\gamma)))$	$E_\infty \le C(1-\gamma)\log(1/(1-\gamma))$	no
Adaptive IE-VI	$1-1/(4\Lambda_{i^\star})$	$O((\bar{R}_E K_\epsilon + \log(2\bar{R}_E))\log(8/(1-\gamma)))$	every Markov surrogate; $\bar{R}_E$ unknown	no

upper bound needed to substitute R_E into MoCoVI's guarantee. There is no universal constant-factor equivalence between these error measures.

Rakhsha et al. [2024, Thm. 2] additionally analyze noisy queries, obtaining the factor $3\eta_{\mathrm{KL}}\alpha_d^{\mathrm{bd}}$ and an additive query-error term. Here α_d^{bd} uses approximation coefficients bounded by their prescribed $w_{\max}$, rather than the unrestricted span above, and the target is V^π for Policy Evaluation or $V^\star$ for Control. Their Control result bounds the loss of the returned policy, rather than the Policy-Evaluation value error, and its displayed denominator requires $3\eta_{\mathrm{KL}} < 1$. These distinctions prevent treating that theorem as the same R_E-based contraction statement for both tasks.

The DDVI factor uses eigenvalues ordered by magnitude and the unit relaxation setting [Lee et al., 2025, Theorem 3.1 and Corollary 3.2]. For A-VC / M-VC, $q := \sqrt{(1-\gamma)/(1+\gamma)}$; the factors are the asymptotic rates for irreducible reversible Policy Evaluation with the prescribed parameters [Goyal and Grand-Clément, 2021, Theorems 5.1 and 5.5]. They do not supply a uniform one-step maximum-norm bound. Anderson VI and general PID-VI have no such general bound in the cited papers; PID-VI does include a spectral acceleration result for its PD specialization on reversible chains. The detailed assumptions and results are discussed below. The IE-VI rows follow Proposition 13 and Theorem 2; the Adaptive IE-VI row follows Theorem 29.

M.2 Operator splitting and model correction

OS-VI [Rakhsha et al., 2022] is the closest related method to IE-VI. Its Policy-Evaluation update solves $\hat{L}^\pi U = r^\pi + \gamma(\mathcal{P}^\pi - \hat{\mathcal{P}}^\pi)V_k$, giving $\|V_{k+1} - V^\pi\|_\infty \le R_E\|V_k - V^\pi\|_\infty$. Proposition 13 identifies this update as one IE-VI correction round in the limit $h \to \infty$. OS-VI solves a surrogate MDP with a corrected reward at each outer iteration; it is not a single explicit Bellman sweep.

MoCoVI [Rakhsha et al., 2024] instead corrects the transition model by minimizing KL divergence from $\hat{\mathcal{P}}$ subject to matching true-model expectations of measurement functions, then solves the corrected-model planning problem. The measurement functions are updated from past value iterates. Their span is used in the error analysis, not to restrict the representation of the computed value function. The exact-query Policy-Evaluation rate depends on q_{MC}, while the noisy-query and Control guarantees have the distinctions described above.

When exact moment correction is well defined and the measurement set contains the target value, that value satisfies the corrected Bellman equation. This fixed-point property does not imply convergence from arbitrary initial iterates or a finite-KL guarantee for every Markov surrogate: a surrogate that assigns zero probability to possible true transitions can have infinite KL error, and its support can prevent exact moment matching. Conversely, failure of the rate condition does not mean MoCoVI diverges. Each corrected model is still a Markov model with the original bounded rewards, so its planned values remain bounded; the original paper explicitly distinguishes this property from OS-VI's possible divergence. IE-VI retains the surrogate transition kernel, corrects rewards, and chooses finite h and depth m to guarantee contraction for every Markov surrogate under Theorem 2. OS-VI and MoCoVI also have RL counterparts, OS-Dyna and MoCoDyna; the corresponding IE-VI construction is IE-DYNA (Appendix K).

The matrix-splitting view of VI acceleration goes back further. Bertsekas [1995] derives an extrapolation $x := F(x + \tilde{\gamma}d) = F(x) + \tilde{\gamma}Qd$ along a fixed direction d for the linear iteration $x := F(x) = h + Qx$, with $\tilde{\gamma}$ chosen by residual minimization. When d is an eigenvector for a dominant simple eigenvalue, the remaining spectrum determines the asymptotic rate through the second-largest eigenvalue modulus. That work also gives a multidimensional invariant-subspace extension. DDVI [Lee et al., 2025] develops rank-s deflation for Bellman iteration, including the inverse factor $(I - \alpha\gamma E_s)^{-1}$ associated with its deflation matrix E_s. These methods act on the exact transition operator's spectral structure; IE-VI instead uses a surrogate operator to approximate an implicit step.

M.3 Model-based RL with surrogate bias

Planning with a fixed surrogate can introduce value bias, but this is only the planning component of the full Dyna architecture, which also updates values from real experience and learns its model. MBPO [Janner et al., 2019] analyzes discrepancies between true and model-based policy returns. Its k-branched construction follows a data-collecting policy π_D under true dynamics before branching, then a new policy π under learned dynamics for k steps; branch times have weights proportional to γ^t. The quantity $\eta^{\mathrm{branch}}[\pi]$ below is the corresponding branched return, not the return of a stationary policy run solely in the learned model.

Theorem 35 (Janner et al. 2019, Thm. 4.2, shared-reward specialization)**.** *Let $0 < \gamma < 1$, use a common initial distribution and reward function with $|r| \le r_{\max}$, and let $d_t^{\pi_D}$ be the true state distribution at time t under π_D. A sufficient specialization of the original error assumptions is*

$$\sup_t \mathbb{E}_{x\sim d_t^{\pi_D}}\left[\sup_a D_{\mathrm{TV}}\Big(\mathcal{P}(\cdot \mid x,a), \hat{\mathcal{P}}(\cdot \mid x,a)\Big)\right] \le \epsilon_m, \qquad \sup_x D_{\mathrm{TV}}(\pi_D(\cdot \mid x), \pi(\cdot \mid x)) \le \epsilon_\pi.$$

The supremum over actions makes this model-error assumption sufficient for the source's expected-error condition. Then

$$\eta[\pi] \ge \eta^{\mathrm{branch}}[\pi] - 2r_{\max}\Big[\frac{\gamma^{k+1}\epsilon_\pi}{(1-\gamma)^2} + \frac{\gamma^k + 2}{(1-\gamma)}\epsilon_\pi + \frac{k}{(1-\gamma)}(\epsilon_m + 2\epsilon_\pi)\Big].$$

Despite its decreasing and increasing terms, the penalty in this bound is minimized at $k = 0$, as the authors explain immediately after their Theorem 4.2. Their Theorem 4.3 motivates useful positive rollout lengths by instead controlling model error under the updated policy's state distribution. MBPO therefore controls model-induced return discrepancy through limited model rollouts; it does not establish the fixed point of a single frozen Bellman iteration. IE-VI's exact-operator analysis addresses a different quantity: contraction toward the true Bellman solution, with $\hat{\mathcal{P}}$ used inside the implicit-step solver. The return-gap bound and the per-step modulus $1 - 1/(4\bar{R}_E)$ measure different quantities. Appendix K adapts the separation between the target equation and surrogate planning to RL updates, without asserting a stochastic convergence theorem.

M.4 Value-iteration acceleration without a surrogate

Other methods accelerate VI without using an approximate model. Anderson VI [Geist and Scherrer, 2018] computes a coefficient vector $\alpha^{k+1} \in \mathbb{R}^{m+1}$ by minimizing $\|\sum_i \alpha_i (T v_{k-m+i} - v_{k-m+i})\|$ (typically in ℓ_2; the paper allows other norms) under $\sum \alpha_i = 1$ and forms $v_{k+1} = \sum_i \alpha_i^{k+1} T v_{k-m+i}$. This mixes the Bellman images of the current estimate and m preceding estimates. For affine Policy Evaluation it is also one Bellman step after affine extrapolation; that equivalence does not generally hold for the nonlinear Control operator.

DDVI [Lee et al., 2025] constructs a rank-s deflation matrix from eigenvectors or Schur vectors of $\mathcal{P}^\pi$ and uses a matrix-splitting update. With exact deflation and relaxation parameter $\alpha = 1$, its Policy-Evaluation error is $O((\gamma|\lambda_{s+1}| + \zeta)^k)$ for every $\zeta > 0$, with constants depending on the instance and ζ [Lee et al., 2025, Corollary 3.2]; eigenvalues are ordered by decreasing modulus. This is the source's special $\tilde{O}$ convention, not a one-step supremum-norm contraction. The rank-1 Control extension has the analogous asymptotic factor $\gamma|\lambda_2|$, with the same arbitrarily small slack, when the optimal policy is unique; here λ_2 belongs to $\mathcal{P}^{\pi^\star}$ [Lee et al., 2025, Thm. 4.1].

First-order accelerated VI [Goyal and Grand-Clément, 2021] uses $v - T(v)$ in Nesterov and Polyak momentum updates. For exact-model Policy Evaluation with an irreducible reversible Markov chain and the prescribed gains, A-VC and M-VC have asymptotic spectral factors $1-q$ and $(1-q)/(1+q)$, respectively, where $q = \sqrt{(1-\gamma)/(1+\gamma)}$ [Goyal and Grand-Clément, 2021, Theorems 5.1 and 5.5]. These factors have inverse-square-root horizon scaling, but the stated asymptotic bounds allow arbitrarily small slack in the exponential base and instance-dependent transients. The S-AVI variant falls back to a VI step when an accelerated candidate fails its residual-decay check [Goyal and Grand-Clément, 2021, Algorithm 1 and Theorem 7.1]. Its guarantee is $O((\gamma')^k)$ for a chosen $\gamma' \in [\gamma, 1)$, retaining the VI rate when $\gamma' = \gamma$.

PID-VI [Farahmand and Ghavamzadeh, 2021] casts the VI iterate sequence as a dynamical system and adds proportional and integral feedback on the Bellman residual together with a derivative term on the iterate sequence: the PD update is $V_{k+1} = T^\pi V_k + \kappa_d (V_k - V_{k-1})$ [Farahmand and Ghavamzadeh, 2021, Eq. 10], the PI update introduces an integral state $z_{k+1} = \beta z_k + \alpha \,\mathrm{BR}(V_k)$ and feeds back $K_I[\beta z_k + \alpha \,\mathrm{BR}(V_k)]$, and the PID update combines the two. The PD update preserves T^π's fixed point directly [Farahmand and Ghavamzadeh, 2021, § 3]; for PI and PID the augmented (z, V)-dynamics has $(0, V^\pi)$ as its fixed point, so the value component converges to V^π when the augmented dynamics is stable [Farahmand and Ghavamzadeh, 2021, § 3]. The paper gives a closed-form rate for the PD variant on reversible Markov chains [Farahmand and Ghavamzadeh, 2021, Proposition 6, Appendix E], and empirical results with tuned gains on more general instances. PID feedback differs from extrapolation, spectral deflation, and Nesterov-style first-order acceleration; the PD update has the form of Polyak momentum on the value-function trajectory, an analogy that Goyal and Grand-Clément [2021] use to motivate Momentum Value Iteration and which Farahmand and Ghavamzadeh [2021, § 7] note coincides with the PD update.

Lee and Ryu [2023] introduce Anchored VI (Anc-VI), $V_{k+1} = \beta_{k+1} V_0 + (1 - \beta_{k+1}) T V_k$ with $\beta_j = 1/\sum_{i=0}^{j} \gamma^{-2i}$ [Lee and Ryu, 2023, § 2], which proves an $O(1/k)$ rate on the Bellman residual $\|TV_k - V_k\|_\infty$ for both the consistency operator T^π and the optimality operator $T^\star$ [Lee and Ryu, 2023, Theorems 1 and 2]; the rate does not degrade as $\gamma \to 1$. For $\gamma < 1$ bounded, the corresponding distance-to-solution rate is $O(\gamma^k)$ with a constant strictly worse than plain VI [Lee and Ryu, 2023, § 2.2]. The approximate-evaluation variant Apx-Anc-VI replaces TV_k by $TV_k + \epsilon_k$ and bounds the Bellman residual by the noise-free $O(1/k)$ term plus a $\max_i \|\epsilon_i\|_\infty/(1-\gamma)$ piece [Lee and Ryu, 2023, Theorem 6]; this additive-error model also accommodates repeated use of an approximate Bellman operator by taking $\epsilon_i = (\hat{T} - T)V_i$. The theorem bounds the effect of those errors but does not supply a surrogate-specific bias correction.

These methods modify Bellman iteration through extrapolation, deflation, Nesterov or Polyak momentum [Goyal and Grand-Clément, 2021], PID feedback [Farahmand and Ghavamzadeh, 2021], or Halpern anchoring [Lee and Ryu, 2023], without requiring an approximate transition model. Goyal and Grand-Clément [2021] give a lower bound for the class $v_0 = 0$, $v_{k+1} \in \mathrm{span}\{v_0, \ldots, v_k, Tv_0, \ldots, Tv_k\}$: an n-state instance has $\|v_k - v^\star\|_\infty \geq \gamma^k/(1+\gamma)$ for $1 \leq k < n$ [Goyal and Grand-Clément, 2021, Theorem 6.3]. The hard instance has one action, so the bound also applies to general Policy Evaluation within this class; the reversible-chain acceleration results rely on additional structure. IE-VI uses surrogate solves and additional model information

that need not satisfy the span condition. Its acceleration guarantees therefore do not contradict this bound. Combining these approaches with IE-VI, for example Anderson on the outer loop, is a deferred extension.

M.5 Optimization as ODE discretization

The variational view of Wibisono et al. [2016], which generalizes the ODE for Nesterov's accelerated gradient method of Su et al. [2016], treats accelerated optimization methods as discretizations of the Euler–Lagrange equation of a Bregman Lagrangian functional on a convex set in $\mathbb{R}^d$: a continuous-time curve X_t whose Euler–Lagrange equation has matching convergence rate $O(e^{-\beta_t})$ under an ideal-scaling condition. Their introduction states that *naive Euler discretization* of this curve does not retain the continuous-time rate and that the auxiliary-sequence Nesterov technique is one replacement. IE-VI uses the corresponding construction for a different ODE: the Bellman residual flow $\dot{V} = \mathrm{BR}(V) = T^\pi V - V$ on $\mathcal{B}(\mathcal{X})$, whose explicit-Euler discretization at step $h = 1$ is VI. The Bellman residual flow has an analogous stability issue: the two-state example in Section B.2 has stiffness ratio $(1+\gamma)/(1-\gamma)$; IE-VI adopts *implicit* Euler instead of auxiliary-sequence Nesterov, with iterative defect correction approximating the implicit step through tractable planning in $\hat{\mathcal{P}}$. Wibisono et al. [2016] study convex $f : \mathbb{R}^d \to \mathbb{R}$ with a single fixed point at the optimum; IE-VI studies the linear (or accretive nonlinear) Bellman generator on a Banach space of value functions. The integrator-design comparison transfers; the flow and its associated discrete updates do not.

M.6 Numerical-analysis grounding

The implicit-Euler step $(I + hL^\pi)^{-1}$ used by IE-VI is the resolvent of the Bellman generator, and the iterative defect correction that approximates it through successive surrogate planning problems is the numerical-analysis technique formalized by Stetter [1978]. Stetter's *defect correction principle* starts from an operator equation $Fx = y$ and an approximate inverse $\tilde{G}$ of F; the iteration $x_{i+1} = x_i - \tilde{G}Fx_i + \tilde{G}y$ (Stetter's version A) converges to the true solution $x^\star$ when $I - \tilde{G}F$ is a contraction. IE-VI's inner loop is the Stetter-A iteration with $F = K(h) = I + hL^\pi$, $y = V_k + hr^\pi$ (the implicit-Euler right-hand side at outer iterate V_k), and $\tilde{G} = \hat{K}(h)^{-1}$; the contraction operator is $Y^\pi = -h\hat{K}(h)^{-1}D^\pi$ with $D^\pi = L^\pi - \hat{L}^\pi = \gamma(\hat{\mathcal{P}}^\pi - \mathcal{P}^\pi)$ (Lemma 16). The fixed-point identity in Proposition 15 is the Stetter-A fixed-point statement. The Theorem 1 resolvent–defect decomposition is the truncation analysis of the Stetter-A iteration after m inner rounds.

The implicit-step contraction in supremum norm for every $h > 0$ follows from the Markov resolvent bound in Proposition 6. It is consistent with backward Euler's A-stability, but scalar A-stability and a positive-real-part spectrum alone do not establish contraction in this particular norm. The representation $K(h)^{-1} = h^{-1}\int_0^\infty e^{-t/h}e^{-tL^\pi}\,dt$ is the Laplace-transform formula for the resolvent; $e^{-tL^\pi} = \lim_{n\to\infty}(I + (t/n)L^\pi)^{-n}$ is the corresponding linear exponential formula. Fixed-h resolvent iteration for $L^\pi V = r^\pi$ has the algebraic form of the proximal-point method of Rockafellar [1976]. His convergence theory assumes maximal monotonicity in a Hilbert space, whereas the present guarantees follow from accretivity and resolvent contraction in the supremum-norm Banach space. For the nonlinear Control generator $I - T^\star$, Lemma 20 supplies the relevant accretivity argument; the semigroup reference is Crandall and Liggett [1971]. Appendices A and D give the precise lemmas, and Appendix L discusses the higher-order scope boundary.

M.7 Deferred extensions and scope boundaries

The exact-operator proofs in this paper specialize to bounded value functions on a measurable state space under supremum norm. Three extensions are deferred to follow-up work, with the closest references already in Section 6 and Appendix L.

Weighted L^p extension. Weighted-L^p error analyses in approximate dynamic programming use concentrability coefficients, as developed by Munos [2007] and Munos and Szepesvári [2008] and extended by Scherrer et al. [2015]. These results motivate a weighted-norm extension of the IE-VI resolvent–defect analysis, which is left to follow-up work.

Higher-order discretizations. The body construction uses backward Euler because the proof needs maximum-norm contraction at every $h > 0$ and positivity of the surrogate resolvent. Diagonally implicit Runge–Kutta, Rosenbrock, Padé-table, and spectral-deferred-correction schemes require separate conditions on the stage matrices and the sign structure of the resulting subproblems before one or both facts are preserved; whether they retain the interpretation in terms of surrogate-MDP planning that IE-VI relies on is open. Appendix L states this scope boundary.

IE-DYNA convergence. The IE-DYNA algorithm (Algorithm 5) keeps IE-VI's separation between the fixed-point equation and surrogate planning, but replaces the full defect by a single-sample defect-corrected reward target. An analogue of Proposition 15 for IE-DYNA is left to follow-up; the empirical curves in Figure 1 (b) characterize the algorithm's behavior on the displayed Cliffwalk setting.

N Experiments

The experiments in Figure 1 compare stable-hit ranges under surrogate error for practical one-round Adaptive IE-VI and OS-VI in Control, and compare IE-DYNA with Q-Learning and OS-Dyna in RL. Figure 1 tests these behaviors on 6×6 stochastic Cliffwalk variants. Panel (a) plots stable-hit query indices for finite-MDP planning; panel (b) plots normalized all-state policy-value L_1 error under an 80k nominal-sample replay-tape budget and 60 held-out evaluation seeds per algorithm. These comparisons leave surrogate-only planning uncharged; the additional studies in Section N.13 also report surrogate work and solver time; Appendix L gives the cost-model details.

Figure 1 (a) reports the earliest recorded query at which the normalized value error of the output policy is below 10^{-3} and remains so at every subsequent recorded output. At $\gamma = 0.99$, exact-model VI reaches the threshold at 49 queries. OS-VI has no stable hit for $\lambda \geq 0.70$, while Adaptive IE-VI reaches a stable hit at every displayed λ, using at most 320 recorded queries. Figures N.6 and N.7 examine the practical one-round residual-stall schedule in Appendix J, using $h_{\text{init}} = 10^6$, $h_{\min} = 1$, and reference setting $(\beta, S) = (0.1, 3)$. Figure N.6 varies the threshold and hit criterion; Figure N.7 varies β and S. The adaptive complexity theorems concern exact surrogate planning, and no termination bound is proved here for the practical stall schedule. Figure 1 (b) uses the fixed-step RL variant and held-out evaluation protocol specified below and in Appendix K.4.

The rest of this appendix specifies the environments, perturbation families, target definitions, reporting metrics, sensitivity checks, and additional finite-MDP configurations behind these plots.

N.1 Planning implementations

Figure 1 (a) uses the practical one-round residual-stall schedule with $h_{\text{init}} = 10^6$, $h_{\min} = 1$, $\beta = 0.1$, and $S = 3$, under a budget of 1000 recorded outer attempts. The appendix Policy-Evaluation comparisons use the same schedule, with budgets stated in their captions, and 64-bit floating-point direct linear algebra:

$$\big((1+h)I - h\gamma \widehat{P}^{\pi}\big)w = \mathrm{BR}^{\pi}(V), \qquad V_{\text{new}} = V + hw.$$

There is no iterative stopping tolerance or iteration cap for this linear system. The batched implementation uses a direct linear solver; the scalar implementation additionally falls back to least squares if the direct solver raises a linear-algebra exception. The accepted high-discount Cliffwalk Policy-Evaluation rerun uses the retained batched path.

The retained practical Control implementation used in Figure 1 (a) and the sensitivity studies in Figures N.6 and N.7 uses warm-started surrogate value iteration in 64-bit arithmetic. With its default settings, each inner computation starts from the current outer value estimate and stops at the first iterate whose surrogate Bellman update differs by less than 10^{-10} in maximum norm, or after 5000 sweeps. This is an absolute successive-iterate tolerance, not a prescribed value-error tolerance. The inner solver supplies a policy greedy for its shifted surrogate problem, with first-index tie breaking. The historical practical Control path can record a rejected candidate's policy while retaining the preceding value estimate. On tolerance success the routine returns the tested iterate without applying that Bellman update; at the cap it returns the last updated iterate. Earlier matched checks reproduced all 41 archived stable-hit counts in the high-discount Cliffwalk Policy-Evaluation comparison and all 114 first- and stable-hit comparisons in the Control threshold-sensitivity study.

The practical plots retain their recorded query-index convention: initialization is excluded and each active outer attempt increments the index, including a rejected candidate. In the batched implementation, the candidate true-model products are formed before the guard is checked; extra arithmetic for inactive batch members is not charged to active trajectories. The scalar path can reject before forming those products and still increments its attempt counter. Thus these indices describe the recorded algorithmic budget, not instrumented physical operator totals. The common-solver study in Section N.13.1 instead counts actual operator applications, including initialization.

The displayed practical panels are regenerated from the accepted reruns identified in Section N.13.4. The preceding matched-count statements describe earlier verification against the archive; the accepted reruns include the local differences reported in the Maze and threshold-sensitivity panels. These disclosures identify the retained numerical implementation and the historical matched checks; they do not establish the exact historical solver build or unavailable historical target arrays. In particular, the schedule-sensitivity configurations are reconstructed from saved figure metadata, and the stated default inner tolerance and cap do not prove historical overrides. The separately configured incumbent-restoration experiment is reported in Section N.13.5.

The retained experiment package also includes a separate Cliffwalk verification of the restarted Adaptive IE-VI implementation, with $\gamma = 0.99$, residual target parameter $\epsilon = 10^{-3}$, and a reported query budget of 1000. The configuration uses one deterministic instance with seed 0 and a support-uniform corruption grid with 19 settings.

The frozen Control implementation starts from $V_0 = 0$, doubles the error hypothesis $\Lambda_i = 2^i$ after an unsuccessful epoch, and restarts each epoch from V_0. It uses

$$h_i = [(1-\gamma)(2\Lambda_i - 1)]^{-1}, \qquad m = \left\lceil \log_2 \frac{8}{1-\gamma} \right\rceil = 10,$$

and at most $\lceil 4\Lambda_i K_\epsilon \rceil$ outer updates per epoch, where

$$K_\epsilon = \max\left\{1, \log(1/\epsilon) + 2\log\frac{1+\gamma}{1-\gamma}\right\}.$$

It checks the true Bellman residual after each completed outer update and stops when

$$\|T^\star V - V\|_\infty \le \epsilon \frac{1-\gamma}{1+\gamma} \|T^\star V_0 - V_0\|_\infty,$$

or when no further complete update fits within the reported query budget. Thus it checks residuals more frequently than the epoch-end checks in Algorithm 3.

Each correction round solves the shifted surrogate MDP by policy iteration in the frozen implementation. Policy evaluation uses a direct linear solve of $(I - \gamma_h \widehat{P}^\pi)U = \tilde{r}^\pi$. Policy improvement uses first-index tie breaking and stops when the policy is unchanged, subject to a cap of 200 iterations. Policies are warm-started between correction rounds and reset at an epoch restart. The routine's nominal 10^{-8} tolerance argument is unused; it does not supply a certified value-error tolerance. The archived OS-VI configuration instead specifies five surrogate policy-iteration steps per update. The frozen solver follows that fixed-step rule without an early policy-stability stop.

A frozen-source CPU, 64-bit verification reproduced all 57 archived stable-hit counts across the 19 settings and all recorded query indices of that restarted Cliffwalk comparison. All 9840 adaptive correction-round planning computations reached policy stability within five iterations, without hitting the cap. Their largest observed shifted-Bellman residual was 7.28×10^{-12}, giving a residual-based value-error bound of at most 3.68×10^{-10}. These measured bounds are diagnostics, not the solver's stopping tolerances. The verification checks the remaining query budget before computing a complete outer update, avoiding discarded candidate work beyond the budget. The artifact preserves the frozen source, diagnostics, and both sets of traces. Full policy trajectories are not identical to the archive, and the exact historical solver build remains unidentified; the reproduced result is that archived stable-hit comparison.

The primary practical Policy-Evaluation and Control curves use the carry-forward implementation: step-size reductions retain the current accepted value rather than restoring the value with the best observed residual. The Maze Control panels in Figures N.8 and N.9 retain the restarted implementation.

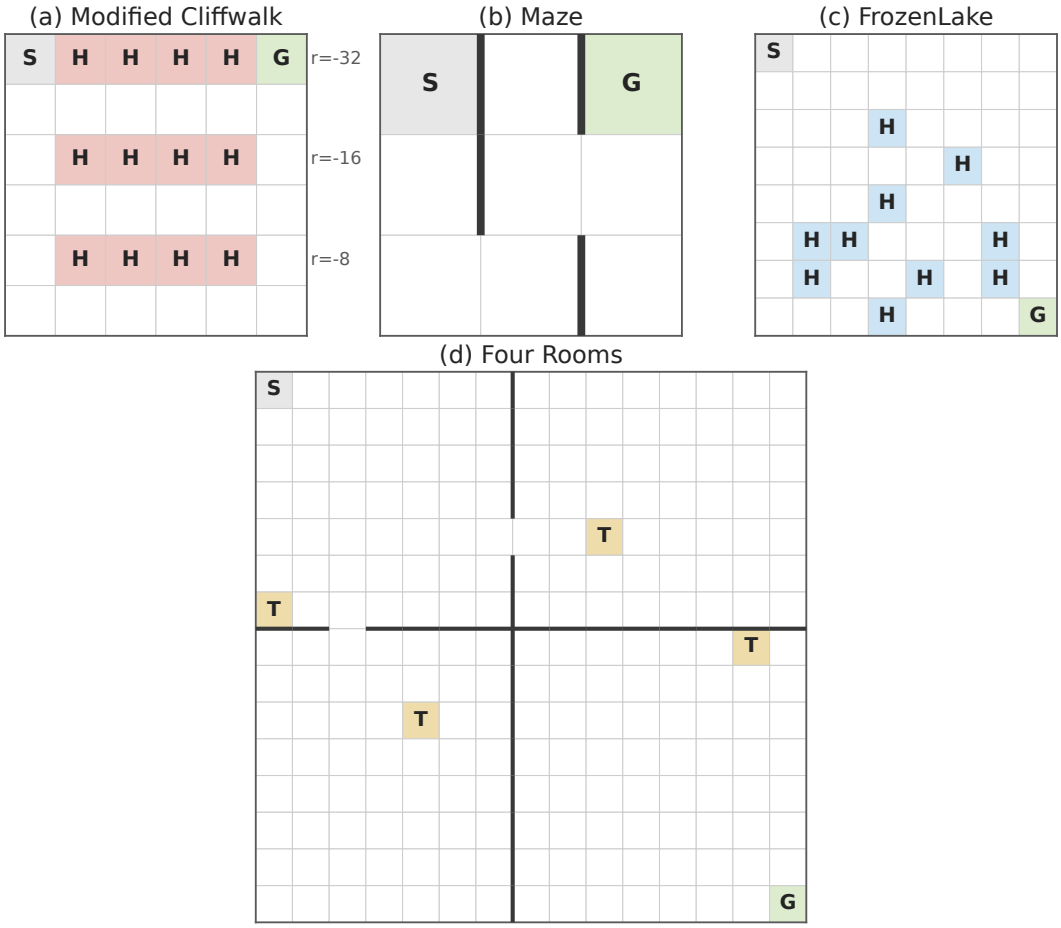


Figure N.1: The finite-MDP schematics identify the start, goal, absorbing hazard, hole, and trap states, and wall geometry used in the appendix experiments. Panel (a) is Modified Cliffwalk: S is the top-left start, G is the top-right goal with reward 20, and H cells are absorbing hazards with rewards -32, -16, and -8 on rows 0, 2, and 4. Panel (b) is Maze; G marks the reward target, and black segments mark blocked adjacencies. Panel (c) is FrozenLake, with H marking absorbing holes. Panel (d) is Four Rooms; T cells are absorbing traps. Cliffwalk, Maze, and Four Rooms use intended-action probability 0.9 with the remaining probability split over the other cardinal actions; FrozenLake uses intended probability $2/3$ with perpendicular slips $1/6$ each.

N.2 Finite-MDP environments

Modified Cliffwalk. The Cliffwalk planning panels use a 6×6 modified Cliffwalk MDP with four cardinal actions and intended-action probability 0.9. The start state is the top-left cell and the goal is the top-right cell. With probability 0.1, one of the other three directions is selected uniformly. Boundary collisions leave the agent in place. The goal state gives reward 20. The middle four cells of rows 0, 2, and 4 are absorbing hazard cells with rewards -32, -16, and -8 on those rows; all remaining nonterminal states have reward -1. This is the finite-MDP Cliffwalk geometry used for Figure 1.

Maze. Additional appendix configurations use a 3×3 maze with top-left start, top-right goal, and the same four-action intended-action dynamics. Attempted moves across blocked adjacencies leave the state unchanged. The blocked adjacencies are

$$\{\{0,1\},\{3,4\},\{1,2\},\{7,8\}\},$$

using row-major state numbering. Reward 1 is assigned to actions whose intended target is the goal state. This environment is an additional finite-MDP stress test and does not define a separate theoretical setting.

FrozenLake. The appendix FrozenLake Control panel uses an 8×8 grid with four cardinal actions, absorbing hole states, and an absorbing goal in the lower-right cell. The intended action has probability $2/3$, and the two perpendicular slips have probability $1/6$ each. Boundary collisions leave the state unchanged. The default hole cells are

$$(2,3),(3,5),(4,3),(5,1),(5,2),(5,6),(6,1),(6,4),(6,6),(7,3),$$

with zero-indexed row-column coordinates. The goal state gives reward 1 and all other states give reward 0.

Four Rooms. The appendix Four Rooms panels use a 15×15 grid with four cardinal actions and two internal walls crossing at the grid midpoint. Each wall has one doorway. The random instance is fixed with seed 0, giving a vertical doorway at row 4, a horizontal doorway at column 2, and absorbing traps at $(4,9)$, $(6,0)$, $(7,13)$, and $(9,4)$, using zero-indexed row-column coordinates. The goal is the lower-right cell. The intended action succeeds with probability 0.9; the remaining probability is split uniformly across the other three actions. Boundary and wall collisions leave the agent in place. The goal reward is 1, trap reward is -1, and other rewards are 0.

N.3 Surrogate error families

The main perturbation family is the support-uniform, or smoothed, surrogate:

$$\hat{\mathcal{P}}(\cdot \mid x,a;\mathcal{P},\lambda) = (1-\lambda)\mathcal{P}(\cdot \mid x,a) + \lambda \mathcal{U}\{y : \mathcal{P}(y \mid x,a) > 0\}.$$

The notation $\mathcal{U}(A)$ denotes the uniform distribution over the finite set A. The endpoint $\lambda = 0$ gives the true model, while $\lambda = 1$ replaces each transition row by the uniform distribution over its true support. The scalar λ is an experiment parameter for setting the surrogate error; it is not an input of the IE-DYNA update in Algorithm 5.

The self-loop surrogate used in the appendix comparisons is:

$$\hat{\mathcal{P}}(\cdot \mid x,a;\mathcal{P},\lambda) = (1-\lambda)\mathcal{P}(\cdot \mid x,a) + \lambda\, \mathbf{1}\{\cdot = x\}.$$

For table-style finite-MDP summaries, the coarse perturbation grids are

$$\begin{aligned} &\lambda \in \{0, 0.1, 0.5, 0.8, 1\} && \text{for Cliffwalk,} \\ &\lambda \in \{0, 0.1, 0.5, 0.7, 1\} && \text{for Maze.} \end{aligned}$$

Curve figures use the denser λ-grids stated in their captions. Figure 1 displays stable-hit curves over the planning λ-grid and RL curves at support-uniform corruption $\lambda = 0.8$. The body RL panel compares Q-Learning, OS-Dyna, and IE-DYNA.

N.4 Targets and metrics

The following definitions apply to Figure 1 and the associated sensitivity and environment comparisons. The additional studies use the separate definitions in Section N.13.

For Policy Evaluation, the target is $V^{\pi^\star}$, where $\pi^\star$ is the optimal policy of the true MDP and then held fixed for evaluation. The reported error is the normalized value error

$$e_q^{\mathrm{PE}} := \frac{\|V_q - V^{\pi^\star}\|_1}{\|V^{\pi^\star}\|_1}.$$

For Control, the reported error is the normalized value error of the recorded output policy π_q:

$$e_q^{\mathrm{ctrl}} := \frac{\|V^{\pi_q} - V^\star\|_1}{\|V^\star\|_1}.$$

In Figure 1 (a), VI records its true-model greedy policy, while OS-VI and Adaptive IE-VI record the policy returned by their surrogate-planning update, including a rejected candidate in the practical adaptive path. These policies are evaluated on the true MDP offline; that evaluation is excluded from the planning-query axis. A policy returned by surrogate planning need not be greedy for the true MDP at the recorded value iterate. The other studies retain the policy construction specified by their

own protocols. For the RL panel, π_t is the recorded policy at nominal sample budget t, evaluated on the true MDP. The plotted error is

$$e_t^{\mathrm{RL}} := 100\,\frac{\|V^{\pi_t} - V^\star\|_1}{\|V^\star\|_1}.$$

State-action pairs are sampled uniformly from the generative replay tape. At $\lambda = 0.8$, each visited empirical transition row is mixed with the uniform distribution over its observed successor support, with weights 0.2 and 0.8, respectively; unvisited rows start uniformly over all states.

Each method is tuned separately on seeds 20–29 by mean integrated error through 40000 nominal samples, then evaluated on seeds 250–309. The fixed grid contains 36 constant, per-visit and global-index learning-rate schedules; OS-Dyna batches are $10, 30, 100$, while IE-DYNA uses batches $10, 30$, $J \in \{1, 3\}$ and

$$h \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 3, 10, 30\}.$$

Curves retain every evaluation at 2000-sample intervals through 80000 nominal samples, using the last completed update within each budget. Bands are 95% pointwise percentile intervals from 20000 resamples of the 60 whole seed trajectories, using the same resampled seed indices for every method. Statistics are computed before the logarithmic display transform.

The reference optimal value and each recorded policy value are computed by tabular policy iteration and linear-system evaluation.

For the planning and sensitivity comparisons, targets are computed by solving the true tabular MDP to residual tolerance 10^{-10}, with value-error budget 10^{-8}, tight enough that target error is negligible relative to the plotting threshold. These tolerances concern reference-target construction, not the inner surrogate solves.

N.5 Query axes and hit times

The planning comparisons report query indices for work against the true model, with study-specific counting conventions given in Section N.1 and Section N.13.1. A true-model application means $V \mapsto \mathcal{P}^\pi V$ for Policy Evaluation and $V \mapsto \{\mathcal{P}^a V\}_{a\in\mathcal{A}}$ for Control. The practical plots record outer attempts under the convention above; these indices need not equal physical application counts. Surrogate-only planning is not charged, as discussed in Appendix L. RL plots use environment samples.

Finite-MDP planning panels report two convergence summaries from the same trajectory. For a displayed error trajectory e_q, either e_q^{PE} or e_q^{ctrl}, the first-hit time

$$q_{\mathrm{FH}}(\tau) := \min\{q \geq 1 : e_q < \tau\}$$

is the earliest query at which the iterate falls below the threshold τ. The stable-hit (or last-passage) time

$$q_{\mathrm{SH}}(\tau) := \min\{q \geq 1 : e_{q'} < \tau \text{ for all } q' \geq q\}$$

is the earliest query after which the error stays below τ for the rest of the run. Both are read directly from the trajectory; no smoothing is applied. The two coincide when the error is non-increasing after the first crossing.

In these definitions, q and q' range over the recorded output indices of the trajectory. For the practical schedule in Figure 1 (a), these are outer-attempt indices $1, 2, 3, \ldots$, excluding initialization, as described in Section N.1. The stable-hit condition is retrospective over recorded outputs until the run terminates; it is distinct from the online Bellman-residual stopping test.

Panel (a) of Figure 1 reports q_{SH} at $\tau = 10^{-3}$; the stable-hit form is used because OS-VI's iterates oscillate in the high-λ regime, where q_{FH} registers transient dips that the iterates leave on the next query. The threshold is normalized. For the Control instance in Figure 1 (a), $\|V^\star\|_1 \approx 5.49 \times 10^4$, so $\tau = 10^{-3}$ corresponds to absolute L1 policy-value error of approximately 55, or 1.5 per state.

N.6 FrozenLake control gamma comparison

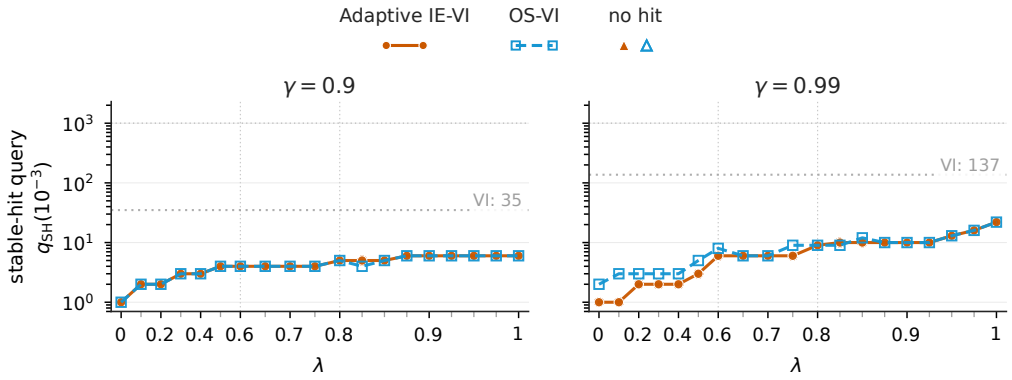


Figure N.2: FrozenLake control under the support-uniform model-error family. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$ using the marker convention of Figure 1. Both Adaptive IE-VI and OS-VI reach the threshold for every displayed λ; the exact-model VI baselines are 35 and 137 queries.

Figure N.2 reports a finite-MDP case where the high-λ OS-VI failure seen on Cliffwalk does not appear. Under support-uniform perturbations, FrozenLake does not generate the high-λ OS-VI loss of stable threshold crossing seen in the Cliffwalk Control panel: OS-VI and Adaptive IE-VI have finite stable-hit times for every displayed λ, and their curves nearly overlap. The figure is kept in the appendix because it documents environment dependence without supporting the separation in Figure 1.

N.7 Four Rooms control gamma comparison

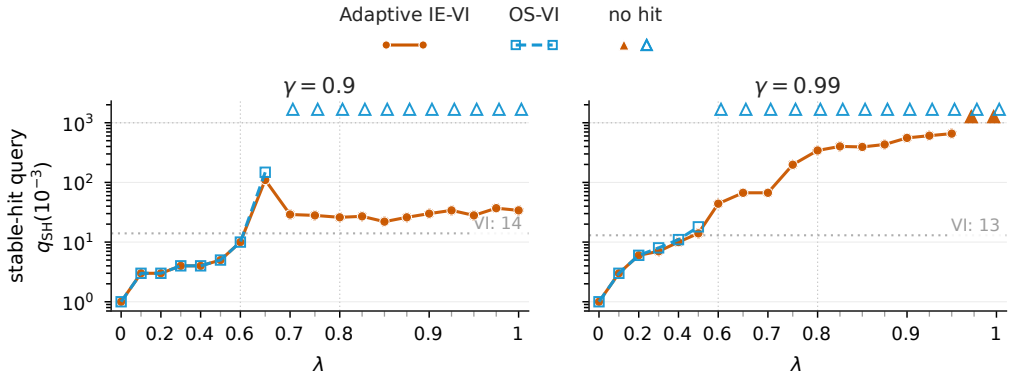


Figure N.3: Four Rooms control under the support-uniform model-error family. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$. Adaptive IE-VI stays finite farther into the high-λ region than OS-VI: OS-VI fails from $\lambda = 0.7$ and $\lambda = 0.6$, while Adaptive IE-VI stays finite through all displayed λ at $\gamma = 0.9$ and through $\lambda = 0.95$ at $\gamma = 0.99$.

Figure N.3 reports a Control setting other than Cliffwalk in which the high-λ OS-VI failure appears. At $\gamma = 0.9$, Adaptive IE-VI stays finite across the full displayed axis even though OS-VI loses stable threshold crossing from $\lambda = 0.7$ onward. At $\gamma = 0.99$, the separation starts earlier for OS-VI, while Adaptive IE-VI stays finite through $\lambda = 0.95$. The last two λ values in the $\gamma = 0.99$ panel also indicate that Four Rooms is a harder Control instance for Adaptive IE-VI than FrozenLake.

N.8 Maze policy-evaluation gamma comparison

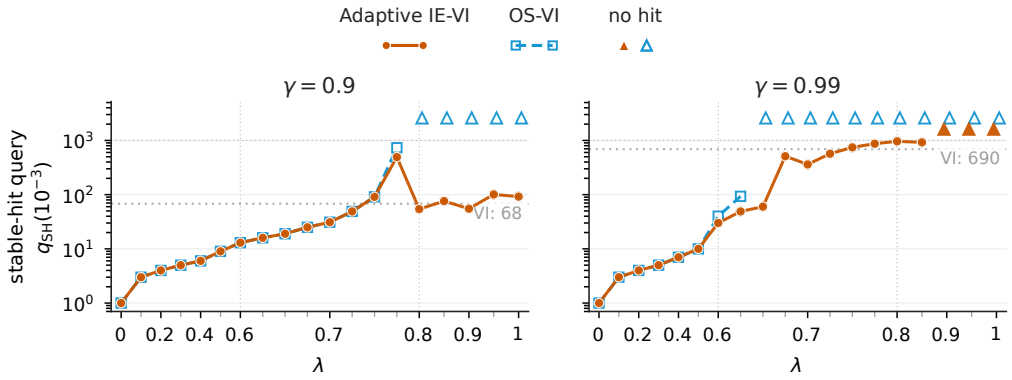


Figure N.4: Maze policy evaluation under support-uniform model error. Panels report $q_{\mathrm{SH}}(10^{-3})$ on a shared λ-grid for $\gamma = 0.9$ and $\gamma = 0.99$. Raising the discount shifts the no-hit region left: OS-VI fails from $\lambda = 0.8$ at $\gamma = 0.9$ and from $\lambda = 0.65$ at $\gamma = 0.99$, while Adaptive IE-VI stays finite through all displayed λ at $\gamma = 0.9$ and through $\lambda = 0.85$ at $\gamma = 0.99$.

Figure N.4 applies the stable-hit format to fixed-policy evaluation on Maze. The target policy is the true-MDP optimal policy $\pi^\star$, held fixed after the true target $V^{\pi^\star}$ is computed. The recorded query index follows the practical accounting convention in Section N.1; planning computations using only the surrogate are uncharged under the same convention as the Control panels. The discount change has two visible effects: the exact-model VI baseline increases from 68 to 690 queries, and the OS-VI no-hit region moves from the high-λ tail to the middle of the displayed axis. Adaptive IE-VI remains finite across the full axis at $\gamma = 0.9$ and remains finite through $\lambda = 0.85$ at $\gamma = 0.99$.

N.9 Four Rooms policy-evaluation gamma comparison

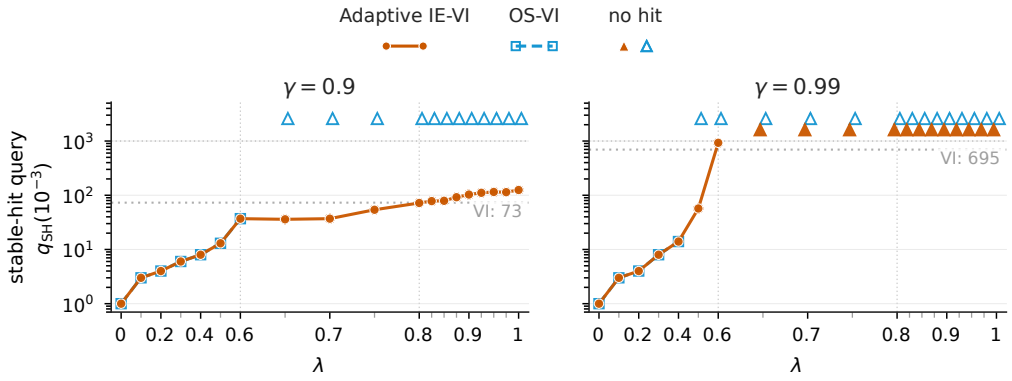


Figure N.5: Four Rooms policy evaluation under support-uniform model error. Panels report $q_{\mathrm{SH}}(10^{-3})$ for the same environment as Figure N.3. At $\gamma = 0.9$, Adaptive IE-VI reaches the threshold across the displayed axis while OS-VI fails from $\lambda = 0.65$. At $\gamma = 0.99$, Adaptive IE-VI stays finite through $\lambda = 0.6$, while OS-VI fails from $\lambda = 0.5$.

Figure N.5 repeats the fixed-policy evaluation protocol of Figure N.4 on the Four Rooms instance used in Figure N.3. The target policy is computed once from the true MDP and then held fixed. At $\gamma = 0.9$, Adaptive IE-VI reaches the threshold across the displayed axis even when OS-VI diverges in the high-λ tail. Raising the discount to $\gamma = 0.99$ shifts the no-hit region left: Adaptive IE-VI remains finite only through $\lambda = 0.6$, while OS-VI loses stable threshold crossing from $\lambda = 0.5$. The VI baseline grows from 73 to 695 queries under the same threshold.

N.10 Threshold and criterion sensitivity

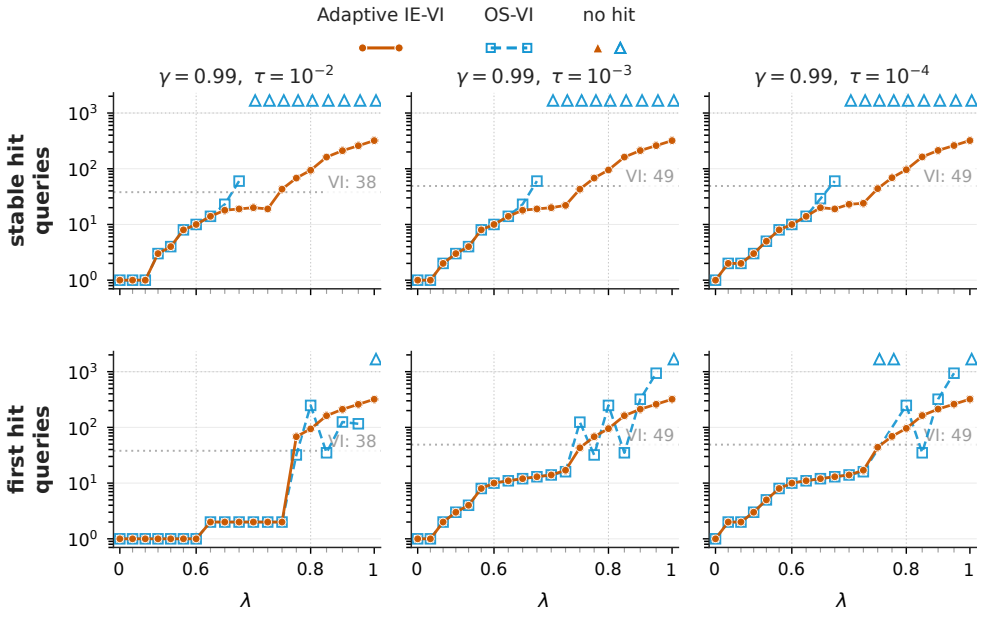


Figure N.6: Threshold and criterion sensitivity for the practical one-round control schedule on Cliffwalk at $\gamma = 0.99$. Columns sweep $\tau \in \{10^{-2}, 10^{-3}, 10^{-4}\}$; rows report $q_{\mathrm{SH}}(\tau)$ and $q_{\mathrm{FH}}(\tau)$. The stable-hit panel at $\tau = 10^{-3}$ uses the same configuration as Figure 1 (a). The first-hit row includes transient OS-VI threshold touches in the high-λ regime.

Figure N.6 evaluates the practical one-round Control schedule at $\tau \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ under both convergence criteria. The practical adaptive method has a stable hit at every displayed λ for all three thresholds, while OS-VI has no stable hit for $\lambda \geq 0.7$. The OS-VI failure is a property of the iterates, not of the criterion: under q_{FH}, OS-VI's reported query at $\lambda = 0.8$ jumps to 248 and at $\lambda = 0.85$ drops to 35 in every first-hit panel, but the underlying trajectory leaves the threshold band on the next query and continues to oscillate for the rest of the 1000-query budget. The stable-hit form discards both the spike and the trough.

N.11 Practical adaptive schedule sensitivity

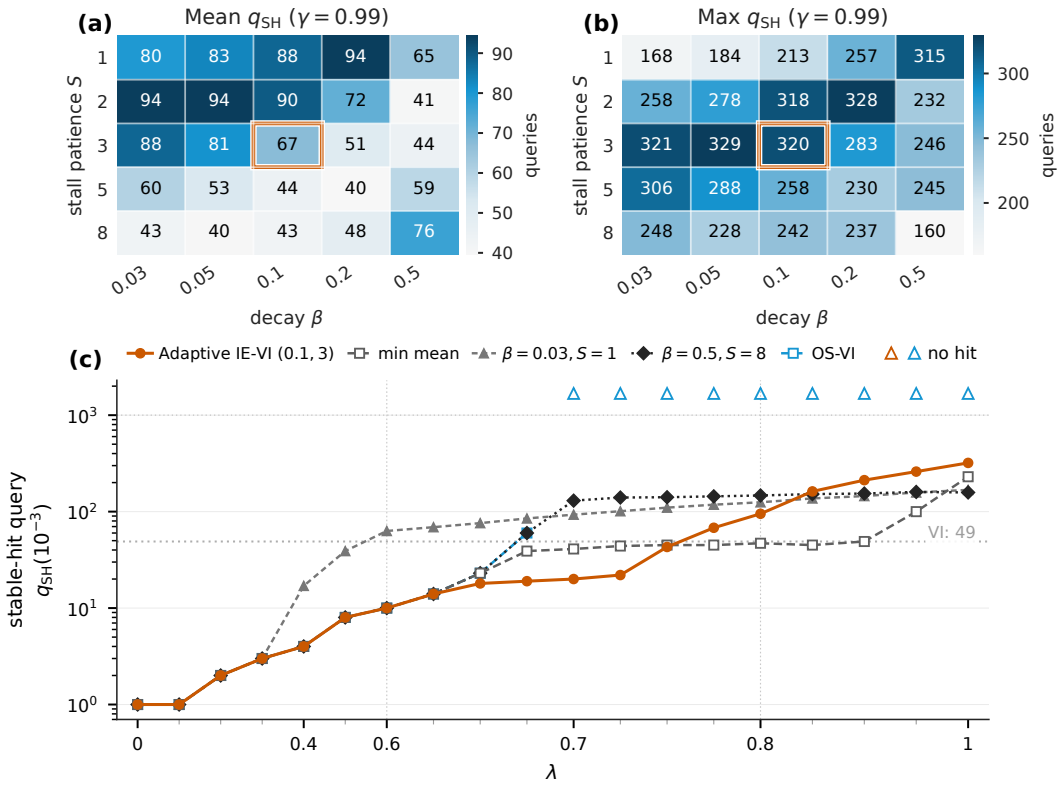


Figure N.7: Sensitivity of the practical one-round control schedule to h-decay factor β and stall patience S at $\tau = 10^{-3}$. Heat maps show the parameter grid, with the reference setting $(0.1, 3)$ outlined. Panel (c) compares representative practical schedules with OS-VI on the same stable-hit scale; triangles mark OS-VI settings with no stable hit within budget. The reference schedule is the one used in Figure 1 (a).

Figure N.7 fixes the criterion at $q_{\mathrm{SH}}(10^{-3})$ for Cliffwalk Control at $\gamma = 0.99$ and varies β and S in the practical one-round schedule. The outlined $(0.1, 3)$ setting is the practical schedule used in Figure 1 (a). The sweep shows how these parameters affect the practical schedule's stable-hit query counts. The endpoint choices h_{init} and h_{min} are not swept here; Appendix J gives their interpretation: $h_{\mathrm{min}} = 1$ is the unit Bellman-residual-flow time scale, and $h_{\mathrm{init}} = 10^6$ is a large-h proxy for the OS-VI limit.

N.12 Additional stable-hit panels

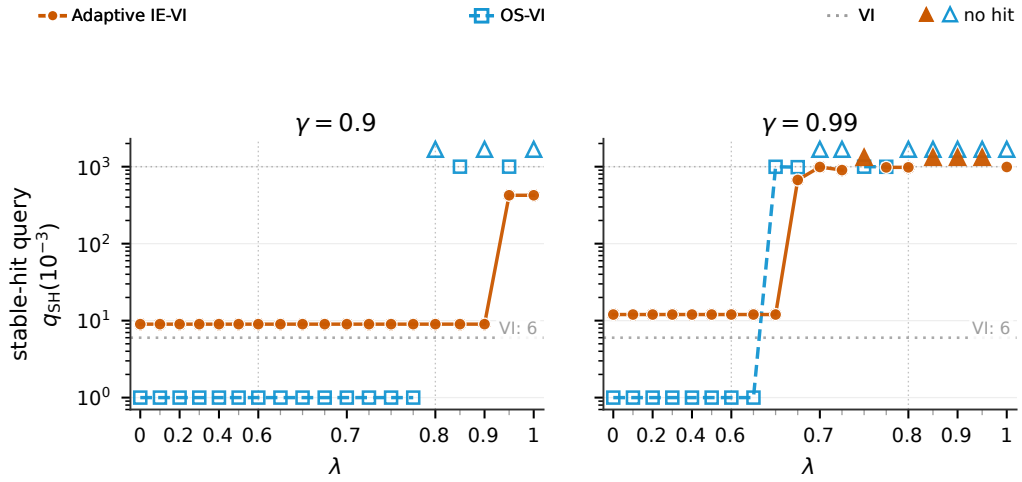


Figure N.8: Maze control under support-uniform model error. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$. At $\gamma = 0.9$, Adaptive IE-VI reaches the threshold at all displayed λ values, while OS-VI has three no-hit points, at $\lambda = 0.8, 0.9, 1$, and otherwise runs close to the query budget in the high-λ tail. At $\gamma = 0.99$, both corrected methods have no-hit regions in the high-λ range, with first no-hit points at $\lambda = 0.7$ for OS-VI and $\lambda = 0.75$ for Adaptive IE-VI.

Figure N.8 adds the Maze Control companion to the Maze Policy Evaluation panel in Figure N.4. The exact-model VI baseline is $q_{\mathrm{SH}} = 6$ for both discounts because the greedy policy reaches the normalized policy-value threshold quickly on this small MDP. The support-uniform perturbation separates the corrected methods only in the high-λ range; the $\gamma = 0.99$ panel has the larger sensitivity to the surrogate as the horizon grows. These first no-hit points describe the displayed grid; the hit sets need not be monotone in λ or nested across methods.

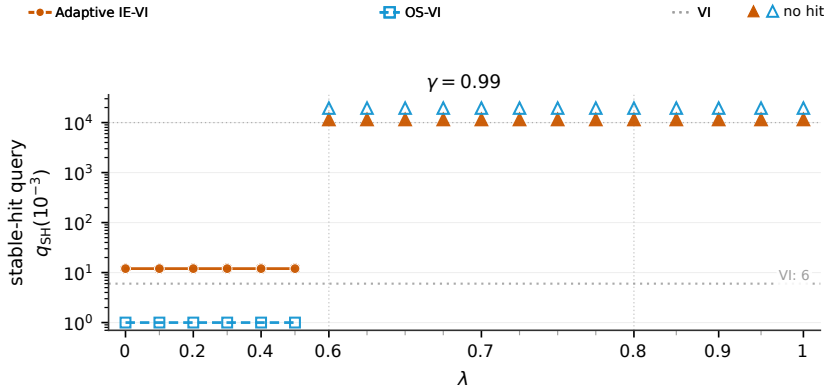


Figure N.9: Maze control under self-loop model error at $\gamma = 0.99$. The panel reports $q_{\mathrm{SH}}(10^{-3})$ over the same piecewise λ-grid as the support-uniform panels. Both OS-VI and Adaptive IE-VI lose stable threshold crossing from $\lambda = 0.6$, while the exact-model VI baseline remains $q_{\mathrm{SH}} = 6$.

Figure N.9 indicates that the self-loop perturbation is harder for the Maze Control instance than support-uniform perturbations at the same discount. Once $\lambda \geq 0.6$, neither corrected method has a stable hit within the 10000-query budget.

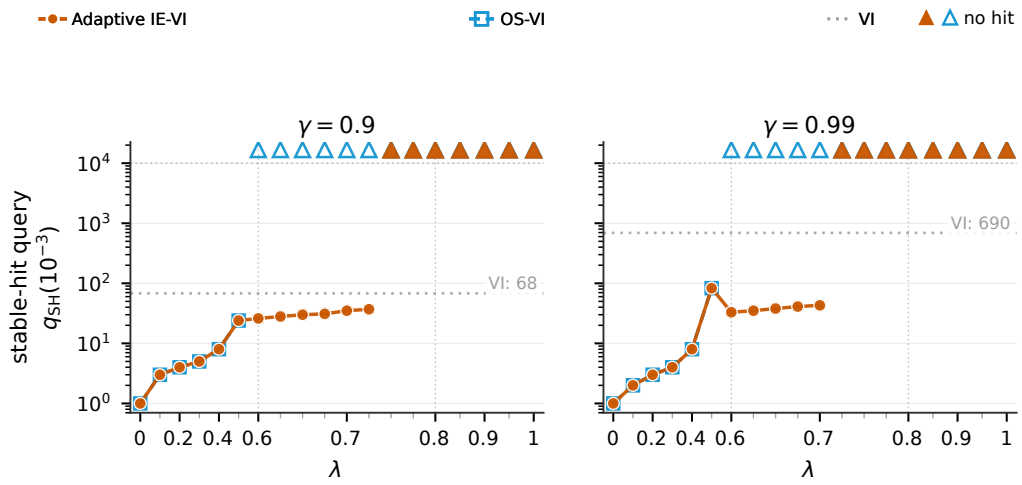


Figure N.10: Maze policy evaluation under self-loop model error. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$. OS-VI loses stable hits from $\lambda = 0.6$ in both panels; Adaptive IE-VI reaches farther, with first no-hit points at $\lambda = 0.75$ for $\gamma = 0.9$ and $\lambda = 0.725$ for $\gamma = 0.99$.

Figure N.10 complements the support-uniform Maze Policy Evaluation panel in Figure N.4. The self-loop surrogate shifts the no-hit region left for both corrected methods, while exact-model VI remains finite through the full displayed axis.

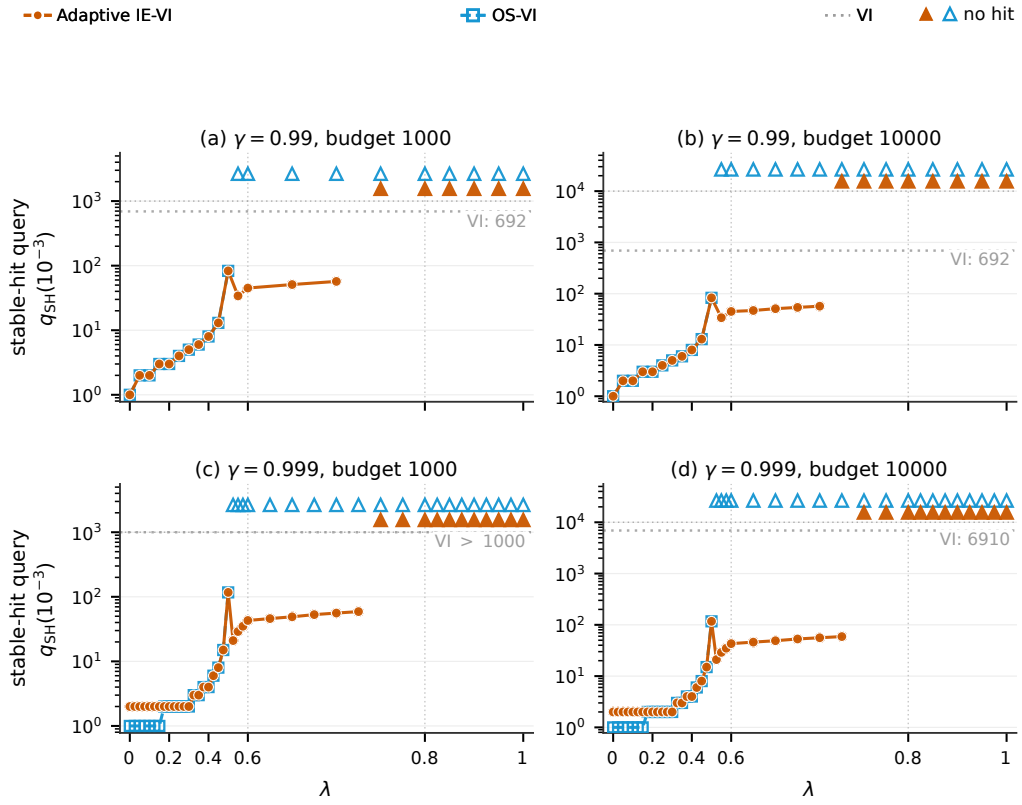


Figure N.11: Cliffwalk policy evaluation under self-loop model error. Columns compare 1000- and 10000-query budgets; rows correspond to $\gamma = 0.99$ and $\gamma = 0.999$. Triangles mark method-grid points with no stable hit within the displayed budget. At $\gamma = 0.99$, increasing the budget does not change the corrected-method no-hit boundary: OS-VI fails from $\lambda = 0.55$, while Adaptive IE-VI fails from $\lambda = 0.725$. At $\gamma = 0.999$, exact-model VI needs 6910 queries, so the 1000-query panel has no VI stable hit.

Figure N.11 reports fixed-policy Cliffwalk comparisons under self-loop model error. The higher discount increases the VI query scale but leaves the corrected-method no-hit transition near the same λ values: OS-VI loses stable hits from $\lambda = 0.525$, while Adaptive IE-VI stays finite through $\lambda = 0.725$ at the displayed grid.

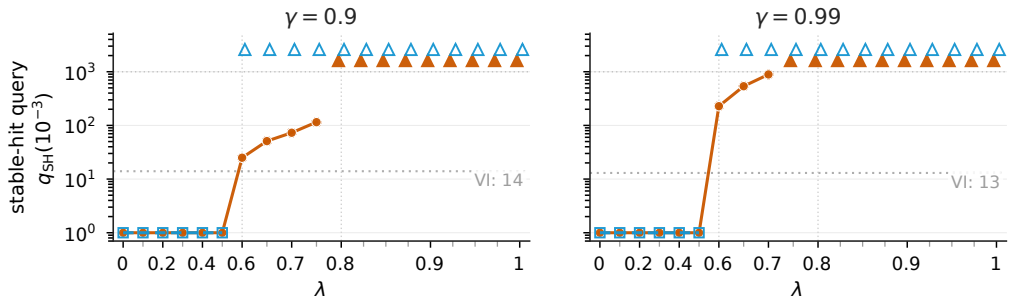


Figure N.12: Four Rooms control under self-loop model error separates Adaptive IE-VI from OS-VI before the high-λ tail. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$ on the same grid as Figure N.3. OS-VI loses stable hits from $\lambda = 0.6$ in both panels; Adaptive IE-VI stays finite through $\lambda = 0.75$ at $\gamma = 0.9$ and through $\lambda = 0.7$ at $\gamma = 0.99$.

Figure N.12 gives the self-loop counterpart to the support-uniform Four Rooms Control panel. The exact-model VI baseline is $q_{\mathrm{SH}} = 14$ at $\gamma = 0.9$ and 13 at $\gamma = 0.99$. The self-loop surrogate moves the OS-VI no-hit boundary to $\lambda = 0.6$, while Adaptive IE-VI remains finite through $\lambda = 0.75$ at $\gamma = 0.9$ and through $\lambda = 0.7$ at $\gamma = 0.99$.

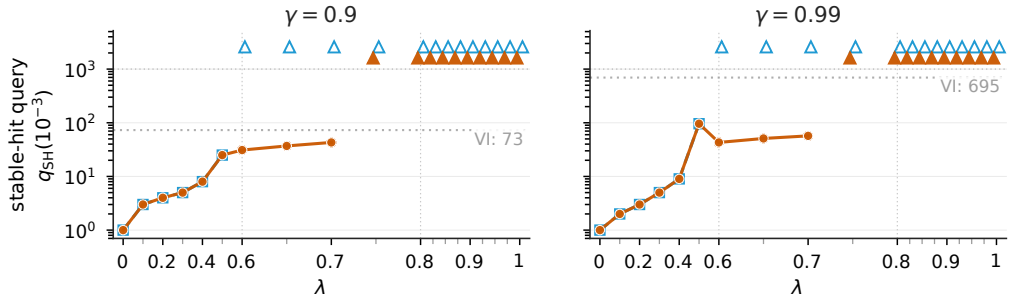


Figure N.13: Four Rooms policy evaluation under self-loop model error has the same no-hit boundary for Adaptive IE-VI at both displayed discounts. Panels report $q_{\mathrm{SH}}(10^{-3})$ for $\gamma = 0.9$ and $\gamma = 0.99$. OS-VI loses stable hits from $\lambda = 0.6$, while Adaptive IE-VI stays finite through $\lambda = 0.7$ in both panels.

Figure N.13 adds the self-loop counterpart to the Four Rooms Policy Evaluation panel in Figure N.5. The exact-model VI baseline grows from 73 to 695 queries when γ changes from 0.9 to 0.99, but the corrected-method no-hit boundaries remain unchanged on this grid.

N.13 Additional cost and horizon studies

The following studies measure surrogate computation, horizon dependence, and solver time in addition to true-model queries. Their protocols are specified separately from the main comparison because they use different inner solvers and evaluation criteria.

N.13.1 Common-solver corruption and accuracy study

The additional control sweep uses the 36-state, four-action Modified Cliffwalk instance with $\gamma = 0.99$ and support-preserving corruption

$$\widehat{P} = (1-\lambda)P + \lambda U_{\mathrm{support}(P)}.$$

The uniform distribution is taken separately on each transition row's support. Intended actions have probability 0.9; the remaining probability is shared by the other actions. State rewards are -1 on ordinary states, $+20$ at the absorbing goal, and -32, -16, and -8 at the absorbing hazard states in the three hazard rows. The full 19-setting grid appears in Tables N.1 and N.2.

OS-VI and Adaptive IE-VI use the same warm-started surrogate VI solver, first-index tie breaking, and a 5,000-sweep cap. The solver stops when

$$\|\widehat{T}U - U\|_\infty \le (1-\beta)\delta_{\mathrm{solve}},$$

where β is the inner discount: γ for OS-VI and γ_h for Adaptive IE-VI. This certifies value error at most δ_{solve}. We compare $\delta_{\mathrm{solve}} = 10^{-8}$ and 10^{-2}. Each run uses the full budget of 1,000 true applications. Its stable hit is the first output whose offline-evaluated policy error

$$\frac{\|V^\pi - V^\star\|_1}{\|V^\star\|_1 + 10^{-12}}$$

is strictly below 10^{-3} and remains so for the rest of that run. Offline policy evaluation is excluded from the algorithmic counts. All 19 Adaptive IE-VI settings have a stable hit; OS-VI has ten hits and nine no-hits at each tolerance. A no-hit denotes failure to meet this criterion within the budget, not a numerical-divergence termination.

Actual true and surrogate applications are counted, including initialization and checks, with cached work counted only once. The Adaptive IE-VI implementation charges its initial application to the zero value vector. It starts at $h = 10^6$ and divides h by ten, down to a floor of one, after three non-improving residuals. A divergence-guard rejection triggers the same reduction immediately and terminates with a failure status if the reduced step reaches the floor. Each run also has a cap of 5,000 attempted outer updates. It carries accepted iterates forward without restoring the value that attained the best residual. All arithmetic uses 64-bit floating point. This study uses one correction round and the common surrogate value-iteration solver specified above. Figure 1 (a) uses the same practical schedule with the retained numerical solver and outer-attempt counting convention described in Section N.1; the present study uses a common inner solver and counts actual operator applications.

We report weighted cost $C_\kappa = N_P + \kappa N_{\widehat{P}}$, where κ is the cost of a surrogate application relative to a true application. VI uses 49 true calls and no surrogate calls throughout this sweep. When $N_P < 49$, the break-even value is

$$\kappa^\star = \frac{49 - N_P}{N_{\widehat{P}}}; \qquad C_\kappa < 49 \text{ for } 0 \le \kappa < \kappa^\star.$$

Equality gives a tie. For example, at $\lambda = 0.7$ and inner accuracy 10^{-2}, Adaptive IE-VI uses $N_P = 21$ and $N_{\widehat{P}} = 2{,}925$, so $\kappa^\star = 28/2925 \approx 0.00957$ (Tables N.1 and N.2). Tables N.3 and N.4 give the full accuracy comparison. The median, across the 19 settings, of the tight/practical surrogate-work ratio for Adaptive IE-VI is 2.613275. This is a deterministic study of one instance, so no across-seed uncertainty applies.

N.13.2 Horizon protocol and sampling interpretation

We optimize a policy at $H = 640$ and hold the policy, transition kernel, rewards, and initialization fixed across

$$H \in \{10, 20, 40, 80, 160, 320, 640\}.$$

Thirty seeds, 202609170000–202609170029, generate nested empirical transition samples, paired across methods and horizons. The two budgets are 80 and $16H$ samples per policy row, totaling 36 times those amounts. These offline samples are excluded from planning counts.

Both IE-VI schedules use the measured error

$$E_\infty = \gamma \max_s \sum_{s'} |P(s,s') - \widehat{P}(s,s')|.$$

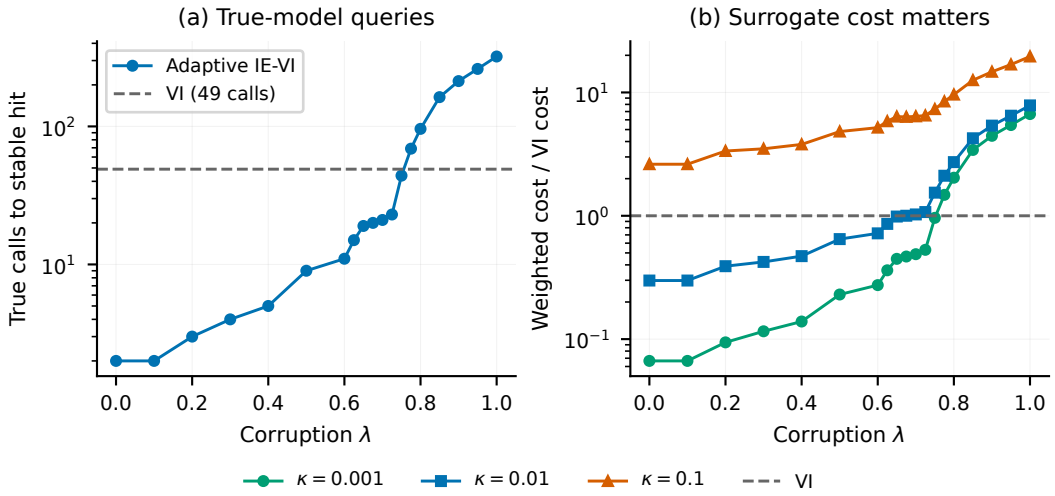


Figure N.14: True-query and weighted cost of Adaptive IE-VI on the deterministic Cliffwalk control sweep, with inner value-function accuracy 10^{-2}. $C_\kappa = N_P + \kappa N_{\widehat{P}}$ is normalized by VI's 49 true calls. Counts are taken at the retrospective stable policy-error hit defined in Section N.13.1; the full table includes OS-VI.

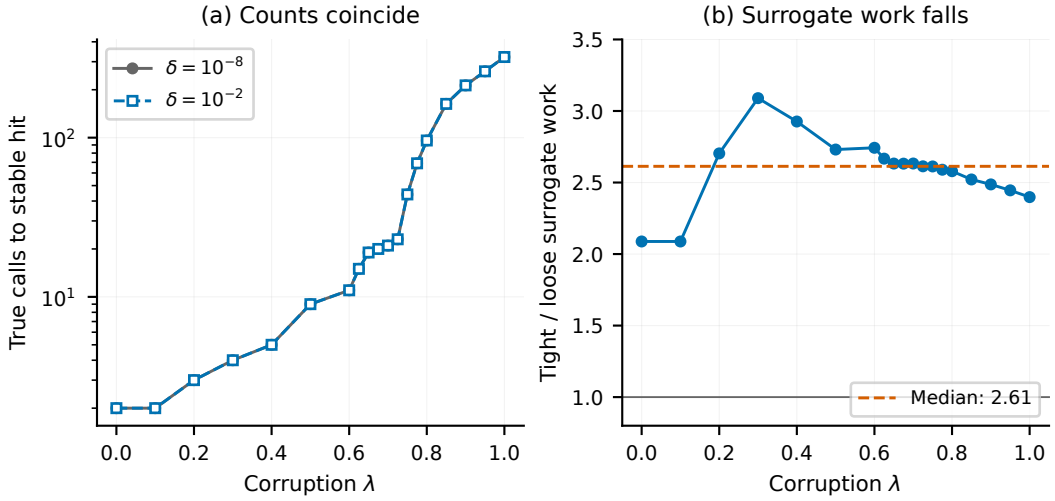


Figure N.15: Inner value-function accuracy on the same control sweep. Adaptive IE-VI true-call counts coincide at 10^{-8} and 10^{-2}, while the median pointwise reduction in surrogate applications is $2.61\times$. The sweep is deterministic.

Table N.1: Full deterministic corruption sweep at inner value-function accuracy 10^{-2}. Counts are full operator applications at the retrospective stable hit. A dash means no stable hit within the complete 1,000-true-call budget; it never means zero work. VI uses $N_P = 49$ and $N_{\widehat{P}} = 0$ throughout. The common-solver and initialization conventions are specified in Section N.13.1.

λ	OS-VI N_P	OS-VI $N_{\widehat{P}}$	Adaptive IE-VI N_P	Adaptive IE-VI $N_{\widehat{P}}$
0	1	1,263	2	1,263
0.1	1	1,263	2	1,263
0.2	2	1,311	3	1,612
0.3	3	1,422	4	1,671
0.4	4	1,551	5	1,807
0.5	8	1,996	9	2,271
0.6	10	2,159	11	2,430
0.625	14	2,459	15	2,711
0.65	23	3,095	19	2,931
0.675	60	5,203	20	2,904
0.7	—	—	21	2,925
0.725	—	—	23	2,953
0.75	—	—	44	3,147
0.775	—	—	69	3,449
0.8	—	—	96	3,745
0.85	—	—	163	4,527
0.9	—	—	213	5,076
0.95	—	—	261	5,641
1	—	—	321	6,374

Table N.2: Weighted costs for the same full sweep at inner value-function accuracy 10^{-2}. The two rightmost columns are $C_\kappa/49$ for Adaptive IE-VI. Strict improvement over VI requires $0 \le \kappa < \kappa^\star$; equality gives a tie. "None" means that the true-call count already exceeds VI's count.

λ	$\kappa^\star$	Cost ratio, $\kappa = .001$	Cost ratio, $\kappa = .01$
0	0.037213	0.0666	0.2986
0.1	0.037213	0.0666	0.2986
0.2	0.028536	0.0941	0.3902
0.3	0.026930	0.1157	0.4227
0.4	0.024350	0.1389	0.4708
0.5	0.017613	0.2300	0.6471
0.6	0.015638	0.2741	0.7204
0.625	0.012541	0.3614	0.8594
0.65	0.010235	0.4476	0.9859
0.675	0.009986	0.4674	1.0008
0.7	0.009573	0.4883	1.0255
0.725	0.008805	0.5297	1.0720
0.75	0.001589	0.9622	1.5402
0.775	None	1.4786	2.1120
0.8	None	2.0356	2.7235
0.85	None	3.4189	4.2504
0.9	None	4.4505	5.3829
0.95	None	5.4417	6.4778
1	None	6.6811	7.8518

Table N.3: Full inner-accuracy comparison for Adaptive IE-VI. The true-call count agrees at both certified value-function accuracies. The last column is the ratio of surrogate applications at 10^{-8} to those at 10^{-2}; its median over all 19 settings is 2.613275.

λ	N_P both accuracies	$N_{\widehat{P}}$ 10^{-8}	$N_{\widehat{P}}$ 10^{-2}	Ratio
0	2	2,637	1,263	2.088
0.1	2	2,637	1,263	2.088
0.2	3	4,358	1,612	2.703
0.3	4	5,163	1,671	3.090
0.4	5	5,288	1,807	2.926
0.5	9	6,201	2,271	2.731
0.6	11	6,665	2,430	2.743
0.625	15	7,230	2,711	2.667
0.65	19	7,716	2,931	2.633
0.675	20	7,643	2,904	2.632
0.7	21	7,701	2,925	2.633
0.725	23	7,717	2,953	2.613
0.75	44	8,221	3,147	2.612
0.775	69	8,931	3,449	2.589
0.8	96	9,659	3,745	2.579
0.85	163	11,414	4,527	2.521
0.9	213	12,623	5,076	2.487
0.95	261	13,794	5,641	2.445
1	321	15,285	6,374	2.398

Table N.4: Full inner-accuracy comparison for OS-VI, paired with Table N.3. True-call counts agree at both accuracies wherever a stable hit exists. The same nine settings have no stable hit at either accuracy.

λ	N_P both accuracies	$N_{\widehat{P}}$ 10^{-8}	$N_{\widehat{P}}$ 10^{-2}
0	1	2,638	1,263
0.1	1	2,638	1,263
0.2	2	2,715	1,311
0.3	3	2,945	1,422
0.4	4	3,182	1,551
0.5	8	4,088	1,996
0.6	10	4,462	2,159
0.625	14	5,049	2,459
0.65	23	6,384	3,095
0.675	60	10,981	5,203
0.7	—	—	—
0.725	—	—	—
0.75	—	—	—
0.775	—	—	—
0.8	—	—	—
0.85	—	—	—
0.9	—	—	—
0.95	—	—	—
1	—	—	—

The conservative schedule is the main text's sufficient choice, archived as "Eq16." The refined schedule chooses h and m together from the theoretical bound: it sets $\Lambda = \max\{1, R_E\}$ and, for positive model error, selects the smallest positive integer m satisfying

$$\left(\frac{4\Lambda}{R_E}\right)^m \geq 8e(m+1)\min\{H, \Lambda\}.$$

It then uses

$$u = \frac{m+1}{4\Lambda m}, \qquad h = \frac{u}{(1-\gamma)(1-u)}.$$

At zero model error, the implementation uses the conservative schedule. This integer search implements the constructive condition in Theorem 27, rather than its explicit sufficient ceiling formula. OS-VI and both IE-VI schedules factor their shifted surrogate matrix once and reuse the LU factors; for each counted true call, the implementation applies the surrogate once and solves one linear system using those factors.

Let V^π denote the fixed policy's value at the current discount. The target is

$$\frac{\|V - V^\pi\|_\infty}{\|V^\pi\|_\infty} \leq 10^{-3}$$

for 50 consecutive outer updates. All 1,680 runs attain it. Reported counts are taken at the first update of that qualifying streak; the subsequent 49 updates are a verification tail, adding $49m$ true calls for IE-VI, and are excluded. Tables N.5 and N.6 give medians and quartiles for all horizons and methods. The displayed figure bands are interquartile ranges. Log–log slopes fit the seven median points; their 95% intervals use 4,000 bootstrap resamples of entire paired seed trajectories (Table N.7). Under the fixed budget, the same sample prefix is used at every horizon, so model error changes with H only through γ.

The archived diagnostic timings replay the predetermined hit counts, including factorization but excluding reference-error checks, the verification tail, and surrogate sampling. They are separate from the stopping-inclusive timing study in Section N.13.3. Their five repetitions per seed are nested timing measurements, not additional independent surrogate seeds.

The growing budget is motivated by finite-tabular concentration: for fixed state/action dimensions and confidence, empirical transition error is $O(n^{-1/2})$ [Weissman et al., 2003]. Thus n proportional to H gives $E_\infty = O(H^{-1/2})$ and a sublinear query bound up to logarithmic factors. The stronger polylogarithmic regime follows under this argument from a sufficient per-row budget of order $H^2/(\log H)^2$. This calculation concerns surrogate accuracy; it does not include model-construction samples in the planning bound.

N.13.3 Timing protocol

The fixed simulator has 100×100 states, $\gamma = 0.995$, and phase seed 20260729. The true transition composes 128 local advection–diffusion microsteps; surrogates use 4, 8, 16, or 32 steps. All planning computations use ordinary VI. One-round IE-VI uses $h = 1024$, the smallest candidate in $\{128, 256, 512, 1024, 2048, 4096\}$ meeting the target within six outer rounds at every fidelity on a separate 24×24 pilot. This pilot is excluded from timing.

For each inner planning problem, VI stops when the shifted-Bellman residual is at most $10^{-6}\|b\|_\infty$, with b the shifted reward. Both methods stop at true residual

$$\frac{\|r + \gamma PV - V\|_\infty}{\|r\|_\infty} \leq 10^{-5}.$$

VI's 1,868 true calls comprise 1,867 updates and the final check. IE-VI uses $P0 = 0$ without a call and charges one true call after each of its six updates, reusing it in the next correction. Its surrogate totals—7,182, 7,149, 7,155, and 7,156, in increasing order of surrogate microsteps—include defect construction and all inner checks (Table N.10).

Ten randomized-order repetitions use order seeds 21260729–21260738. No outliers are removed. These are repetitions of one fixed problem. Times are summarized by mean and standard error; speedups are means of paired per-repetition ratios, with Student-t 95% intervals (Table N.8). These

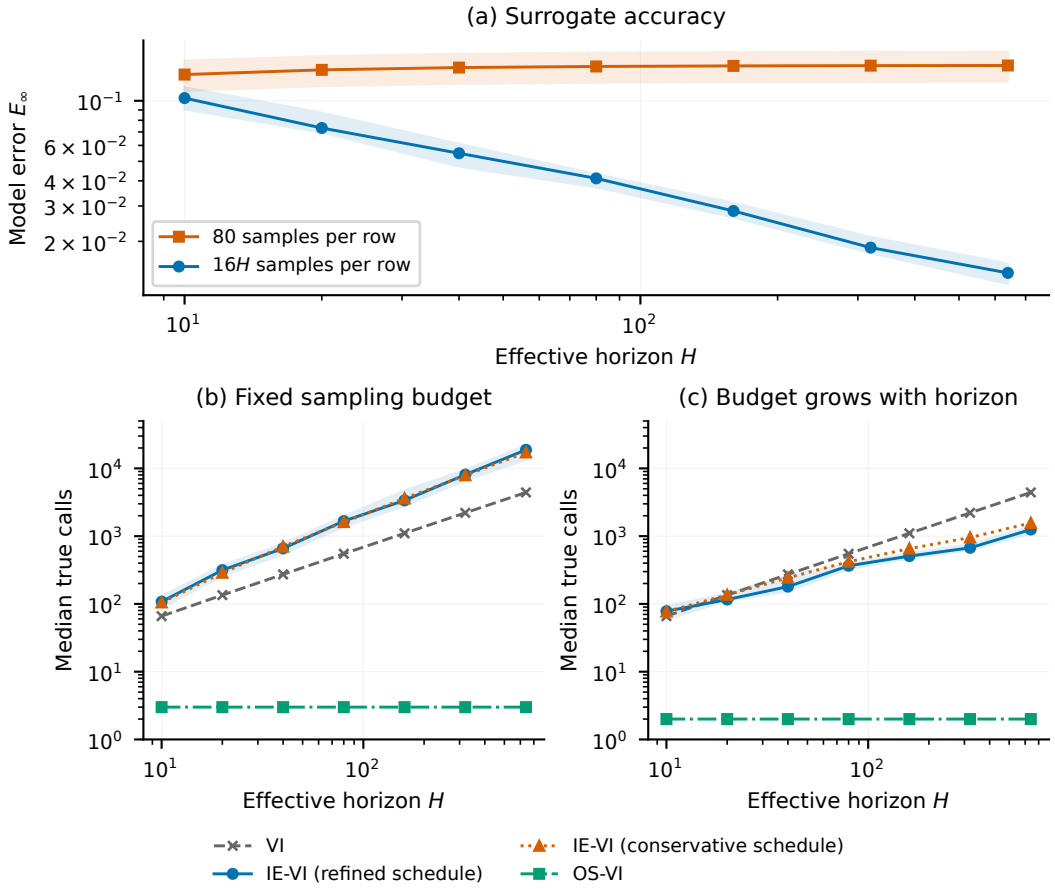


Figure N.16: Policy evaluation with 80 or $16H$ surrogate samples per row. Curves are medians over 30 paired surrogate seeds; the shaded bands are interquartile ranges for model error and IE-VI counts under the refined schedule. Counts exclude offline sampling and the verification tail. The conservative and refined schedules are the two theory-derived choices of h and m specified in Section N.13.2.

Table N.5: All seven horizons with a fixed budget of 80 samples per row. Query entries are median N_P [25th, 75th percentile] over 30 paired surrogate seeds. Counts start at the first of 50 qualifying outer updates and exclude the remaining verification tail and offline samples. The target and both IE-VI parameter schedules are specified in Section N.13.2.

H	VI	OS-VI	Median E_∞
10	66[66, 66]	3[3, 3]	0.135000
20	135[135, 135]	3[3, 3]	0.142500
40	273[273, 273]	3[3, 3]	0.146250
80	550[550, 550]	3[3, 3]	0.148125
160	1,102[1,102, 1,102]	3[3, 3]	0.149062
320	2,208[2,208, 2,208]	3[3, 3]	0.149531
640	4,418[4,418, 4,418]	3[3, 3]	0.149766

H	IE-VI (conservative schedule)	IE-VI (refined schedule)
10	105[84, 133]	108[88, 128]
20	288[240, 344]	315[260, 370]
40	702[576, 819]	660[545, 770]
80	1,610[1,330, 1,880]	1,668[1,386, 1,944]
160	3,597[2,992, 4,191]	3,372[2,808, 4,690]
320	7,896[6,576, 9,216]	8,078[6,727, 9,429]
640	17,173[14,313, 20,046]	18,808[13,496, 21,952]

Table N.6: All seven horizons with a growing budget of $16H$ samples per row. Query entries are median N_P [25th, 75th percentile] over 30 paired surrogate seeds. Counts start at the first of 50 qualifying outer updates and exclude the remaining verification tail and offline samples. The target and both IE-VI parameter schedules are specified in Section N.13.2.

H	VI	OS-VI	Median E_∞
10	66[66, 66]	2[2, 3]	0.103125
20	135[135, 135]	2[2, 2.75]	0.073229
40	273[273, 273]	2[2, 2]	0.054844
80	550[550, 550]	2[2, 2]	0.041146
160	1,102[1,102, 1,102]	2[2, 2]	0.028337
320	2,208[2,208, 2,208]	2[2, 2]	0.018627
640	4,418[4,418, 4,418]	2[2, 2]	0.013943

H	IE-VI (conservative schedule)	IE-VI (refined schedule)
10	77[70, 91]	78[59.5, 92]
20	136[128, 164]	116[112, 141]
40	243[209.25, 279]	180[154, 265]
80	420[382.5, 447.5]	365[327.5, 383.75]
160	654.5[605, 723.25]	507.5[470, 553.75]
320	954[888, 1,077]	672.5[630, 758.75]
640	1566.5[1381.25, 1738.75]	1,251[1102.5, 1390.5]

Table N.7: Full-range slopes from ordinary least-squares fits of log median versus log H, over $H = 10$–640. Brackets are paired-seed bootstrap 95% intervals from 4,000 resamples. Degenerate intervals for VI and fixed-budget model-error slopes follow the design; they do not imply independent environment replications. These intervals differ from the interquartile bands in the horizon figure.

Budget	Quantity	Slope	95% interval
80 per row	E_∞	0.022	[0.022, 0.022]
80 per row	VI	1.0097	[1.0097, 1.0097]
80 per row	OS-VI	0	[0, 0]
80 per row	IE-VI (conservative schedule)	1.2133	[1.1985, 1.2164]
80 per row	IE-VI (refined schedule)	1.216	[1.1992, 1.2192]
$16H$ per row	E_∞	−0.4844	[−0.5089, −0.4575]
$16H$ per row	VI	1.0097	[1.0097, 1.0097]
$16H$ per row	OS-VI	0	[−0.0627, 0]
$16H$ per row	IE-VI (conservative schedule)	0.7175	[0.6906, 0.7474]
$16H$ per row	IE-VI (refined schedule)	0.6635	[0.6326, 0.6866]

intervals use an independence approximation despite possible correlated desktop noise. All 50 runs reach the stopping target.

The primary comparison measures solver time, including iterations and convergence checks. It excludes operator construction, warm-up/JIT, initial vector and threshold preparation, reward and reference construction, pilot calibration, imports, startup, and file I/O. A separate comparison adds the recorded preparation time of the true operator and, for IE-VI, its selected surrogate operator (Table N.9). These preparation measurements include coefficient construction, kernel warm-up, and any JIT compilation or cache loading encountered. Operators were prepared before randomized solver execution. This sum still excludes the other preparation costs listed above and is not end-to-end experiment-generation time. Reference errors are assessed afterward. No OS-VI timing baseline is included.

N.13.4 Compute and artifact

The new studies ran on an AMD Ryzen AI Max+ 395 with 16 physical cores, 32 logical processors, 128 GiB RAM, and Windows 11 Pro 10.0.26200. Each process used one numerical-library thread

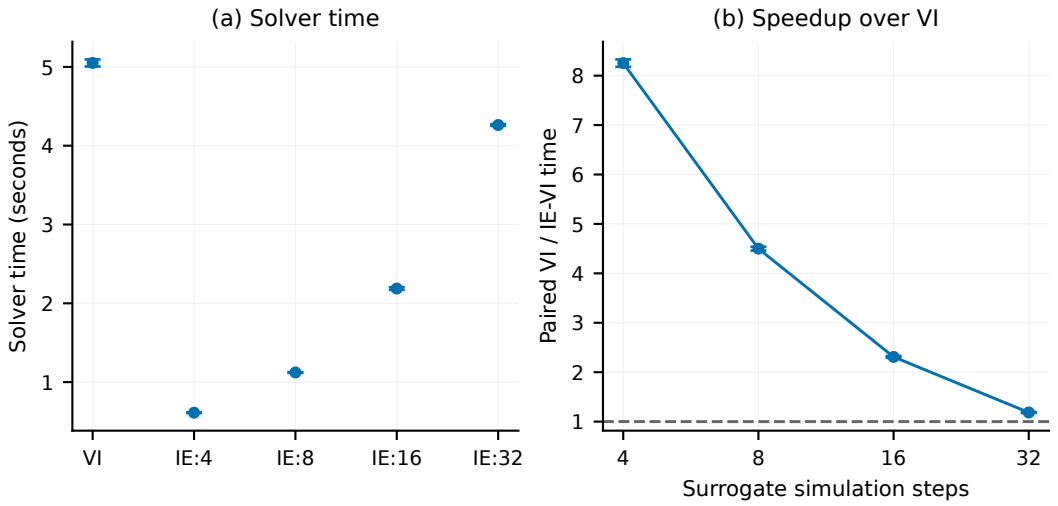


Figure N.17: Time and paired speedup relative to VI on one 10,000-state simulator. Points show means and error bars one standard error over ten randomized-order repetitions. Timings include convergence checks and exclude setup, warm-up, reference construction, and pilot calibration. Tables N.8 and N.10 gives operator counts and paired 95% intervals.

Table N.8: Solver-only time and paired speedup over ten randomized-order repetitions of one fixed 10,000-state simulator. Time is mean ± standard error in seconds. Speedup is the mean of per-repetition VI/IE-VI ratios with a Student-t 95% interval. Setup, warm-up, reference construction, pilot selection, and other preparation are excluded. No OS-VI timing baseline was run.

Method	Solver seconds	Paired speedup [95% interval]
VI	5.05024 ± 0.04468	—
IE-VI, 4 steps	0.61198 ± 0.00380	$8.2538[8.0848, 8.4229]$
IE-VI, 8 steps	1.12240 ± 0.00451	$4.4998[4.4124, 4.5871]$
IE-VI, 16 steps	2.18741 ± 0.01749	$2.3093[2.2655, 2.3532]$
IE-VI, 32 steps	4.26290 ± 0.01151	$1.1847[1.1613, 1.2082]$

Table N.9: Recorded operator-preparation plus solver time for the same ten repetitions as Table N.8. Each method is charged its measured true-operator preparation; IE-VI is also charged preparation of its selected surrogate. Preparation includes coefficient construction and kernel warm-up/JIT encountered in the recorded execution order. Other preparation and experiment-generation costs are excluded. Time is mean ± standard error in seconds; speedup is the mean of paired VI/IE-VI ratios with a Student-t 95% interval.

Method	Preparation + solver seconds	Paired speedup [95% interval]
VI	5.31649 ± 0.04731	—
IE-VI, 4 steps	0.88008 ± 0.00593	$6.0412[5.9497, 6.1328]$
IE-VI, 8 steps	1.39112 ± 0.00837	$3.8220[3.7547, 3.8894]$
IE-VI, 16 steps	2.45936 ± 0.02140	$2.1623[2.1232, 2.2015]$
IE-VI, 32 steps	4.54040 ± 0.01473	$1.1709[1.1491, 1.1928]$

Table N.10: Charged operator counts and maximum final relative true residual for the timing study in Table N.8. Counts include convergence checks and are identical across the ten repetitions of each method. All 50 runs meet the online relative true-residual target 10^{-5}. These repetitions measure timing variability on one fixed problem, not variation across environments.

Method	N_P	$N_{\widehat{P}}$	Maximum relative residual
VI	1,868	0	9.98242×10^{-6}
IE-VI, 4 steps	6	7,182	4.50529×10^{-6}
IE-VI, 8 steps	6	7,149	4.35107×10^{-6}
IE-VI, 16 steps	6	7,155	4.44089×10^{-6}
IE-VI, 32 steps	6	7,156	4.50428×10^{-6}

and CPU execution. The accepted deterministic sweep used one worker; horizon and timing studies ran sequentially. No exclusive CPU affinity was imposed, and ordinary background activity remained. Software versions were Python 3.13.15, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, Numba 0.67.0, and Matplotlib 3.11.2.

The accompanying reproduction package includes the data and code for the reported experiments, with instructions in `REPRODUCE.md`.

Paths are relative to the repository root. Planning records and outputs are in `datasets/reference/planning/archives/`; the RL data for Figure 1(b) are in `studies/cliffwalk/`. Verified extraction materializes the records into a local cache without changing the archives. From the repository root, `python reproduce.py verify-dataset` verifies the planning archives, and `python reproduce.py render-paper -output-root artifacts/paper` generates the figure and table exports in a new output directory. The commands `python reproduce.py run all` and `python reproduce.py compare all` run the planning experiments and compare their outputs with the reference data.

For the residual-incumbent experiment in Section N.13.5, the package includes the 38 common-solver restoration trajectories used in Table N.11. Its implementation is under `methods/incumbent-restoration/`. Source identities, independently certified reference targets, effective settings, diagnostics, and comparison reports accompany the traces. Historical-build, reconstructed-configuration, and unavailable historical target-array qualifications still apply: fresh execution does not establish identity with every historical trace. Keep accepted archives and retained historical data intact when reproducing the results. Displayed table values are rounded; the numerical exports retain full precision.

N.13.5 Residual-incumbent restoration

We also evaluate a residual-incumbent variant of the practical one-round schedule in Appendix J. It retains the accepted value estimate with the smallest observed true Bellman residual, starting from V_0, and restores that estimate whenever the step size decreases. For Control, it saves the value together with its cached true-model products and associated output policy; the initial policy is greedy with respect to V_0. A strict residual improvement replaces the saved tuple. A reduction after either a stall or a guard rejection restores it before the next surrogate computation. An unchanged step-size floor does not trigger restoration. Cached restoration requires no additional true-model application. The Policy-Evaluation implementation saves the value and its Bellman-residual vector.

This variant uses the practical one-round stall rule; the incumbent construction in Corollary 34 concerns the separate prescribed multi-round epochs. For the experiment here, we use the same deterministic Cliffwalk instance, 19-setting corruption grid, zero initialization, certified surrogate VI solver, and 1000-true-application budget as Section N.13.1. The settings are $h_{\text{init}} = 10^6$, $h_{\min} = 1$, $\beta = 0.1$, and $S = 3$, with inner value-error targets 10^{-8} and 10^{-2}. Counts in Table N.11 use the same retrospective stable policy-error criterion. Each recorded output consists of the value and policy carried after any restoration; the attempted candidate's surrogate work remains charged.

Table N.11: Stable policy-error hit costs for the carry-forward and residual-incumbent variants on the complete common-solver corruption grid. For each variant, N_P coincides at the two inner value-error targets; the two $N_{\widehat{P}}$ columns report surrogate applications at the indicated target δ_{solve}. Counts include initialization and checks, as specified in Section N.13.1. These deterministic runs use one instance.

	Carry-forward			Residual incumbent		
λ	N_P	$N_{\widehat{P}}$		N_P	$N_{\widehat{P}}$	
		10^{-8}	10^{-2}		10^{-8}	10^{-2}
0	2	2637	1263	2	2637	1263
0.1	2	2637	1263	2	2637	1263
0.2	3	4358	1612	3	4358	1612
0.3	4	5163	1671	67	29994	11323
0.4	5	5288	1807	66	29962	11423
0.5	9	6201	2271	66	29962	11459
0.6	11	6665	2430	65	29930	11367
0.625	15	7230	2711	65	29930	11361
0.65	19	7716	2931	65	29930	11372
0.675	20	7643	2904	65	29930	11373
0.7	21	7701	2925	65	29930	11367
0.725	23	7717	2953	65	29930	11361
0.75	44	8221	3147	65	29930	11355
0.775	69	8931	3449	64	29898	11339
0.8	96	9659	3745	64	29898	11333
0.85	163	11414	4527	64	29898	11331
0.9	213	12623	5076	64	29898	11323
0.95	261	13794	5641	64	29898	11317
1	321	15285	6374	63	29866	11297

Restoration raises N_P from 4–44 to 65–67 for λ from 0.3 to 0.75, above VI's 49, and lowers it from 69–321 to 63–64 for $\lambda \geq 0.775$, while using more surrogate applications at every $\lambda \geq 0.3$. We therefore keep carry-forward in the primary studies.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state the scope as discounted MDP planning with an RL algorithmic variant. The theorem-level claims are for exact model-query operators and are stated in Section 4.2, with supporting proofs in Appendix D. IE-DYNA is specified and evaluated empirically, but this draft does not claim a stochastic convergence theorem for it.

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   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

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